

LOW CYCLE FATIGUE CRACK PROPAGATION IN STEEL UNDER VARIOUS CYCLIC LOADING HISTORIES

By

M. Arief Rahman Panjaitan

A dissertation submitted to Nagoya University
in partial fulfillment of the requirements for
the Degree of Doctor of Engineering

Department of Civil Engineering
Graduate School of Engineering, Nagoya University
Nagoya, Japan

ABSTRACT

Low cycle fatigue crack is a crack which propagates by the repetition of large plastic strains. According to the researches on low cycle fatigue, the greatest of total life is consumed in crack propagation phase and only small part of the total life is spent for crack initiation, so that crack propagation manner in low cycle field should be carefully determined. The ability to estimate low cycle fatigue (LCF) crack propagation in steel is an essential aspect for assessment of the LCF lifetime or maintenance intervals. Due to the importance of LCF crack investigation, researches in LCF propagation have been more sophisticatedly conducted and revealed that cyclic J -integral range, ΔJ , is a parameter that correlates to the crack growth rate and is applicable to describe its manner. This dissertation evaluates crack propagation behavior under variable amplitude loading through employing several LCF tests, determines parameters that correspond to the crack growth rate and develops an estimation model to evaluate crack propagation.

In the first part of the work, LCF crack propagation under constant amplitude loadings was experimentally investigated. Since it was already found that ΔJ is a parameter governing the crack growth rate, the ΔJ was also resulted from FE analysis in this study. To ensure the accuracy of estimation model, additional LCF test data from a past study were also considered and collectively plotted with the current investigation on the LCF. Both results indicated that the crack growth rates in the LCF correlate with the ΔJ and were also distributed in the same region within narrow band. According to the results, a formula for the LCF crack growth rate under constant amplitude loadings was proposed.

As the second part of the work, evaluation was conducted under two-steps variable amplitude conditions to simply reveal crack propagation behavior when the amplitudes

were changed. Accordingly, several test cases with the same value of high amplitude and different values of low amplitude loading were employed. The results presented that crack growth rate under high amplitude performed equivalent trend to the crack growth under constant amplitude. Contrarily, under low amplitude loading, the majority of the crack growth rates were noticeably to be present above the regression curve of constant amplitude conditions indicating higher crack extension rates. To investigate accelerated crack growth at low amplitude loading, a correction factor, which is the ratio of the crack growth rate under variable conditions to that under constant amplitude loading was introduced and verified to be correlated with the crack propagation manner.

At the last part, the study examines crack growth rate under random variable amplitude loadings where all amplitudes were un-repetitive to present fully random attributes. The difference on the COD magnitude and COD average were also designated in these test cases. The result clarified the same crack growth manner to the two-steps variable cases. A higher crack growth rates than those under constant amplitude loading was identified when the amplitude was decreased from high to low levels. In contrast, the indistinguishable manner on the crack growth rates to those under constant amplitude were figured out when the amplitudes were escalated from low to high levels. At the final work, the estimation model obtained from two-steps variable was then evaluated to the random loading and resulted the good interpretation due to crack propagation manner. Those findings confirmed that the proposed formula is applicable to the variable amplitude loading.

ACKNOWLEDGEMENT

First and foremost, praises and thanks to the God, Allah SWT, the Almighty, for His showers of blessings throughout my research work to complete the research successfully.

I would like to express my deep gratitude to my research supervisor, Professor Kazuo Tateishi for providing me invaluable guidance throughout this research. He has taught me the methodology to carry out the research and to present the research works as clearly as possible. It was a great privilege and honor to work and study under his guidance.

I also extend my sincere appreciation to Prof. Junji Kato and Dr. Takeshi Hanji of Nagoya University for giving valuable advice in the seminar and dissertation. I am also grateful to Prof. Kengo Anami of Shibaura Institute of Technology for providing constructive comments and his valuable time in reviewing my dissertation.

I am also thankful to Dr. Masaru Shimizu for the many valuable helps, advice and enjoyable discussions. Many thanks go to Ministry of Education, Culture, Sports, Science and Technology (MEXT) of Japan who provided me the scholarship to support my study in Nagoya University.

I am very much thankful to my wife, Ayu Raina Mufida and my daughters, Qahirah and Husna, for their love, understanding, prayers and continuing support to complete this research work. Also I express my thanks to my parents, sister, and brother, for their support and encouragement.

Many thanks also go to all my friends and colleagues, for their knowledge sharing, helped me to deal with the administration in Japan and made my life so happy by their warm friendship during those past years.

Finally, my thanks go to all the people who have supported me to complete the research work and study in Nagoya University.

Table of contents

ABSTRACT	I
ACKNOWLEDGEMENT	III
Table of contents	V
List of figures.....	IX
List of tables	XIII
1. Introduction	1
1.1. Background.....	1
1.2. Previous studies	3
1.2.1. Low cycle fatigue (LCF)	9
1.2.2. Crack growth rate evaluated by EPFM.....	13
1.2.3. Crack closure	14
1.2.4. J -integral	17
1.2.5. Cyclic J -integral range.....	21
1.2.6. Path independence of ΔJ	22
1.3. Research objectives.....	23
1.4. Organization of dissertation contents.....	23
2. Basic Methodology of the Study	27

2.1.	Introduction.....	27
2.2.	Specimen.....	27
2.3.	Crack propagation measurement.....	28
2.4.	FE modeling.....	30
2.5.	Summary.....	36
3.	Constant Amplitude Loading (CA)	39
3.1.	Introduction.....	39
3.2.	Previous study on the constant amplitude loading.....	39
3.3.	Present study on constant amplitude loading.....	42
3.4.	Summary.....	47
4.	Two-steps Variable Amplitude Loading (VA).....	49
4.1.	Introduction.....	49
4.2.	Load cases.....	49
4.3.	Load-COD relationship.....	50
4.4.	Crack propagation behavior	52
4.5.	Amplification factor (AF)	55
4.6.	Estimation model on the AF	58
4.7.	Validation on the estimation to the test results crack growth rate	62
4.8.	Crack propagation analysis	64
4.9.	Summary.....	65
5.	Random Variable Amplitude Loading (RA)	67
5.1.	Introduction.....	67

5.2.	Load cases.....	67
5.3.	Test result arranged with ΔJ	70
5.4.	Summary	77
6.	Summary and Conclusions	79
6.1.	Conclusion	79
6.2.	Recommendation for future study	81
	References	83
	APPENDIX A: Digital image of the crack.....	89
	APPENDIX B: Crack propagation analysis	91

List of figures

Fig. 1.1 Low cycle fatigue damages in steel bridge piers in the Hyogo-ken Nanbu earthquake.....	2
Fig. 1.2 Illustration of crack closure (Elber, 1971)	4
Fig. 1.3 Plasticity area around the crack tip and the plastic envelope formed in the beginning of a crack propagation (Elber, 1971)	4
Fig. 1.4 Domain of the effective stress intensity which was introduced by Elber (1971)	5
Fig. 1.5 Variability of the effective stress intensity affected by the crack closure due to the difference in applied stress range (Elber, 1971)	5
Fig. 1.6 Assignment of the tensile overload in steady cycles.....	6
Fig. 1.7 Retardation in crack propagation due to overload employment in tensile phase	6
Fig. 1.8 Displacement normal to the crack plane (u_y) at minimum load	8
Fig. 1.9 Number of cycles range for low cycle fatigue (Agrawal et al., 2014)	10
Fig. 1.10 Strain-life relationship in ELCF and LCF regimes for hot-rolled carbon steel (Gardner and Davies, 2010).....	10
Fig. 1.11 Idealized elastoplastic stress-strain behavior (Kim, 2015).....	12
Fig. 1.12 Hardening model in material behavior (Kim, 2015)	12
Fig. 1.13 The three loading modes of a crack (Dahlin and Olsson, 2005)	13
Fig. 1.14 Fatigue crack closure mechanisms in steels (Anderson, 2011).....	16
Fig. 1.15 J closed contour in a two-dimensional solid (Anderson, 2011).....	18
Fig. 1.16 Two arbitrary contours Γ_1 and Γ_3 surrounding the crack tip. If such contours are	

linked by Γ_2 and Γ_4 , a specific contour is presented (Anderson, 2011).....	20
Fig. 1.17 Stress-strain curve used to calculate ΔJ (Par Ljustell, 2013)	20
Fig. 1.18 Flow chart of the research	24
Fig. 2.1 Specimens (unit: mm)	28
Fig. 2.2 Experimental program and the observation	29
Fig. 2.3 True stress-strain relationship established with the principle of kinematic hardening rule	30
Fig. 2.4 Assignment of cyclic loading to the 3D-specimen model and the boundary condition of the model.....	31
Fig. 2.5 Fluctuation range where ΔJ was defined	33
Fig. 2.6 Integration path used for independency verification (unit: mm).....	33
Fig. 2.7 The ΔJ independency to integration path.....	34
Fig. 2.8 Loading cycles used for independency verification.....	34
Fig. 2.9 The ΔJ independency to loading cycles	35
Fig. 2.10 Integration path for analysis (unit: mm).....	35
Fig. 2.11 ΔJ -crack length relationships	36
Fig. 3.1 Location of weld metal on the CT specimen.....	40
Fig. 3.2 Crack growth rate and ΔJ under constant amplitude loading at the past study.	41
Fig. 3.3 Load-COD hysteresis loop	42
Fig. 3.4 Load-COD verification	43
Fig. 3.5 Crack propagation behavior and ΔJ distribution under constant amplitude loadings.....	44
Fig. 3.6 ΔJ -crack growth rate relationships under constant amplitude loadings.....	45
Fig. 3.7 Crack length-number of cycles relationships under constant amplitude loadings	46
Fig. 4.1 Loading patterns of two-steps variable amplitude loadings.....	50
Fig. 4.2 Load-COD hysteresis loop under VA-1	51

Fig. 4.3 Crack growth rate behavior	52
Fig. 4.4 Crack growth rate-crack length relationships.....	52
Fig. 4.5 ΔJ -crack growth rate relationships under constant and two-steps variable amplitude loadings.....	54
Fig. 4.6 AF -crack length relationships.....	56
Fig. 4.7 AF - ΔJ ratio relationships under two-steps variable amplitude loadings.....	57
Fig. 4.8 AF - ΔJ relationships under two-steps variable amplitude loadings	59
Fig. 4.9 Cumulative plots of estimation and test results on AF - ΔJ ratio relationships ..	61
Fig. 4.10 Regression process to obtain estimated AF at $0.95 \leq \Delta J_i / \Delta J_{i-1} < 1$	62
Fig. 4.11 Estimated AF at the $\Delta J_i / \Delta J_{i-1} < 1$	63
Fig. 4.12 Estimated AF at the $0.95 \leq \Delta J_i / \Delta J_{i-1} < 1$	63
Fig. 4.13 Verification on the estimation model to the test result crack growth rate (unit: mm/cycle)	64
Fig. 4.14 Crack propagation analysis results	65
Fig. 5.1 Loading patterns of random variable amplitude loading	68
Fig. 5.2 Loading process used to define the ΔJ for RA-3.....	69
Fig. 5.3 The ΔJ for RA-3 defined based on constant amplitude principle	69
Fig. 5.4 Crack growth rate-crack length relationships under CA-1 and random amplitude loadings.....	71
Fig. 5.5 ΔJ -crack growth rate relationships under constant and random amplitude loadings	72
Fig. 5.6 AF - ΔJ ratio relationships under random amplitude loadings.....	73
Fig. 5.7 AF - ΔJ relationships under random amplitude loadings	74
Fig. 5.8 Verification of estimated crack growth rate to the test result (unit: mm/cycle)	75
Fig. 5.9 Crack propagation analysis results	76

List of tables

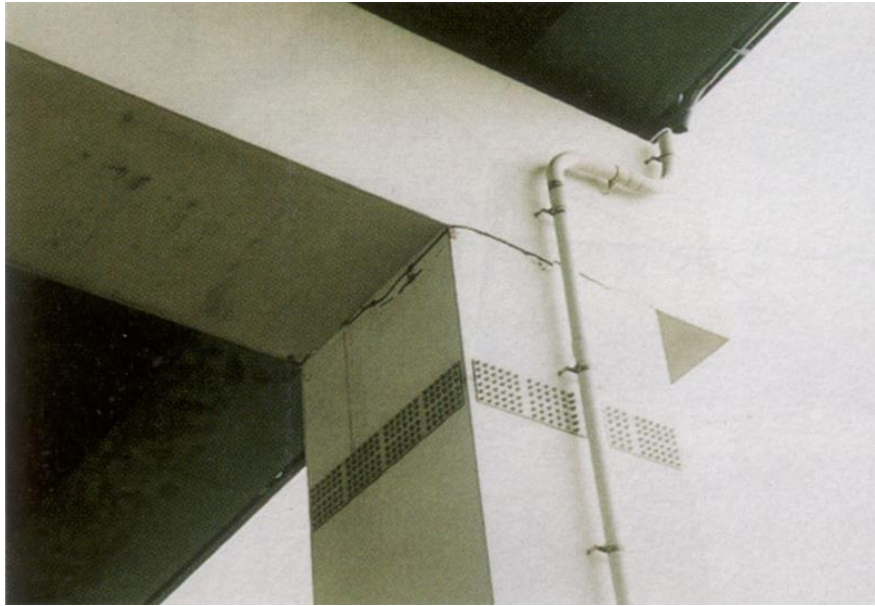
Table 2.1 Mechanical properties	27
Table 3.1 Test matrix for past research work	40

1. Introduction

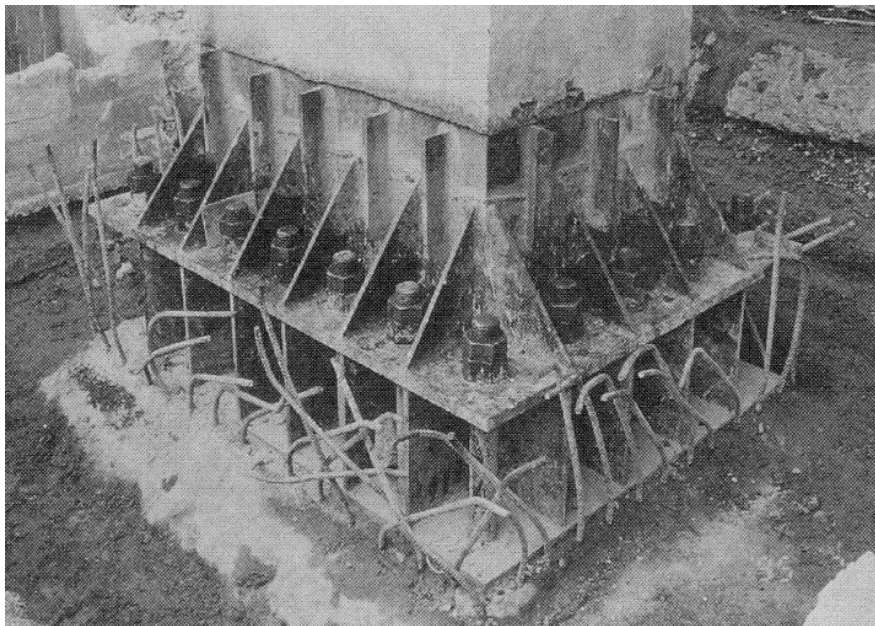
1.1. Background

Low cycle fatigue is a failure mode exhibiting a small number of large plastic strain repetitions to failure. One type of LCF damages is fracture identified on steel bridge components after Hyogo-ken Nanbu earthquake hit Japan in 1996 (Miki, 1997) as shown in **Fig 1.1**. This figure presents LCF failures observed in welded joints in steel piers, i.e., beam-column connection and triangular ribs on the base joint. Related to these failures, several researchers (Sakano et al., 1998; Sakano et al., 2001; Chen et. al, 2007; Park et al., 2015) carried out a series of LCF test on steel-beam column joint and pier base joint and evaluated crack propagation behavior.

Until recently, LCF crack propagation has primarily been investigated under constant amplitude loading (Terao et al., 2015 and Hanji et al., 2017) and evaluations on the variable amplitude loading were mainly conducted in high cycle fatigue (HCF) as conducted by Corbly and Packman (1973), Raghuvir et al. (1996), Trebules et al. (1973), Mc Millan and Pelloux (1967), and Matthews et al. (1971). These researches acknowledged high amplitude stress levels in uniform low amplitude stress conditions. In such studies, crack growth rate was noticeably retarded when amplitude decreased from high to low levels and accelerated when the amplitude rose from low to high levels. However, actual seismic motion are characterized by amplitude changeability from high to low or from low to high levels, and the effect of these alterations on crack growth in



(a) Beam to column connection



(b) Base joints of steel bridge pier

Fig. 1.1 Low cycle fatigue damages in steel bridge piers in the Hyogo-ken Nanbu earthquake

LCF field has not yet been clarified.

To maximize the fatigue resistance of steel member and prevent the failure in steel caused by LCF crack propagation due to seismic motion, it is important to reveal crack

propagation manner under variability of cyclic strain.

1.2. Previous studies

Principally, fatigue life is introduced as summation of crack initiation and crack propagation life. Murakami et al. (1983) discovered that crack propagation phase of LCF holds most the total life of the specimen during large cyclic strain employment than the phase of the crack initiation. Correspondingly, crack propagation life should be accurately estimated so that the function of the steel member can be maximized without the failure on the structure. At the previous research work, crack growth rate was identified to be linearly varied with the stress intensity factor range on log-log scales in the high cycle fatigue region known as Paris-Erdogan model (1963). However, this approach is not applicable to the LCF field, because large scale plastic deformation occurs around the crack tip in this region.

As far, a number studies evaluated crack growth manner due to the abrupt change in amplitude were confirmed in HCF. At the past, Elber (1971) ascertained that in case of the load is in tensile, a crack will not open in the following cycle until a remarkably high tensile load is employed to the specimen as indicated in **Fig. 1.2**. Evaluation on the sample of aluminum-sheet revealed the almost equivalent opening/closing stress value were indicated to be one half than the highest employed stress in R equal to 0 (R is the ratio of minimum to maximum stress of cyclic loading). R equal to zero means the cyclic loading fluctuates from zero to designated tensile stress.

Explanation of the mechanism on crack-closure is described. As depicted in **Fig. 1.3**, at the area around the crack tip, the existence of the plastic zone is indicated. The figure described that when the crack propagation passes this plasticity area (the size is developed to be wider together with crack length), such wake of growing crack will form the deformed material envelope as illustrated in the left side of the figure

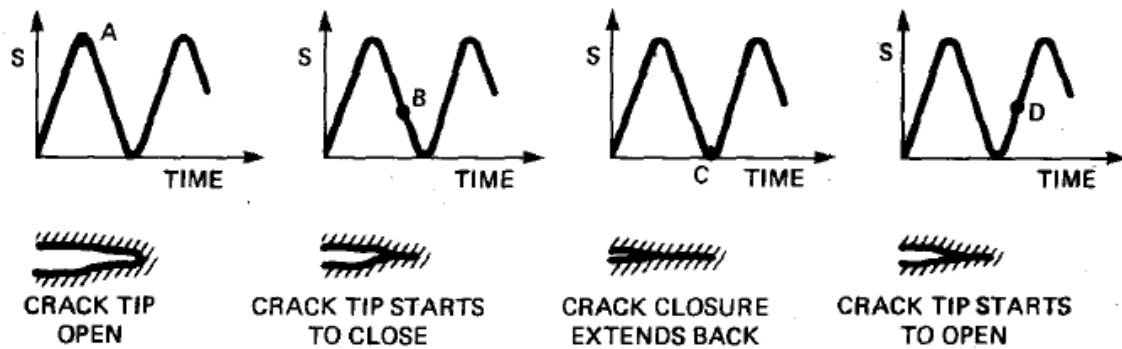


Fig. 1.2 Illustration of crack closure (Elber, 1971)

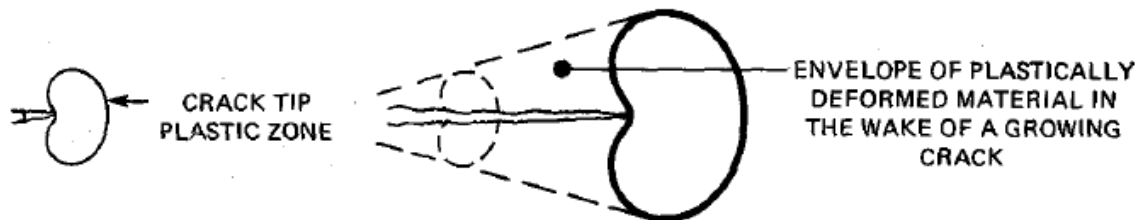


Fig. 1.3 Plasticity area around the crack tip and the plastic envelope formed in the beginning of a crack propagation (Elber, 1971)

In this envelope, the existence of residual tensile deformations are representation of residual compressive stresses. In this process, residual tensile deformation triggers the crack to be close until a remarkably high tensile loading was applied to the specimen, eventhough in tensile portion. Study conducted by Elber (1971) by X-ray-diffraction measurements also ascertained the presence of residual compressive stresses in the beginning of a crack.

Elber also discovered that the crack propagation rate are more satisfyingly to be related to an effective stress-intensity range, $\Delta K_{\text{eff}} = K_{\text{max}} - K_{\text{op}}$, as shown in **Fig. 1.4**, since a crack is open in the phase of the tensile load application, rather than the total ΔK which has the range from minimum to the maximum applied stress. The significance in

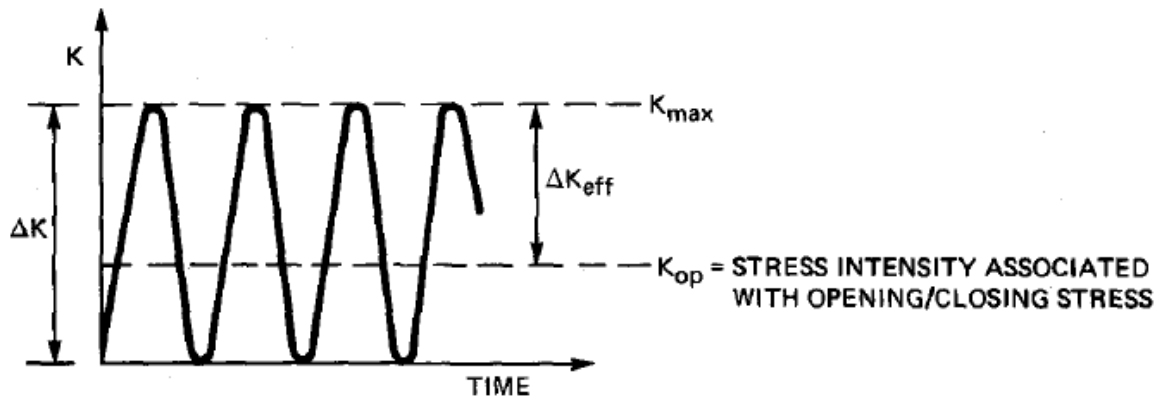


Fig. 1.4 Domain of the effective stress intensity which was introduced by Elber (1971)

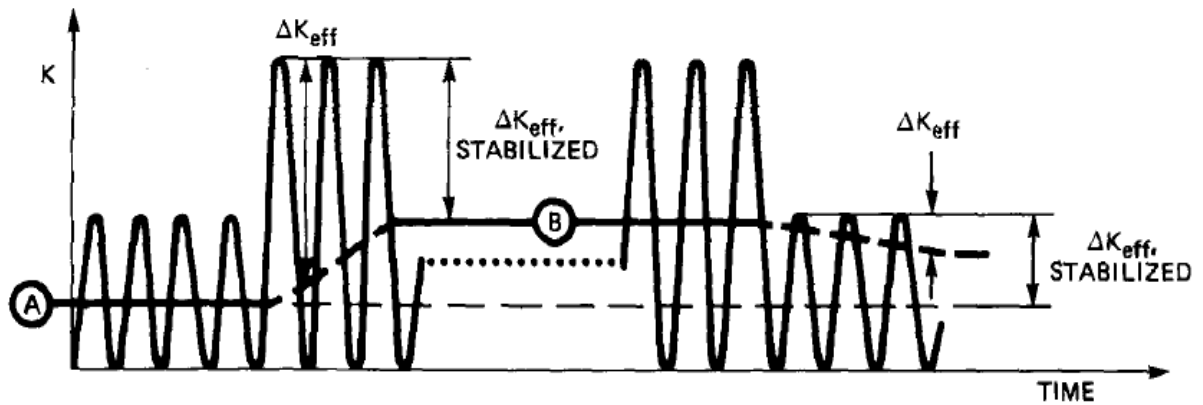


Fig. 1.5 Variability of the effective stress intensity affected by the crack closure due to the difference in applied stress range (Elber, 1971)

explaining evaluated load sequence effects in crack progress was then figured out from the effective stress intensity principle.

Both crack retardation and acceleration effects may be accounted on the crack closure as presented in **Fig. 1.5**. In figure, If A is defined as balanced closure stress intensity associated to the cyclic load at the lower level. The stabilized stress intensity addressed to the cyclic load at the higher level is indicated with B. In the process of transition from low to high, at the high level, the ΔK_{eff} is un-permanently to be higher

than B. Hence, at the high level, acceleration of propagation is obtained. Contrarily, much less value in ΔK_{eff} is indicated for the conversion from high to low level than the A-stabilized value and contribute the retardation in crack. Such behaviors have also been observed experimentally by Elber (1971).

Ling and Schijve (1992) and Shercliff and Fleck, (1990) undertook the tests to evaluate crack propagation behavior when one or more overloads were put in steady cycles as expressed in **Fig. 1.6**. These studies confirmed that crack growth rates of steady cycles behind overload were less than constant cycles without overload as shown in **Fig 1.7**. However, the behavior of the crack extension observed from those study are still

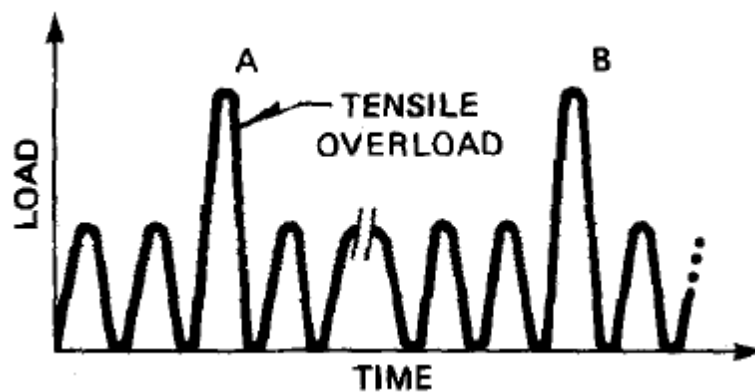


Fig. 1.6 Assignment of the tensile overload in steady cycles

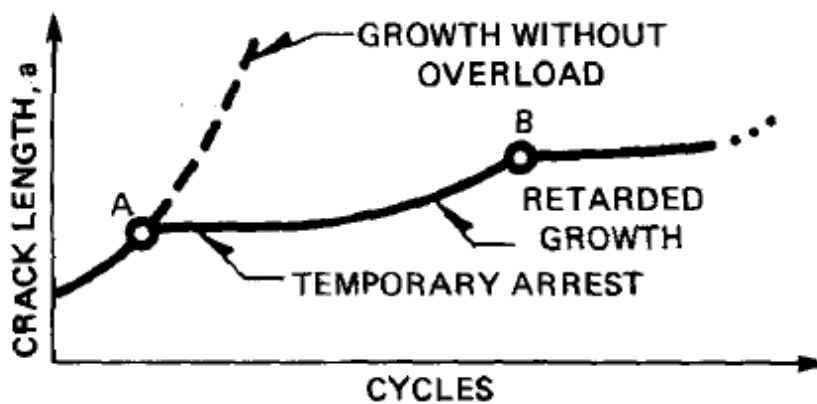


Fig. 1.7 Retardation in crack propagation due to overload employment in tensile phase

needed to be clarified in LCF domain since it is acknowledged that there is the difference on the plasticity area surrounding the crack tip under LCF and HCF in which this distinction may also lead to give the different crack growth manner.

Shahani et al. (2009) evaluated fatigue crack growth rate behavior under variable stress ratio and proposed four prediction models that contained some parameters, i.e., cyclic J -integral, crack tip opening displacement, crack mouth opening displacement and stress intensity factor, to estimate crack growth rate.

Gonzales-Herrera and Zapatero (2008) evaluated CT aluminum specimen model tri-dimensionally with considering different specimen thickness and stress range. This study discovered that crack closure and crack opening mechanism caused abrupt transition contour of deformation at the external side of the crack tip on the specimen slice model as expressed in **Fig 1.8**. This contour area was identified has similar size with other specimen regardless the thickness. Relied on this study, in 2D model, this significant abruptness cannot be identified.

Tateishi and Hanji (2004) introduced LCF strength curves for plain steel material, deposit metal and heat affected zone (HAZ), which were based on local strain amplitude at a cracking point. Accordingly, relationships between local strain amplitude and LCF life were presented in this study.

At the past study, LCF crack behavior in steel under multiaxial loading was also investigated (Itoh and Miyazaki, 2003). Examinations were done with developing correlation between two LCF life values from experiment and analytical study. Equivalent strain parameter resulted from analytical study was confirmed to give good estimation indicated by fatigue life distributed within narrow band in relationship of estimated-experimental fatigue life. Past studies (Tanaka et. al, 2006 and Tanaka et. al, 2003) also revealed crack propagation behavior from hole in tubular specimens subjected to cyclic torsional loading and identified that cyclic J -integral was an appropriate parameter to estimate crack growth. Dong et al. (2016) tested CT specimen made of Q345 steel under constant amplitude conditions and discovered linear relationship between

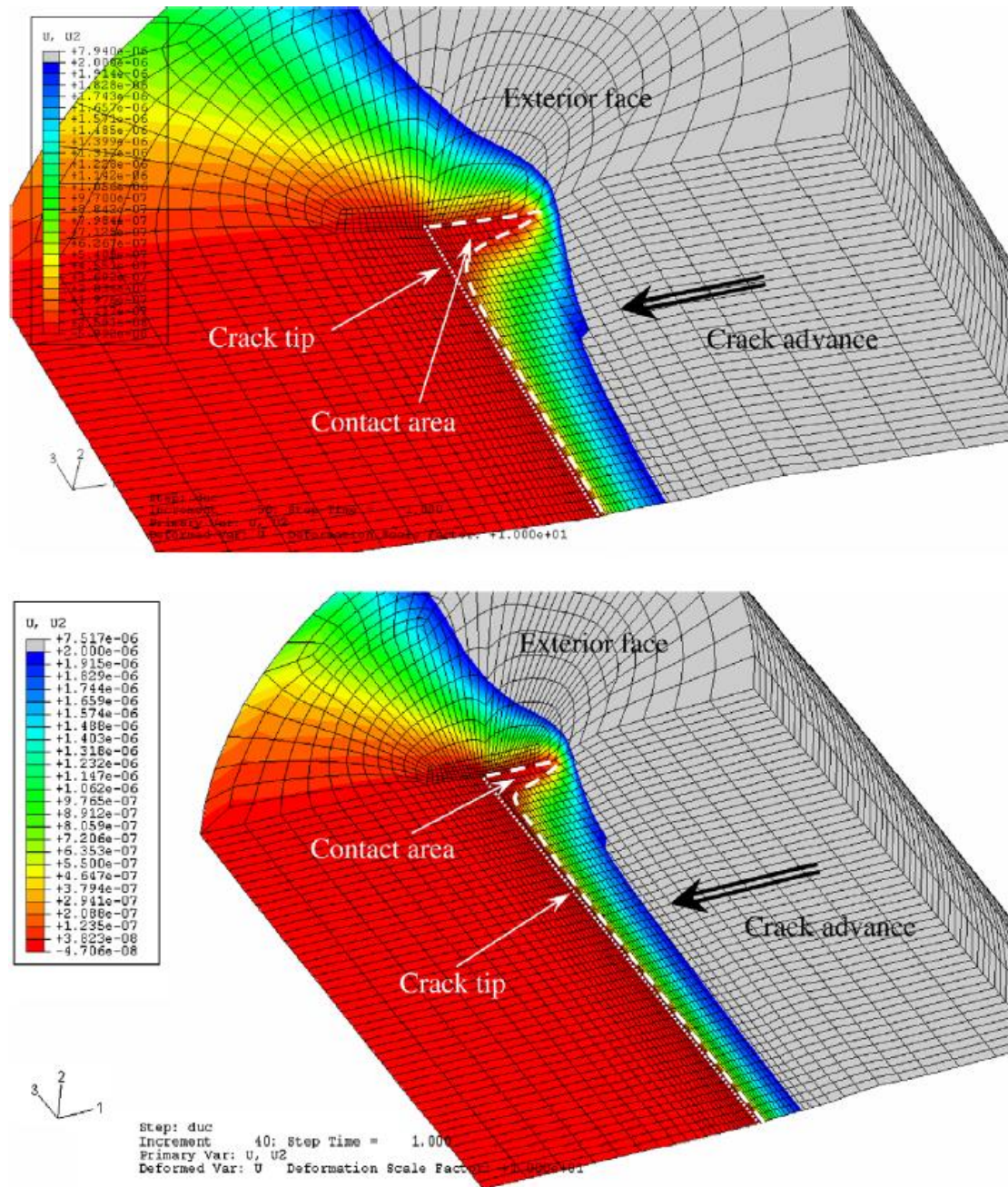


Fig. 1.8 Displacement normal to the crack plane (u_y) at minimum load

crack growth rate and the cyclic crack tip opening displacement (CTOD).

Former research works in low cycle fatigue also found that J-integral calculated from the area under a load–displacement hysteresis loop, called the cyclic J-integral range,

correlates with the crack growth rate (Usami and Kumar, 1996 and Masatoshi, 2002). Jono et al. (1993) carried out the load-controlled elasto-plastic fatigue crack growth tests under variable amplitude loading condition. It was found that crack growth rate can be estimated by cyclic J -integral range and also by fracture mechanics parameter $J_{\max}$ which is the maximum J -integral. Terao et al. (2015) tested CT specimen made of steel and deposit metal under constant amplitude conditions and examined crack propagation rate. Besides, the difference in average of displacement was also considered. This study discovered that crack growth rate correlates with cyclic J -integral range regardless the specimen steel grades and the average of cyclic displacement. Hanji et al. (2017) examined LCF crack propagation rate tendency under constant amplitude loading and established estimation model by acknowledging cyclic J -integral range (ΔJ) parameter. This study also considered ΔJ to reveal crack growth behavior in a corner welded joints. At the past research work, Tanuma and Kobayashi (2002) also applied the cyclic J -integral concept to crack growth estimation under extremely high-strain.

In this present research work, examination on crack growth under constant amplitude conditions was firstly conducted to obtain rigorous formula to estimate crack propagation. Next, crack growth rates under variable amplitude loading particularly when the amplitude loadings were changed, were evaluated to provide prediction model and then its applicability was verified.

1.2.1. Low cycle fatigue (LCF)

LCF crack is a typical crack which propagates at the high stress levels and a low number of cycles to failure. In all fatigue cracks, the weakening material caused by cyclic loading is started from the crack initiation and continued with the crack propagation. LCF is addressed with the fatigue life consumed in the crack propagation phase. As expressed in **Fig. 1.9**, at the stress range from zero to endurance strength, the occurring stress will

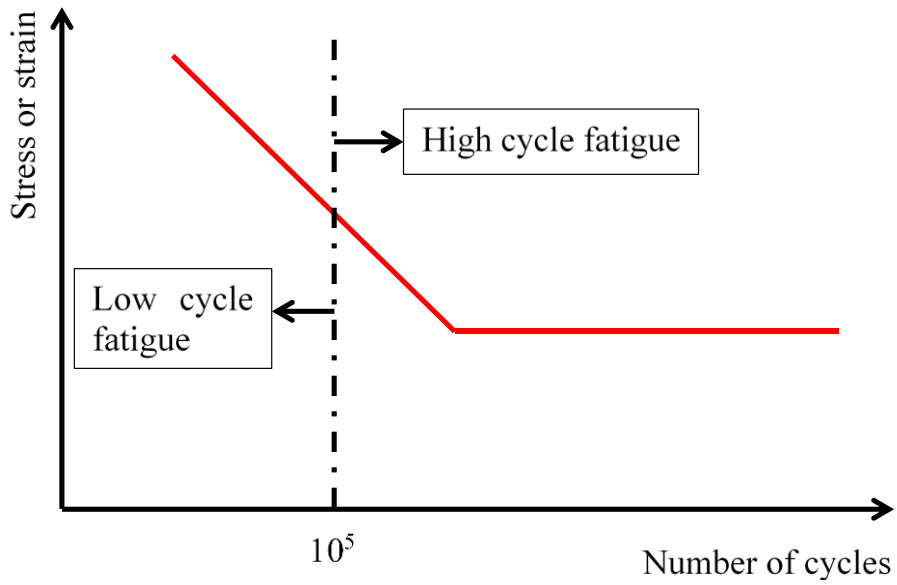


Fig. 1.9 Number of cycles range for low cycle fatigue (Agrawal et al., 2014)

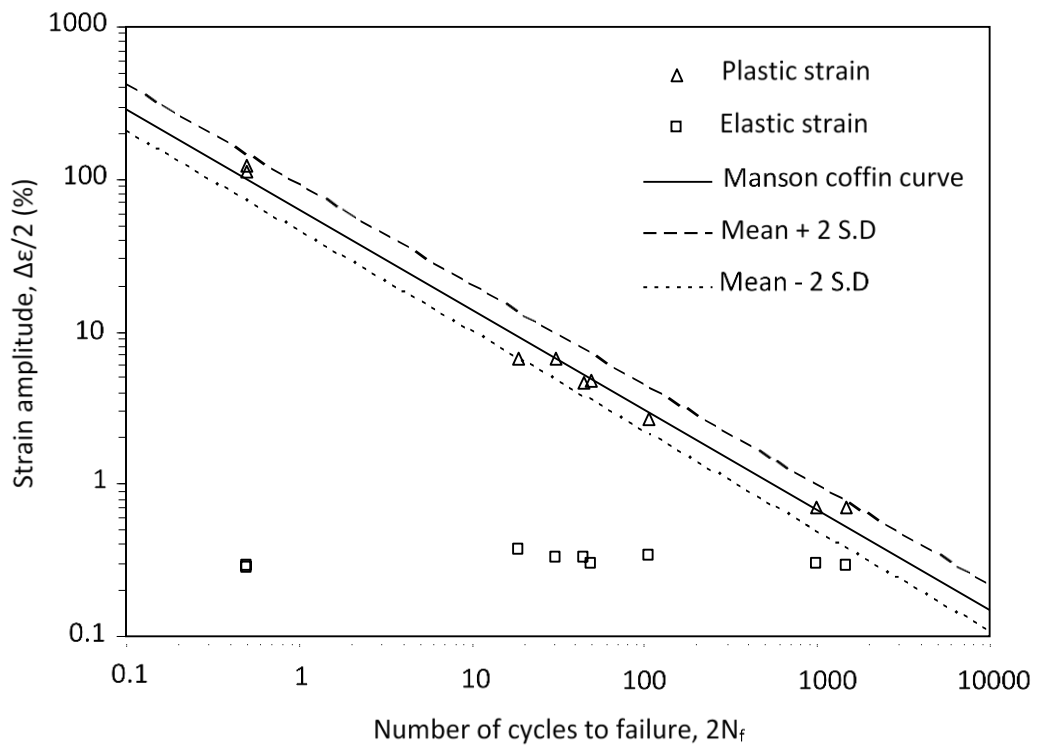


Fig. 1.10 Strain-life relationship in ELCF and LCF regimes for hot-rolled carbon steel (Gardner and Davies, 2010)

not cause the fatigue failure on the steel component, meanwhile when the stress range

over than endurance strength, it is possible to give fatigue failure on steel components. LCF is attributed with the number of cycles less than 10^5 cycles for its fatigue life, meanwhile high cycle fatigue (HCF) addressed its fatigue life over than 10^5 cycles. LCF could also be sub divided into:

- LCF for less than 10^5 cycles to failure.
- LCF for less than 100 cycles to failure.

The second part of the LCF at above classification is extremely low cycle fatigue (ELCF). The discussions on the (ELCF) have been made at the several past studies (Coffin, 1971; Shimada et al., 1987; Komotori and Shimizu; 1998; Chung and Abel, 1988). **Fig. 1.10** presented the difference on the strain-fatigue life behavior between LCF and ELCF. From the figure, the plastic strain-life data from LCF tests scatters approximately on a straight line when plotted on a log-log scale. This observation established the basis of the Coffin-Manson relationship (Coffin, 1954; Manson, 1965). The figure also indicates that ELCF fatigue data do not obey the Manson-Coffin Curve, where such curve was too over to estimate the fatigue life in ELCF. The failure mechanisms are also different in LCF and ELCF. For example, the center of the cylindrical bars in push-pull fatigue test is the common location of the fracture due to ELCF test. Meanwhile, at the same type of the test, the crack often starts from the surface under LCF test (Shimada et al., 1987; Kanvinde and Deierlein, 2005). Related to Manson-Coffin's law (strain-life relationships) application in ELCF, several previous studies (Tateishi et al., 2007, Hatanaka and Fujimitsu, 1984, Kuroda 2002) were carried out to control the disadvantage of formula.

1.2.2. Idealized stress-strain relationship

Generally, analytical work assumed material in idealized elastoplastic stress strain behavior due to employment of the cyclic loading as shown in **Fig. 1.11**. At the application of the tensile load, the stress increases with slope of E (elastic modulus) until

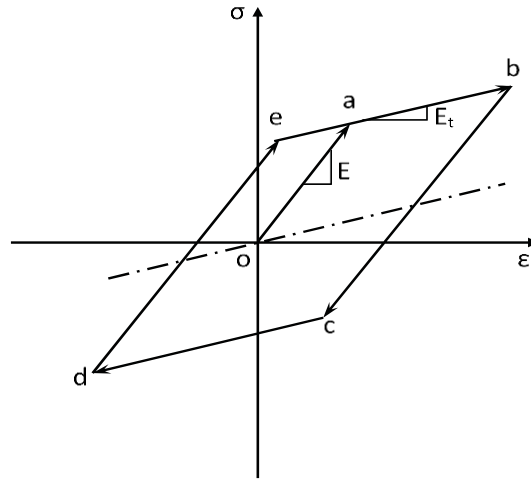


Fig. 1.11 Idealized elastoplastic stress-strain behavior (Kim, 2015)

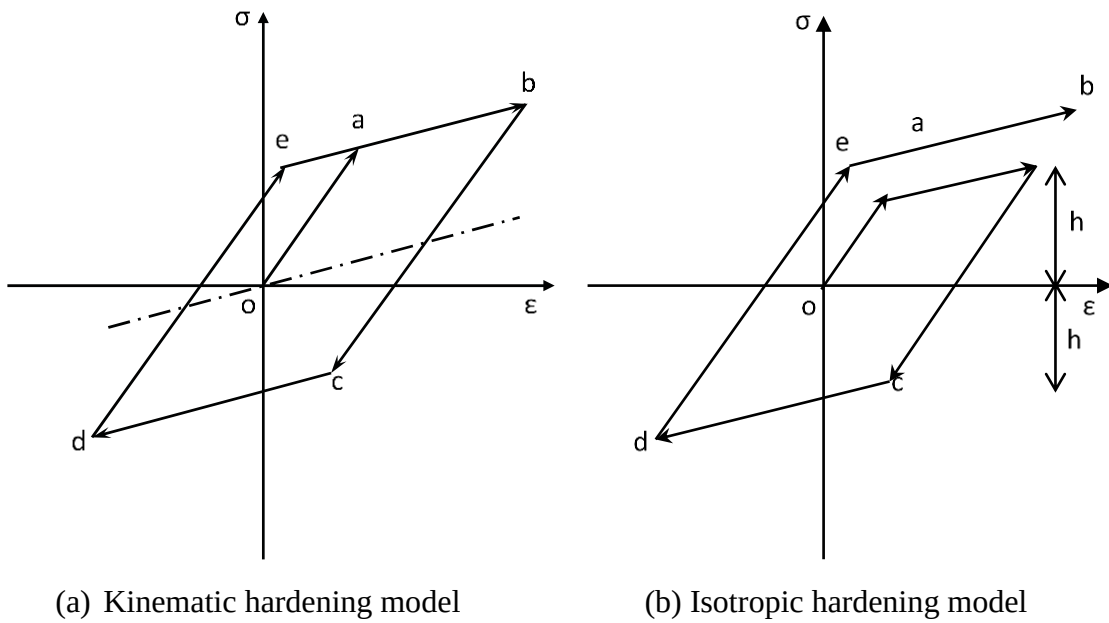


Fig. 1.12 Hardening model in material behavior (Kim, 2015)

yield stress, σ_y (line o-a). When the load is continued, after yielding, the stress experiences the plastic phase with slope E_t (tangent modulus) (line a-b). Upon removing the load, the stress perform elastic unloading with slope E (line b-c). Due to loading in the opposite

direction, the material will eventually yield in that direction (point c). At the strain hardening phase, line a-b or c-d, the specimen requires more force to continuously deform in plastic region.

The material behavior in hardening model is outlined in **Fig. 1.12**. **Fig. 1.12(a)** performs kinematic hardening model. In such model, the elastic range remain constant, where line b-c is equal to line d-e. The model also adheres the principle where the center of the elastic region, symbolized with dash-dot line moves parallelly to the work hardening line. The center of elastic domain is also utilized as an evolution variable of the model.

Fig. 1.12(b) exhibits isotropic hardening model. The model considers that elastic range (yield stress) proportionally increases to the plastic strain. Besides, it also contemplates the yield stress for the reversed loading is equal to the previous yield stress. Due to sustainability of cyclic loading, the plastic strain is used as an evolution variable of the model.

1.2.2. Crack growth rate evaluated by EPFM

Crack growth rate can be simply defined as the difference of the crack length at

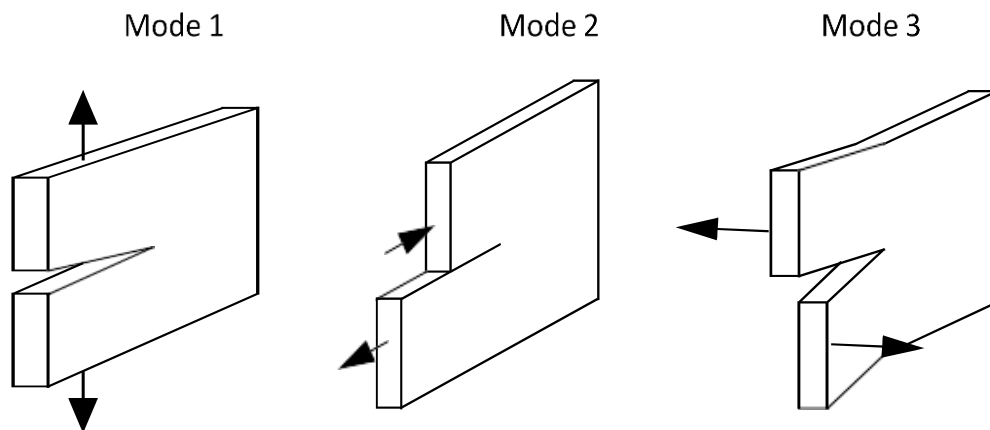


Fig. 1.13 The three loading modes of a crack (Dahlin and Olsson, 2005)

the peak amplitude of the current cycle to the crack length at the one cycle previously. Under large cyclic strains, Elastic Plastic Fracture Mechanics (EPFM) has proven to be an appropriate domain for the characterization of the fatigue process at the area surrounding the crack tip. In EPFM, specimen responses such as stress, strain, and deformation are assigned to evaluate crack growth rate. Such investigation is needed since the details of the fatigue mechanism under large cyclic strains are largely unknown. At the very low loading, EPFM may not be appropriate for fatigue crack growth analyses. Such low loads are not considered in this present study.

Fatigue crack growth behavior was usually evaluated through three types of mode, as expressed in **Fig. 1.13**. In most cases, mode 1 was selected to evaluate the crack growth rate. Mode 2 and 3 were usually employed with the other modes to obtain crack growth under combined loadings.

1.2.3. Crack closure

Crack closure is the specimen behavior where the crack surfaces experiences contact due to cyclic loading application. The closure is identified when the external load acting on the material. At the large cyclic loading, due to surfaces contact, the inflection points where the sustainability of the hysteresis loop is changed, will be indicated in the unloading process.

The crack closure can be simulated by the finite element method. At the 1970s, the sophisticated development of the computer allows the possibilities of using finite element models to evaluate the crack closure caused by plastic deformation at the crack tip. A very simple and small models were applied at the earlier studies but recently more complicated model can be examined due to rapid development of the computer capability. At the past, a related paper on plasticity caused crack closure investigation by the finite element analysis was carried out by Solanki et al. (2004). In such study, the crack is modelled and cyclic load was employed to the component as the basic principle. By

freeing the crack tip node at the several stages in the load cycle, crack growth is established. Consideration of a constitutive model of elastic plastic on the plastically deformed material is firstly applied in tensile load and followed by the cyclic loading employment. The crack closure is able to be evaluated when the first node behind the crack tip was released where the crack face leaves contact (crack opening) and followed by contact at the opposing crack surfaces. Several simulation were provided by several researches (McClung and Sehitoglu, 1989; Solanki et al., 2004; and Gonzales and Zapatero, 2005) and obtained the dependence of the crack closure influence relied on the finite element size. Additionally, at the separate studies, the results were also influenced by the choice of constitutive model (Jiang et al., 2005; Pommier and Bompard, 2000; and Pommier, 2001).

Fig. 1.14(a) expresses compelling evidence for plasticity-induced closure from the experiment was conducted by James and Knott (1985). In that study, they presented crack from the test in a standard specimen and to eliminate a little part of material from the surfaces of the crack approaching 0.5 mm from the tip, they utilized electron machining. They located the specimen back to the machine for test, and identified that accelerated crack propagation is indicated than the prior measurement. They asserted that the plastic wake triggered the closure and decreasing the crack propagation. Contrarily, acceleration in crack propagation is indicated since the residual stress in plastic area had been eliminated by machining.

Fig. 1.14(b) outlines roughness-induced closure which is affected by the microstructure. Relied on the observation, combined condition on the microscopic view is possibly caused by the deflection form of crack in case of un-homogeneity of the material, even though on a global scale, fatigue cracks propagate in pure Mode 1 conditions. Illustration about the typical modes are previously given in **Fig. 1.13**. When the crack is subjected to Mode 2, the crack leads to give different pattern from the Mode 1. The closure form can also be shown in **Fig. 1.14(b)**. The displacement-controlled

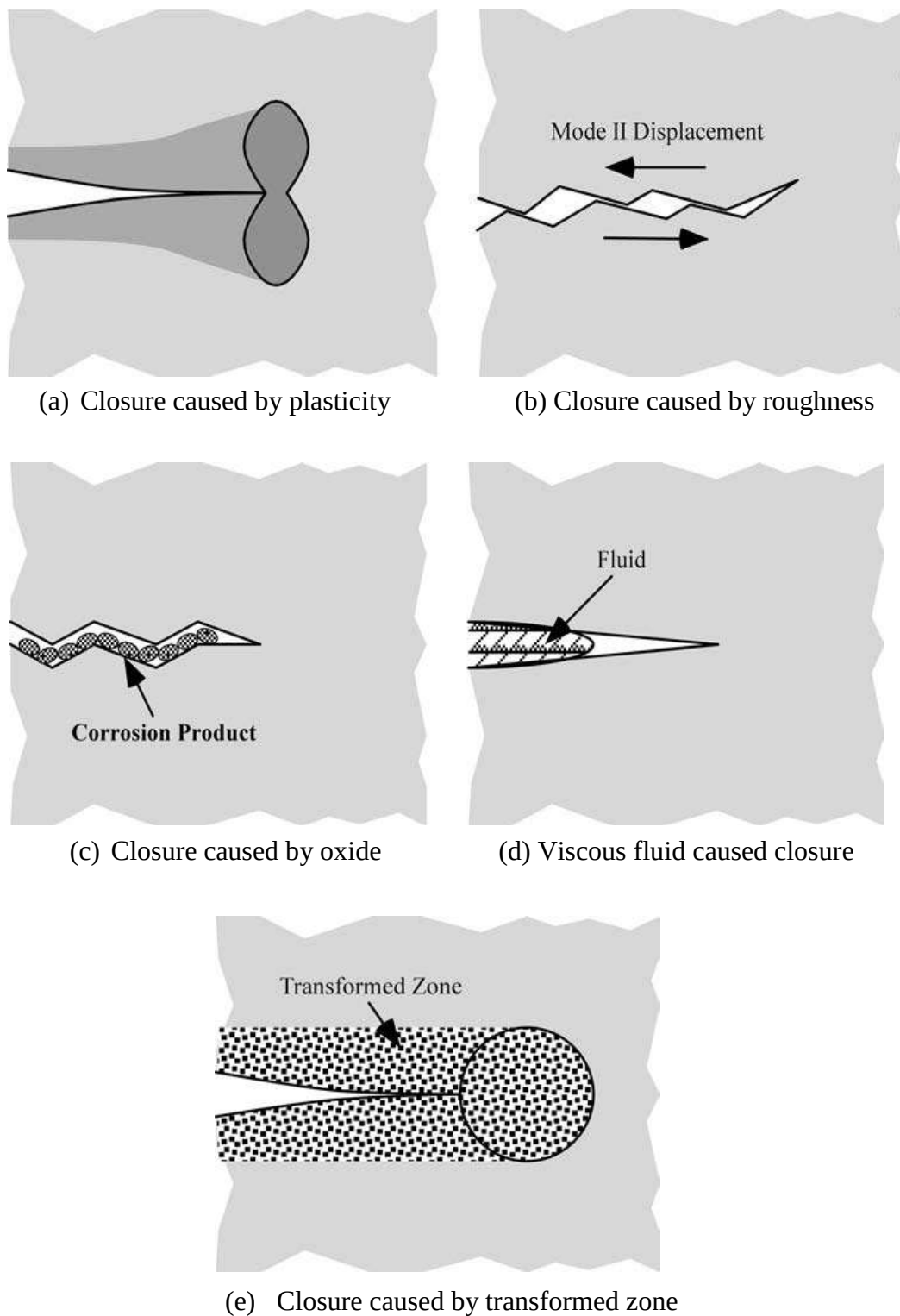


Fig. 1.14 Fatigue crack closure mechanisms in steels (Anderson, 2011)

applied to the specimen causes un-proper condition in both crack surfaces, which yields

crack faces contact when the load is in tensile phase. In case of coarse-grained materials, the higher closure load in fatigue can be produced due to higher degree of surface roughness (Gray et al., 1983).

An aggressive environment is usually corresponded with the oxide-induced closure as shown in **Fig. 1.14(c)**. In this closure, between crack faces, oxide debris or other corrosion products become wedged. In many cases, the crack closure may also be present by a viscous fluid, as presented in **Fig. 1.14(d)**. As presented by the oxide mechanism, between the crack surfaces, such fluid behaves as a block. **Fig. 1.14(e)** outlines the closure caused by transformed zone that wakes up of process zone and causes the closure. From the described typical closures, the microstructure, yield strength, and environment are the determining factors that lead to cause the closure.

The crack wedging is noticeably present on the several of the closure process outlined in **Fig. 1.14** (closure caused by viscous fluid, roughness and product of corrosion). The obstruction of some types is figured out as the reason due to the prevention of the crack closure. In this mechanism, the term residual crack opening is considered to be an appropriate approach to give the description. The closure caused by transformed zone can be classified to be a real mechanism in crack closure since its crack faces received the closure force from the residual stress.

1.2.4. *J*-integral

The *J*-integral is the parameter used to evaluate crack growth in the EPFM field. By such definition, the parameter is capable to specify the crack tip level, although the plasticity area is not limited to a wide zone if it is measured to the dimension of in-plane specimen. In another condition, the plastic zone may not interact with any in-plane dimension for the theory to be applicable.

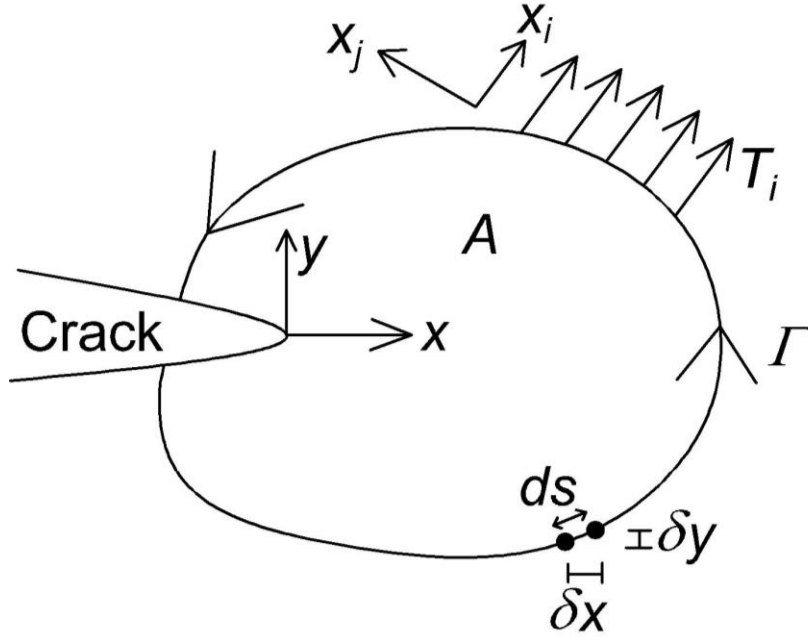


Fig. 1.15 J closed contour in a two-dimensional solid (Anderson, 2011)

Rice (1968) presented the J contour integral. Idealization to the non-linear elastic behavior from the elastic plastic nonlinear condition was conducted in this study. Rice figures out that J , is a path independent line integral and introduced as follows:

$$J = \int_{\Gamma} \left(W \delta y - T_i \frac{\delta u_i}{\delta x} ds \right) \quad (1.1)$$

$$W = \int_0^{\varepsilon_{ij}} \sigma_{ij} \cdot \delta \varepsilon_{ij} \quad (1.2)$$

Where W is the strain energy density, T_i are the traction vector components, u_i are the displacement vector components in x_i -axis, ds is the differential length along the contour Γ surrounding the tip, δx and δy are differential length along the contour in x and y -axis, σ_{ij} and ε_{ij} are stress and strain in the loading process, respectively.

Rice (1968) then employed the divergence theorem to convert **Eq. (1.1)** into area integral as illustrated in **Fig. 1.15**, the equation is constructed as follows:

$$J = \int_A \left[\frac{\delta W}{\delta x} - \frac{\delta}{\delta x_j} \left(\sigma_{ij} \frac{\delta u_i}{\delta x} \right) \right] \delta x \delta y \quad (1.3)$$

in which A is the specific area for computing J , δx_j is the differential coordinate scalar component in x_j -axis.

Employing the strain energy density characteristics, presented by **Eq. (1.2)**, we can change the first expression in brackets of **Eq. (1.3)** to be **Eq. (1.4)**.

$$\frac{\delta W}{\delta x} = \frac{\delta W}{\delta \varepsilon_{ij}} \cdot \frac{\delta \varepsilon_{ij}}{\delta x} = \sigma_{ij} \left(\frac{\delta \varepsilon_{ij}}{\delta x} \right) \quad (1.4)$$

Relationship between strain-displacement (for small strains) is applied and gives,

$$\frac{\delta \varepsilon_{ij}}{\delta x} = \frac{1}{2} \left[\frac{\delta}{\delta x} \left(\frac{\delta u_i}{\delta x_j} \right) + \frac{\delta}{\delta x} \left(\frac{\delta u_j}{\delta x_i} \right) \right] \quad (1.5)$$

in which δx_i is the differential coordinate scalar component in x_i -axis and δx_j is the differential coordinate scalar component in x_j -axis. The x_i and x_j -axis are provided in **Fig. 1.15** to construct definition of strain in respect of the traction.

Eq. (1.4) is applicable when elastic potential refers to W . Substituting correlation of strain-displacement (**Eq. (1.5)** to **Eq. (1.4)**) gives,

$$\frac{\delta W}{\delta x} = \frac{1}{2} \sigma_{ij} \left[\frac{\delta}{\delta x} \left(\frac{\delta u_i}{\delta x_j} \right) + \frac{\delta}{\delta x} \left(\frac{\delta u_j}{\delta x_i} \right) \right] \quad (1.6)$$

Since σ_{ij} is equal to σ_{ji} , **Eq. (1.6)** is derived as follows:

$$\frac{\delta W}{\delta x} = \sigma_{ij} \frac{\delta}{\delta x_j} \left(\frac{\delta u_i}{\delta x} \right) \quad (1.7)$$

Invoking the equilibrium condition:

$$\frac{\delta \sigma_{ij}}{\delta x_j} = 0 \quad (1.8)$$

Eq. (1.8) can be fulfilled, if σ_{ij} is constant, so that:

$$\sigma_{ij} \frac{\delta}{\delta x_j} \left(\frac{\delta u_i}{\delta x} \right) = \frac{\delta}{\delta x_j} \left(\sigma_{ij} \frac{\delta u_i}{\delta x} \right) \quad (1.9)$$

In **Eq. (1.9)**, the term is equivalent to square brackets in second expression in **Eq. (1.3)**. Thus, **Eq. (1.3)** integrand vanishes and $J = 0$ for every closed contour.

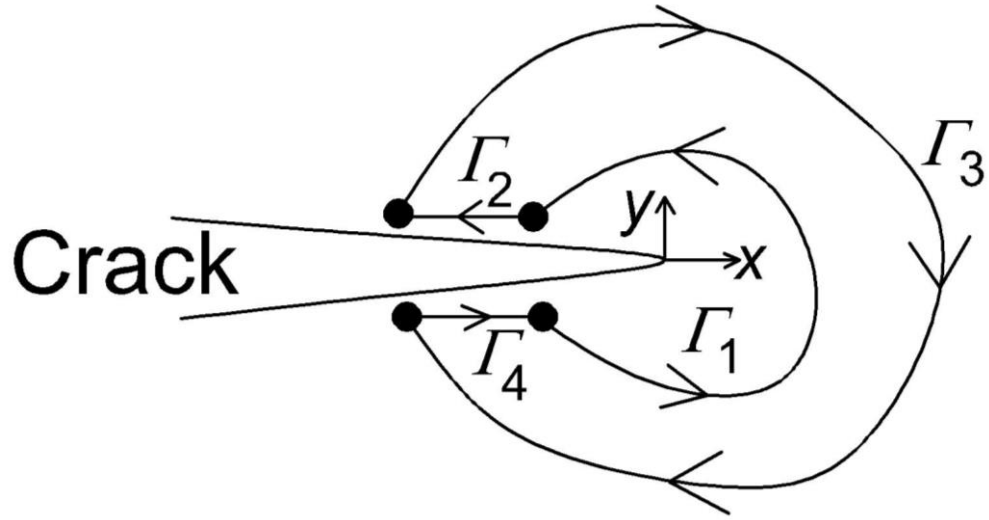


Fig. 1.16 Two arbitrary contours Γ_1 and Γ_3 surrounding the crack tip. If such contours are linked by Γ_2 and Γ_4 , a specific contour is presented (Anderson, 2011)

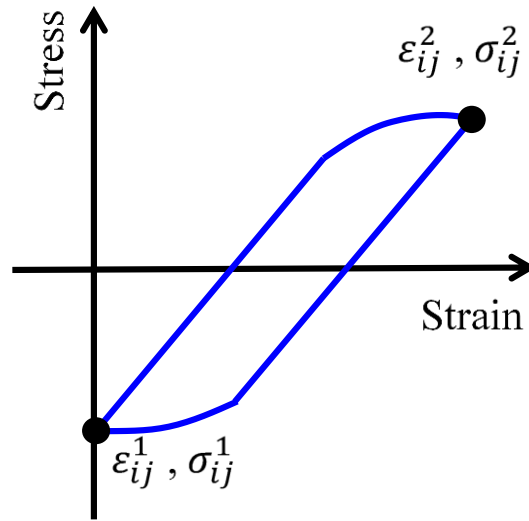


Fig. 1.17 Stress-strain curve used to calculate ΔJ (Par Ljustell, 2013)

Around the tip, suppose two evaluated contours Γ_1 and Γ_3 , as outlined in **Fig. 1.16**. When Γ_1 and Γ_3 are associated by path along the surface of crack (Γ_2 and Γ_4), a closed contour is presented. Accumulation from each path in closed contour is equal to total J and written as follows:

$$J = J_1 + J_2 + J_3 + J_4 = 0 \quad (1.10)$$

On the crack face, the traction vector (T_i) and the differential length in vertical axis (δy) are equal to zero, $T_i = \delta y = 0$. From these results, J -integrals at the particular path, $J_2 = J_4 = 0$ and $J_1 = -J_3$. Here, the equivalent J will be yielded at the any counterclockwise path surrounding the tip. From this result, that J is path-independent can be proven from this derivation.

1.2.5. Cyclic J -integral range

Lamba (1975) firstly introduced the definition of in connection to stress concentration factor (SCF), where SCF is determined as the ratio of maximum stress to the nominal stress (Bhandari, 2008). The definition is established since the stress concentration is indicated at each employment of tensile loading and cyclic stress-strain ranges are also characterized on the material in front of the crack tip. Accordingly, $\Delta\sigma_{ij}$ and $\Delta\varepsilon_{ij}$ is used to result the ΔJ .

A curve of cyclic stress strain in which the initial value is presented with 1 in numbering and final value is symbolized with 2 in numbering is expressed in **Fig 1.17**. The range of the J -integral is introduced by Lamba (1975), Dowling and Begley (1976) and Lambert et al. (1988) as follows:

$$\Delta J = \int_{\Gamma} \left(W' \delta y - \Delta T_i \frac{\delta \Delta u_i}{\delta x} ds \right) \quad (1.11)$$

$$W' = \int_{\varepsilon_{ij}^1}^{\varepsilon_{ij}^2} \Delta \sigma_{ij} \cdot \delta(\Delta \varepsilon_{ij}) = \int_{\varepsilon_{ij}^1}^{\varepsilon_{ij}^2} (\sigma_{ij}^2 - \sigma_{ij}^1) \cdot \delta(\Delta \varepsilon_{ij}) \quad (1.12)$$

In which, Γ is the integration path surrounding the tip, ΔT_i and Δu_i are traction and displacement vector change from the initial point 1 to the final point 2. W' is addressed to the strain energy density.

Generally, in computing process of ΔJ , the loading phase of the cyclic curve from initial point 1 to point 2 is merely involved in the integration of the strain energy density rather than the overall loop. In case of initial point 1 is positioned at zero stress and strain, the ΔJ is equivalent to J . **Eq. (1.3)** is a result of **Eq. (1.1)** to involve the condition in which the stress and strain at point 1 is not zero.

1.2.6. Path independence of ΔJ

In case of W' exhibits the strain energy density, the stresses can be stated by evaluating W' with considering the strains:

$$\Delta\sigma_{ij} = \frac{W'}{\delta(\Delta\varepsilon_{ij})} \quad (1.13)$$

Eq. (1.13) is required for evaluating the path independence of ΔJ . Evaluating ΔJ along a closed contour, Γ , as formerly shown in **Fig. 1.15** and employing divergence theorem to convert **Eq. (1.13)** into an area integral gives:

$$\Delta J = \int_A \left[\frac{W'}{\delta x} - \frac{\delta}{\delta x_j} \left(\Delta\sigma_{ij} \frac{\delta\Delta u_j}{\delta x} \right) \right] \delta x \delta y \quad (1.14)$$

where A is the area defined by Γ . By assuming W' displays the strain energy density, **Eq. (1.13)**, the first term of integrand can be written as:

$$\frac{W'}{\delta x} = \frac{W'}{\delta(\Delta\varepsilon_{ij})} \frac{\delta(\Delta\varepsilon_{ij})}{\delta x} = \Delta\sigma_{ij} \frac{\delta(\Delta\varepsilon_{ij})}{\delta x} \quad (1.15)$$

It can be shown that **Eq. (1.15)** is similar to the second expression in the integrand in **Eq. (1.14)** by employing the strain-displacement relationships for small strains. Hence, for any closed contour, the ΔJ is equal to zero. By considering the contour previously illustrated in **Fig. 1.16**, the path independence of ΔJ was evaluated surrounding the crack tip and yielded $\Delta J_1 = -\Delta J_3$.

In case of proportional loading, an individual variable of crack tip is expressed.

Hence, due to the existence of the proportional loading in the area around the tip, the ΔJ specifically introduces the changes in stress and strain around this zone. Under the monotonic loading, due to large plastic distribution or the large crack propagation, the J integral discontinue to contribute a single-parameter of crack tip condition. Similarly, there is only limited studies expecting ΔJ to specify crack propagation beyond an undoubtable level of deformation. Accordingly, evaluations of ΔJ have to be conducted.

1.3. Research objectives

The purpose of this research is to investigate crack propagation manner in steel due to the employment of plastic strains variability.

To achieve it, the basic characteristic of crack propagation under constant amplitude was firstly revealed and then to obtain crack propagation behavior when the amplitudes were reduced or increased, a simple variable amplitude loading, two-steps variable condition was employed. Applicability of estimation model established from two-steps variable was then evaluated to the random variable amplitude loading.

1.4. Organization of dissertation contents

The flowchart of this study is presented in **Fig. 1.18**. The content of this paper are concisely described as follows.

Chapter 1 describes introduction of this study, including research background, previous studies, the basic literature elaborating LCF characteristics, specimen behavior due to applied cyclic loading, i.e., crack propagation and crack closure, and basic theory of EPFM parameters used to evaluate crack propagation rate, i.e., cyclic J -integral and cyclic J -integral range, research objectives and organization of the dissertation.

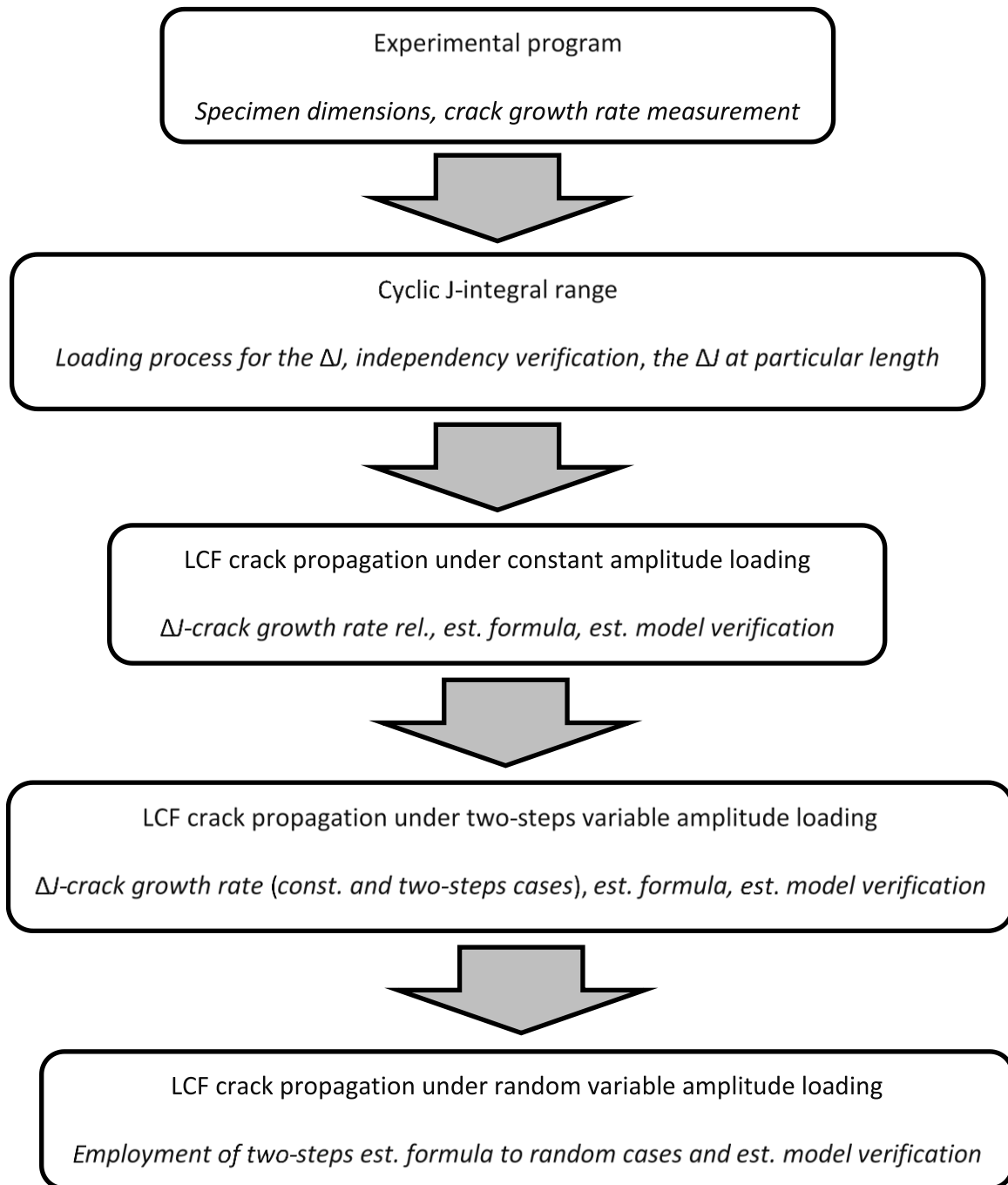


Fig. 1.18 Flow chart of the research

Chapter 2 elaborates the way to carry out the research which consists of explanation about the specimen characteristics, test set-up, the way to apply cyclic loading, and to measure crack length. Besides, this chapter also discusses the method to

define the ΔJ , i.e., loading process and the path consideration to compute ΔJ . In this chapter, the ΔJ independency due to integration path and loading cycle was also verified before calculating ΔJ at the designated crack length.

Chapter 3 elaborates crack propagation behavior under LCF constant amplitude loadings. In this chapter, firstly crack growth rates information from the past research work (Hanji et al. 2017), the study which also examined crack growth rate under constant amplitude loading was briefly explained. Prior to the discussion for variable amplitude loadings, verification on the analysis to test was also done in this typical loading to confirm whether the analysis can satisfyingly figure out the test. Next, the load-COD behavior in entire hysteretic loop and crack growth rate- ΔJ relationship were evaluated. To ensure the accuracy, crack growth rates from Hanji et al. (2017) were collectively plotted with the crack growth rates of present LCF tests to yield estimation formula.

Chapter 4 performs LCF two-steps variable amplitude loading tests in steel. To compare the behavior, two-steps variable's crack growth rates were cumulatively plotted with those under constant amplitude loading. In this chapter, a ratio of crack growth rate of variable amplitude to that under constant amplitude loading, named as 'amplification factor', was presented and identified to be correlated with crack growth rate. Due to the uncomplicated attributes of two-steps cases, estimation model was constructed from these typical loadings. At the end of the work, verifications were made with outlining the value of crack propagation rate and the crack length originated from the analysis and the test.

Chapter 5 discusses crack growth rate behavior under LCF random variable amplitude loadings. In this chapter, random loading's crack growth rate manners were also characterized to that under constant amplitude loading. In such cases, observations were emphasized on the crack growth at the domain of amplitude change, i.e., from high to low levels and from low to high levels. Since the estimation model had been acquired from two-steps variable cases, such formula was also employed to the random variable amplitude loading and then validated.

2. Basic Methodology of the Study

2.1. Introduction

Study methodology which involves the way to do experiment and analysis is elaborated in this chapter. Experimental study describes specimen characteristics, test set up configuration, the loading mode, typical of loading histories, technical measurement of crack opening displacement (COD) and crack length. This chapter also discusses about the way to perform analyses to compute the ΔJ since the crack growth rate was evaluated through the ΔJ . On the analysis, explanation about model characteristics, crack growth simulation on the model and the ΔJ independency due to loading cycle and path was evaluated prior to the ΔJ computation which relied on the applied COD.

2.2. Specimen

The specimen tested in this experiment was a compact tension specimen shown in **Fig. 2.1**. The configurations were arranged based on ASTM E1820-031 (2008). The mechanical properties of tested steel obtained from mill sheet are shown in the **Table 2.1**.

Table 2.1 Mechanical properties

Steel type	Yield strength (N/mm ²)	Tensile strength (N/mm ²)	Elongation (%)
SS400	296	431	32

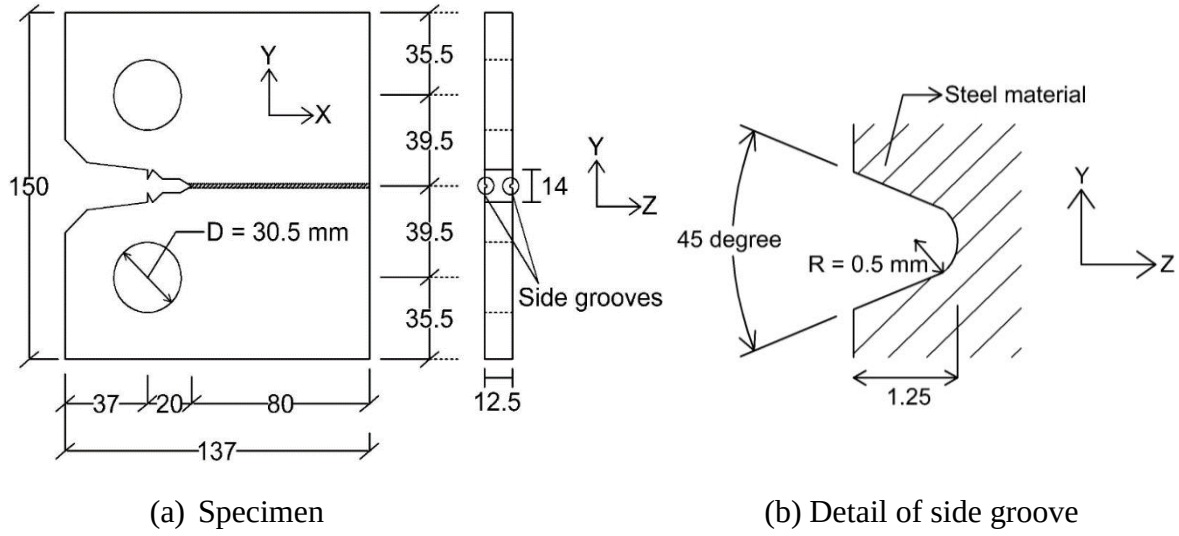


Fig. 2.1 Specimens (unit: mm)

The specimen was cut from a steel plate of SS400 and thickness of all specimens was arranged in 12.5 mm. Side grooves were introduced at both side surfaces of the specimen in order to guide a crack path as straight as possible.

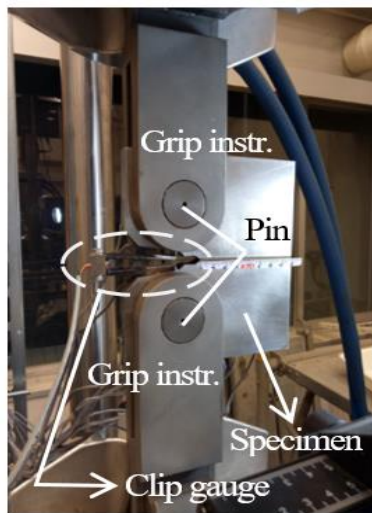
2.3. Crack propagation measurement

A test machine, which is type of Shimadzu EHF-UB300KN-40N, was employed to evaluate specimen behavior under LCF tests. The maximum loading capacity is 300 kN. Experimental set up is presented in **Fig. 2.2(a)**. The specimen was fixed to a testing machine and the cyclic displacement was applied. The cyclic loading was carried out with controlling the crack opening displacement (COD). Three loadings of the COD were applied to the specimens, i.e., constant amplitude, two-steps variable amplitude and random variable amplitude loadings. Detail information on the loading sequence will be given in each chapter later. Loading speed was approximately 0.001mm/sec to 0.05mm/sec.

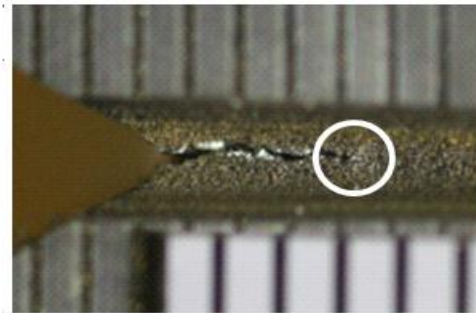
The applied cyclic loading refers to the mode 1 as explained at the **Section 1.2.2**. Due to the given load, at the employment of tensile loading, a crack occurs from the notch

point and propagates along the grooves.

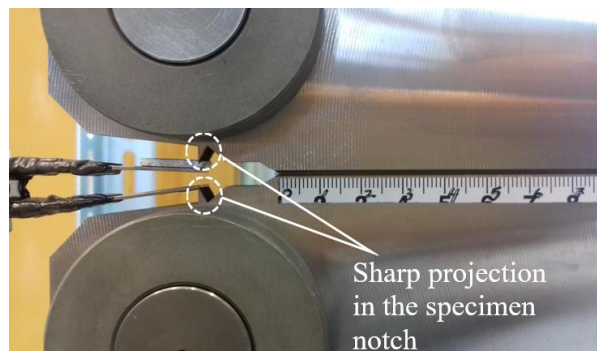
The propagation was observed with a high-resolution microscope at each peak tensile load. The microscope has the magnification of 50 to 100. The crack tip was rigorously identified relied on digital image of crack as shown in **Fig. 2.2(b)**. To measure the crack opening displacement (COD), a clip gauge was mounted at the sharp projection in the specimen notch as illustrated in **Fig. 2.2(c)**. This clip gauge can exactly measure maximum opening of 5.0 mm. The cyclic loadings were continued until the crack growth exhibited insignificant rate. From the observation of the crack extension, crack growth



(a) Test set up



(b) Digital image of crack



(c) Mounted position of the clip gauge

Fig. 2.2 Experimental program and the observation

rate is introduced as:

$$\frac{da}{dN} = \frac{a_i - a_{i-1}}{1 \text{ cycle}} \quad (2.1)$$

where da/dN is the crack growth rate (mm/cycle), a_i is the crack length at an observed cycle (mm), a_{i-1} is that at one cycle prior to the observed cycle (mm).

2.4. FE modeling

In this study, the specimen at the ligament was modeled by fine solid element mesh with 1.0 mm x 0.5 mm x 0.625 mm. Solid element type is eight-node, isoparametric and arbitrary hexahedral elements. Analysis was performed using Abaqus FEA (Ver. 6.13) to obtain ΔJ . A quarter model of the CT specimen was utilized to conduct the analysis. This model was constructed with symmetric supports assigned at the thickness center (XY-plane). The material properties of the model were assigned using the same characteristics listed in **Table 2.1** where kinematic hardening rule based on true stress-strain was applied. **Fig. 2.3** expressed the difference slope between E and E_t (E and E_t are

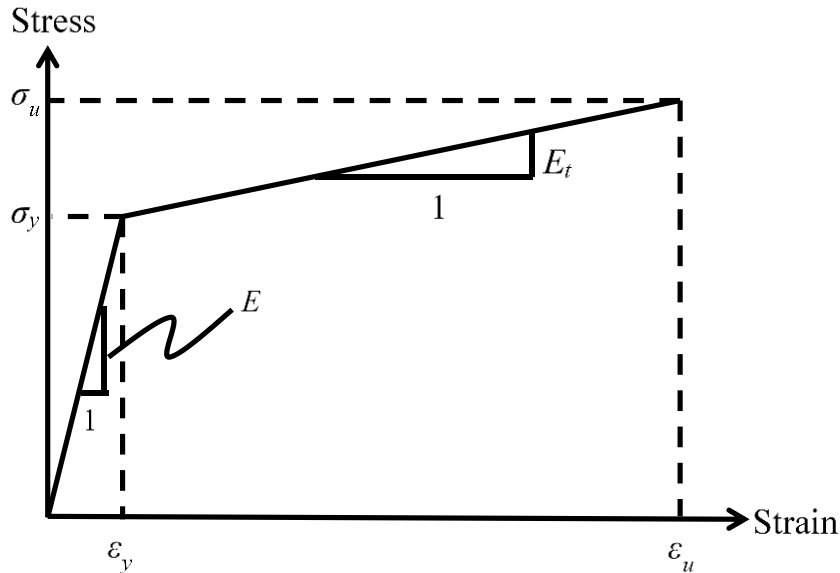


Fig. 2.3 True stress-strain relationship established with the principle of kinematic hardening rule

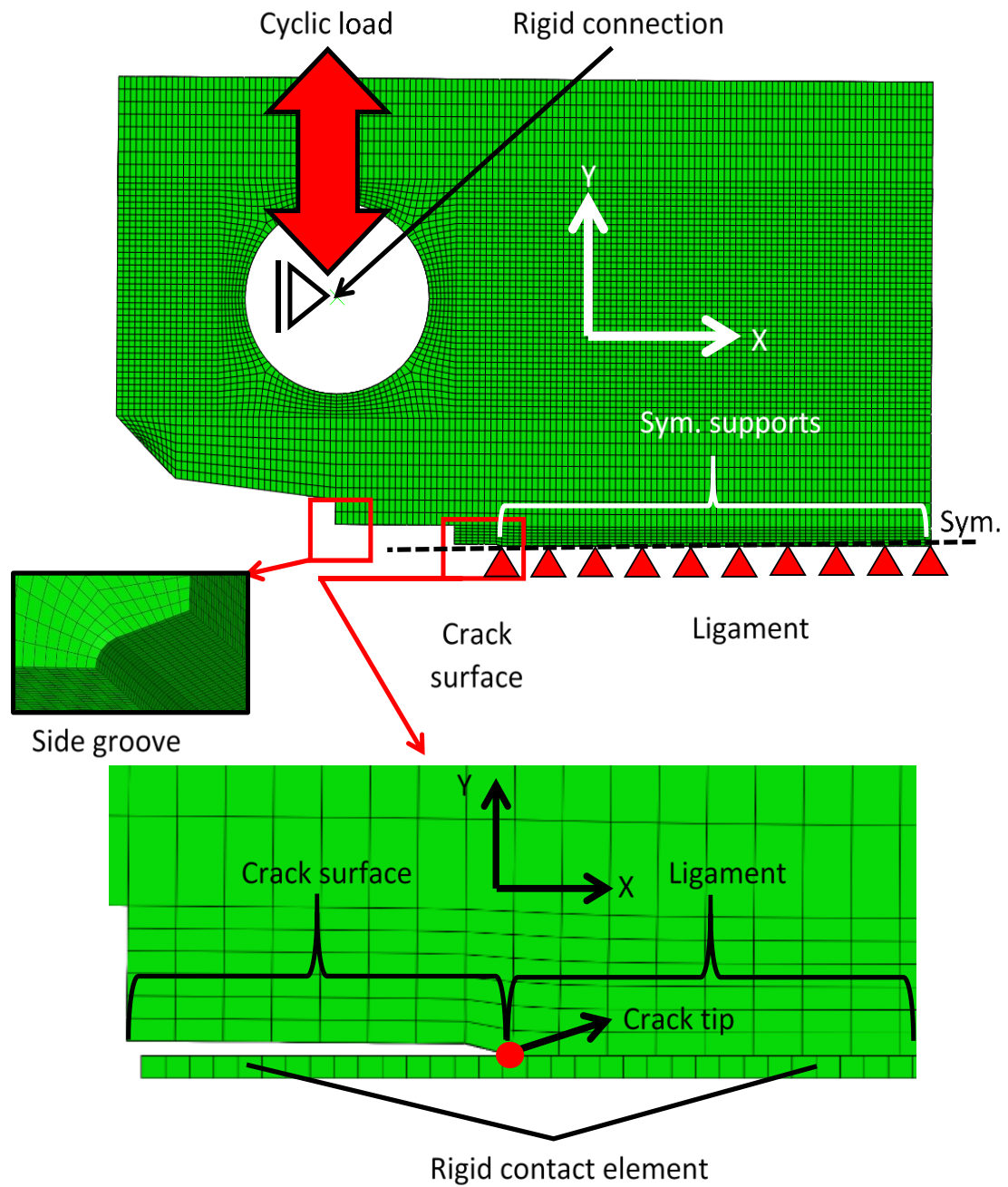


Fig. 2.4 Assignment of cyclic loading to the 3D-specimen model and the boundary condition of the model

the first slope initial elastic and the second slope of plastic phase, respectively). E and E_t was obtained from the mill sheet and can be calculated as follows:

$$E = \frac{\sigma_y}{\varepsilon_y} \quad (2.2)$$

$$E_t = \frac{\sigma_u - \sigma_y}{\varepsilon_u - \varepsilon_y} \quad (2.3)$$

where σ_y is the yield strength (MPa), ε_y is the yield strain (mm/mm), σ_u is the ultimate strength (MPa), ε_u is the ultimate strain (mm/mm).

According to the true stress-strain relationship, E is 200,000 MPa and E_t is 670.23 MPa. Those two moduli were assigned as the mechanical properties of the material.

Fig. 2.4 figures out cyclic loading application to the pin center and to ensure the contact, rigid connection was established between pin and CT specimen. Abaqus standard module was used to conduct the analysis until 1.5 cycles. At the 0-cycle, a ligament was initially supported in the crack opening direction. Then, at the first 0.5 cycle-tensile loading, the crack was modelled by releasing supports from the notch point to crack tip position. Accordingly, several models with different crack length, 8 to 56 mm, were provided. The crack length is fixed until 1.5 cycles in the analysis although the crack propagates in every cycle during the test. The compressive load applied after tension caused crack surface to be pressed and experiences contact. To ensure the closure, a rigid contact element was established along the crack surface-ligament and placed right below the model as shown in **Fig. 2.4**.

The value of ΔJ was calculated based on specimen responses at the final process of cyclic loading as shown in **Fig. 2.5**. The ranges of each parameter shown in **Eq. (1.11)** and **(1.12)** were taken as the difference from the minimum point to the maximum point, during the loading process from 1.0 to 1.5 cycles as shown in figure. The figure also describes the different loading portions to compute ΔJ . **Fig. 2.5(a)** expresses the loading process from 0-COD to tensile-COD and **Fig. 2.5(b)** outlines the process from compressive-COD to tensile-COD. The loading processes presented in such figures were utilized to calculate ΔJ at the different averages of COD.

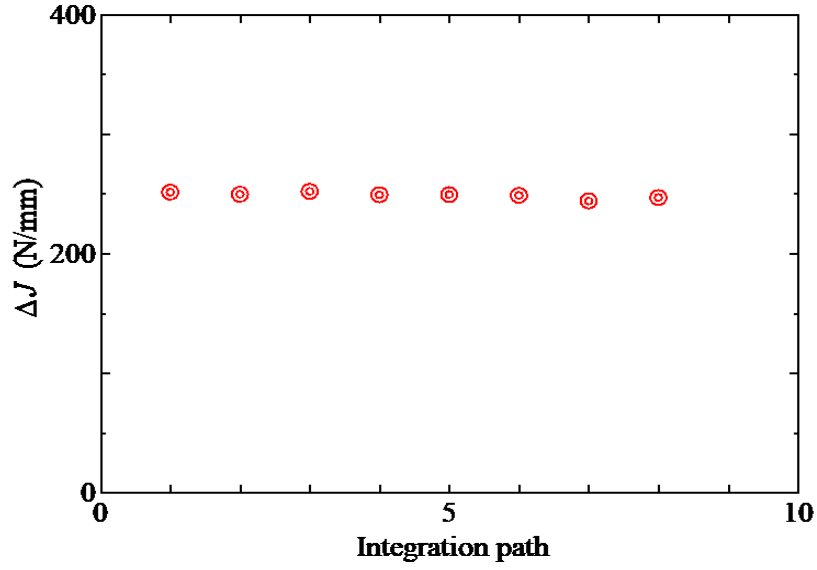


Fig. 2.7 The ΔJ independency to integration path

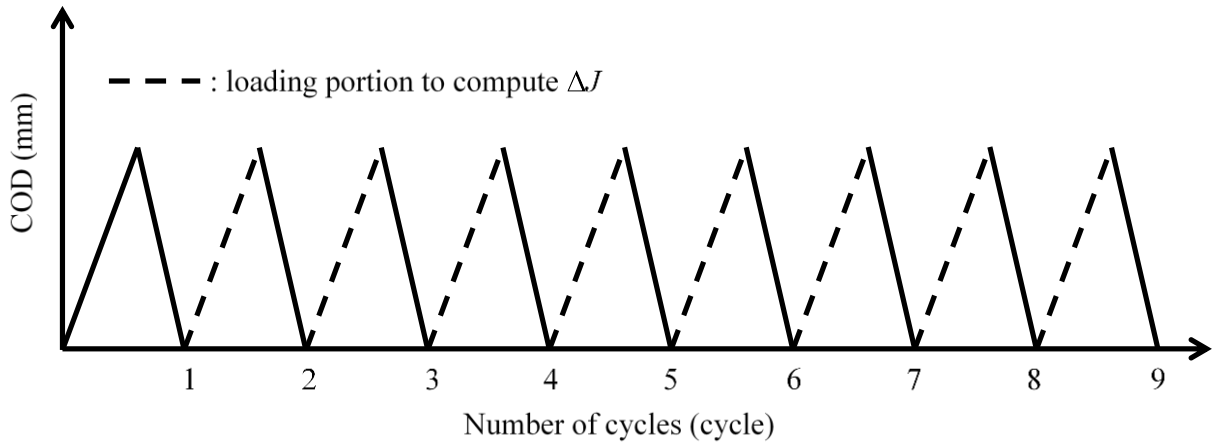


Fig. 2.8 Loading cycles used for independency verification

verification was done for 0.5 mm-COD at 16 mm-long crack. For overall analyses, loading portion of 1.0 to 1.5 cycles was chosen. **Fig. 2.7** provides the ΔJ independency in the integration paths where negligible difference of ΔJ value is obtained.

To figure out independency of ΔJ to loading cycles, at 6th-path number, several patterns of the loading were considered, i.e., 1.0 to 1.5-loading cycle, 2.0 to 2.5-loading cycle, etc. The detail information about considered loading cycles is provided in **Fig. 2.8**.

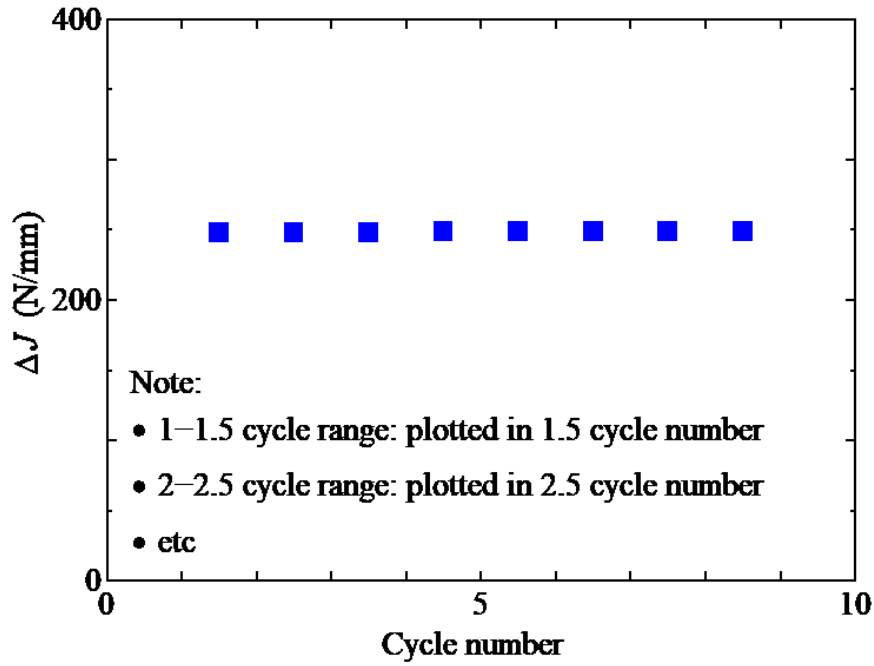


Fig. 2.9 The ΔJ independency to loading cycles

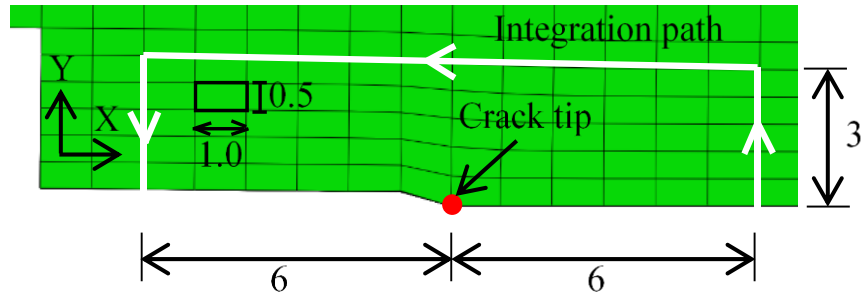


Fig. 2.10 Integration path for analysis (unit: mm)

Similar to the path independency clarification, the analysis was also conducted for 0.5 mm-COD at 16 mm-long crack. As shown in **Fig. 2.9**, un-obvious differences of ΔJ value are confirmed in loading cycles.

An integration path for calculating ΔJ is exemplified in **Fig. 2.10** for the 8 mm-long crack. The same paths were taken for other size cracks. For the efficiency, the portion of 1.0 to 1.5 cycle is selected since another cyclic loading, i.e., response of 2.0 to 2.5 cycle results similar ΔJ value. It has been previously explained at the independency of ΔJ to

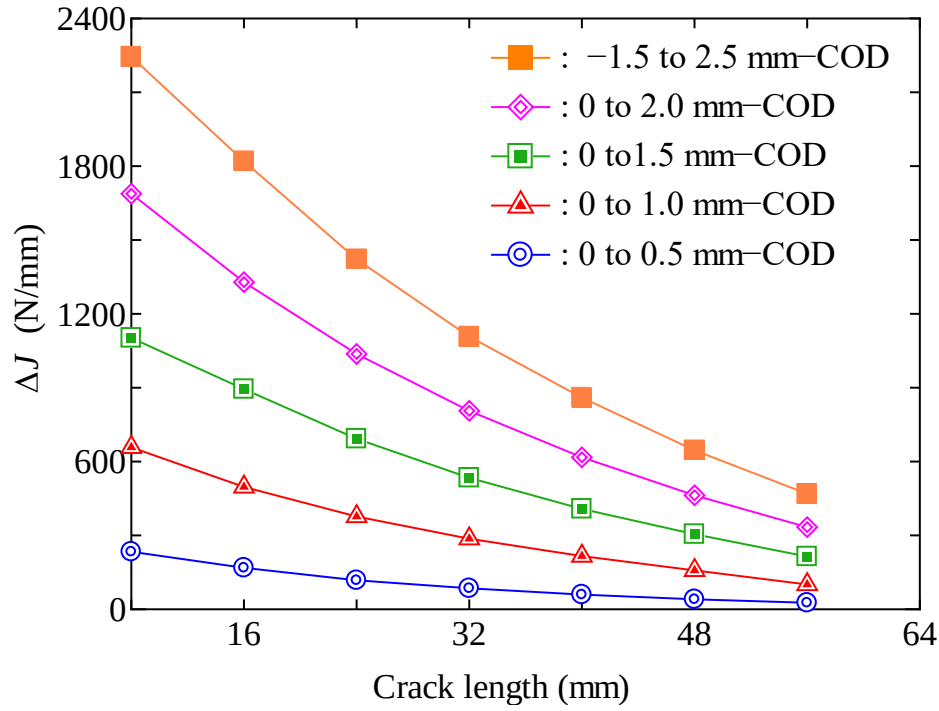


Fig. 2.11 ΔJ -crack length relationships

loading cycles. The analyses were carried out with assigning fixed crack length (ranging from 8 to 56 mm at 8 mm intervals) and fixed constant displacement amplitude (ranging from -1.5 to 2.5 mm at 0.25 mm intervals). In figure, un-symmetrical range of -1.5 to 2.5 mm-COD was considered since the minimum COD more than -1.5 mm is difficult to be reached due to the high compressive load. The obtained ΔJ values are summarized in **Fig. 2.11**. The value of ΔJ for arbitrary crack length and amplitude was linearly interpolated from two nearest points shown in the figure.

2.5. Summary

This chapter comprises the study methodology elaborating the way to carry out experiment and analysis. In the experiment description, explanations about the way to arrange experimental setting up, to employ the load and to measure crack propagation

have been provided. As for the following step, on the analysis section, the independency of ΔJ due to integration path and loading cycle was verified. It can be showed by negligible differences on the obtained ΔJ values. Moreover, the ΔJ computation results related to the crack length are constructed and perform declining manner along with the crack extension due to displacement-controlled employment. The ΔJ obtained from this chapter is utilized as the parameter to examine crack growth rate which will be elaborated later at the following contents in this study.

3. Constant Amplitude Loading (CA)

3.1. Introduction

This chapter evaluates crack growth rate behavior under constant amplitude loading with using the ΔJ . In the previous chapter, the methodology to acquire the crack propagation rate due to applied cyclic loading has been explained. To obtain the accurate description of the behavior, the results under constant amplitude loadings obtained at the previous study were also contemplated in this present study. To describe the specimen responses, load-COD relationships, verification on the analysis to the test result, and crack growth rate- ΔJ under constant amplitude loading (CA) are in orderly performed. At the final part, relying on the cumulative crack growth rate plots from past and current work, estimation model was provided and validation was conducted with showing crack propagation analysis results.

3.2. Previous study on the constant amplitude loading

At the previous research work, Hanji et al. (2017) evaluated LCF crack growth rate under constant amplitude loading using compact tension (CT) specimens. The dimensions of the specimen were established referring to the ASTM standard E1820-08. To assign the crack origin in specimen, an artificial notch was built. To obtain crack propagation behavior, three specimen types were provided to identify crack propagation

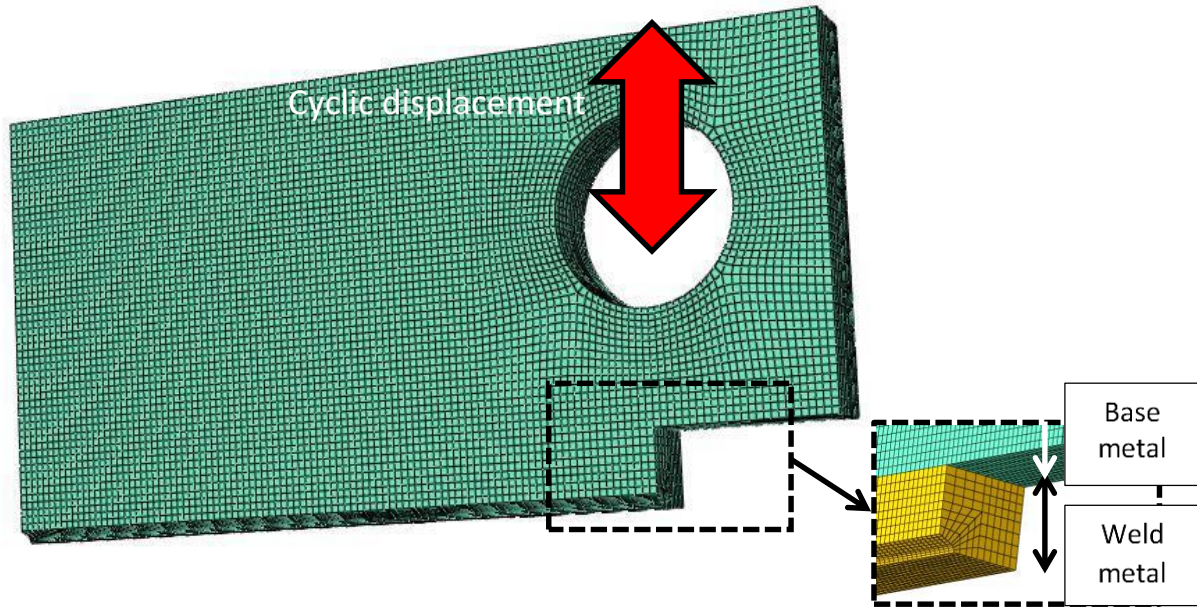


Fig. 3.1 Location of weld metal on the CT specimen

Table 3.1 Test matrix for past research work

Case	1	2	3	4	5	6	7	8	9	10	11
Material	JIS-SM400A			JIS-SM400A				Weld metal			
Displ. range (mm)	0.8	1.5	3.0	0.8	1.5	3.6	1.5	0.4	0.8	1.5	3.0
Min. displ. (mm)	0	0	0	0	0	0	1.5	0	0	0	0

rate in different material, plain steel and weld metal. To fabricate specimen, steel plates of JIS-SM490A and JIS-SM400A (steel grade of 490 and 400 N/mm²) and a weld metal of JIS Z 3183 S501-H were used. As for the welding part, its position can be shown on the **Fig. 3.1**. The materials were originated from the different grades and all specimens had the uniform dimensions of the 60 mm-height, 62.5 mm-width, 12.5 mm-thickness.

Several constant amplitude loadings applied to the CT specimen in different material characteristics are outlined on the **Table 3.1** above. As shown on table, the load

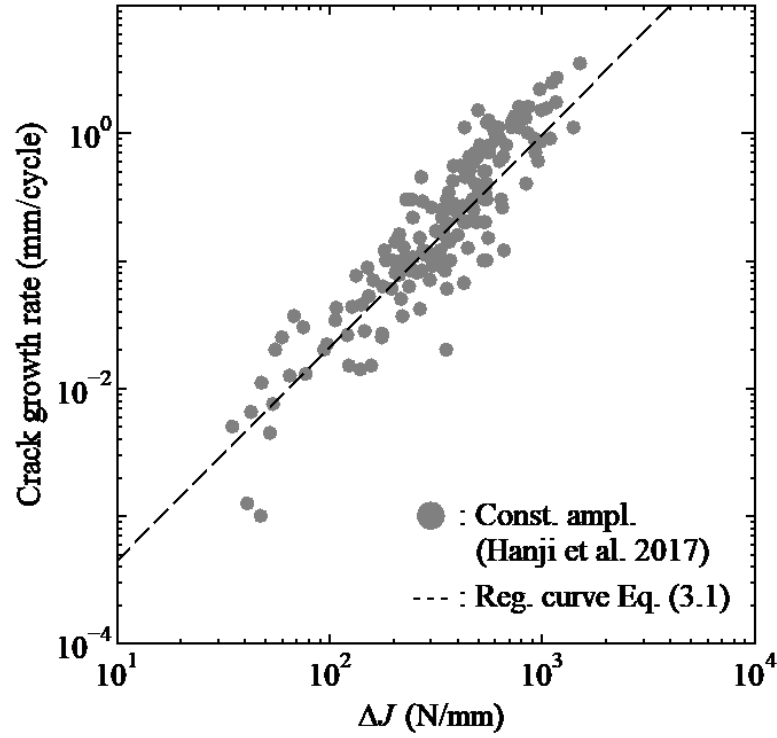


Fig. 3.2 Crack growth rate and ΔJ under constant amplitude loading at the past study

cases also contemplate different average of cyclic displacement. This past study presented cyclic J -integral range, ΔJ , a LCF parameter obtained through 2D FEA. In definition, these ΔJ values were calculated based on plane strain analysis on the considered path where the 2D plane was assumed as middle plane in CT specimen thickness of 3D-model.

The test results were arranged with establishing the relationships between crack growth rate, da/dN and ΔJ . As expressed in **Fig 3.2**, although there were the differences in displacement range and displacement mean, the crack growth rates are dispersed within a relatively narrow area. Additionally, all plots were distributed in the same area, as shown in figure, meaning that the cyclic J -integral range correlates with the crack growth rate regardless of the material. Based on the results, the following formula can be derived assuming that the relationship yields a straight line on log-log scales that can be approached by **Eq. (3.1)** as follows:

$$\frac{da}{dN} = 9.6 \times 10^{-6} \cdot \Delta J^{1.67} \quad (3.1)$$

where da/dN is the crack growth rate (mm/cycle), and ΔJ is cyclic J -integral range (N/mm).

Evaluation of the crack growth performed by Hanji et al. (2017) interpreted that **Eq. (3.1)** is applicable to examine crack extension on the different grade of the steel material.

3.3. Present study on constant amplitude loading

For constant amplitude loadings, two COD amplitudes were taken, CA-1 and CA-2. The maximum CODs for CA-1 and CA-2 were 1.0 mm and 0.5 mm respectively, while

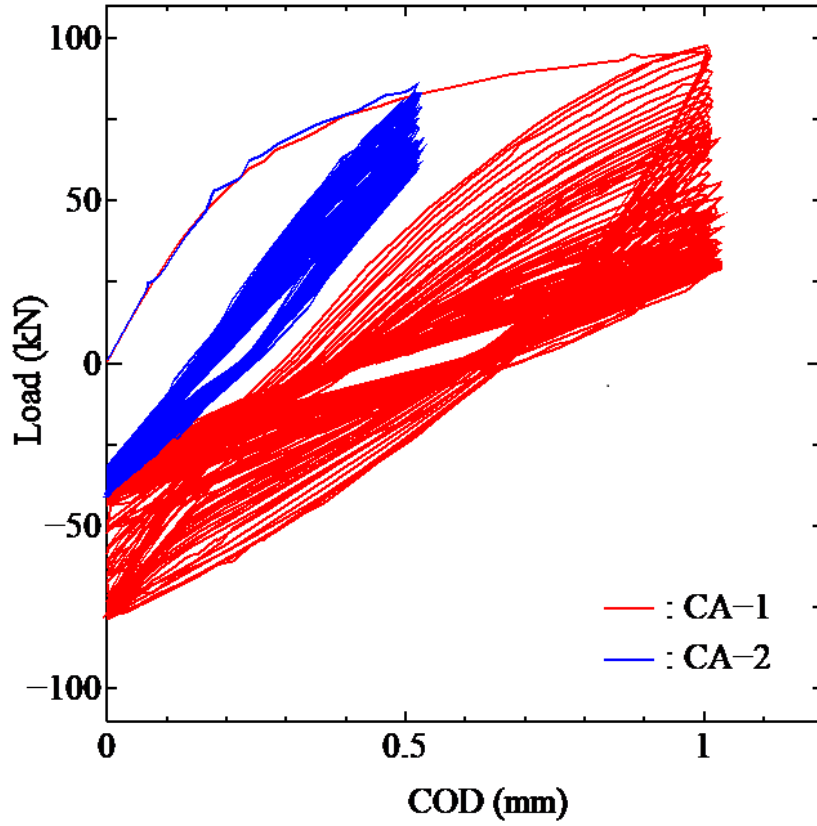


Fig. 3.3 Load-COD hysteresis loop

the minimum COD was set to be 0 mm for both cases.

Fig. 3.3 shows the measured load-COD relationships for CA-1 and CA-2 during entire loading histories. In the graph, the maximum load of each cycle continually decreased because of the crack extension. This behavior is more noticeable as the COD

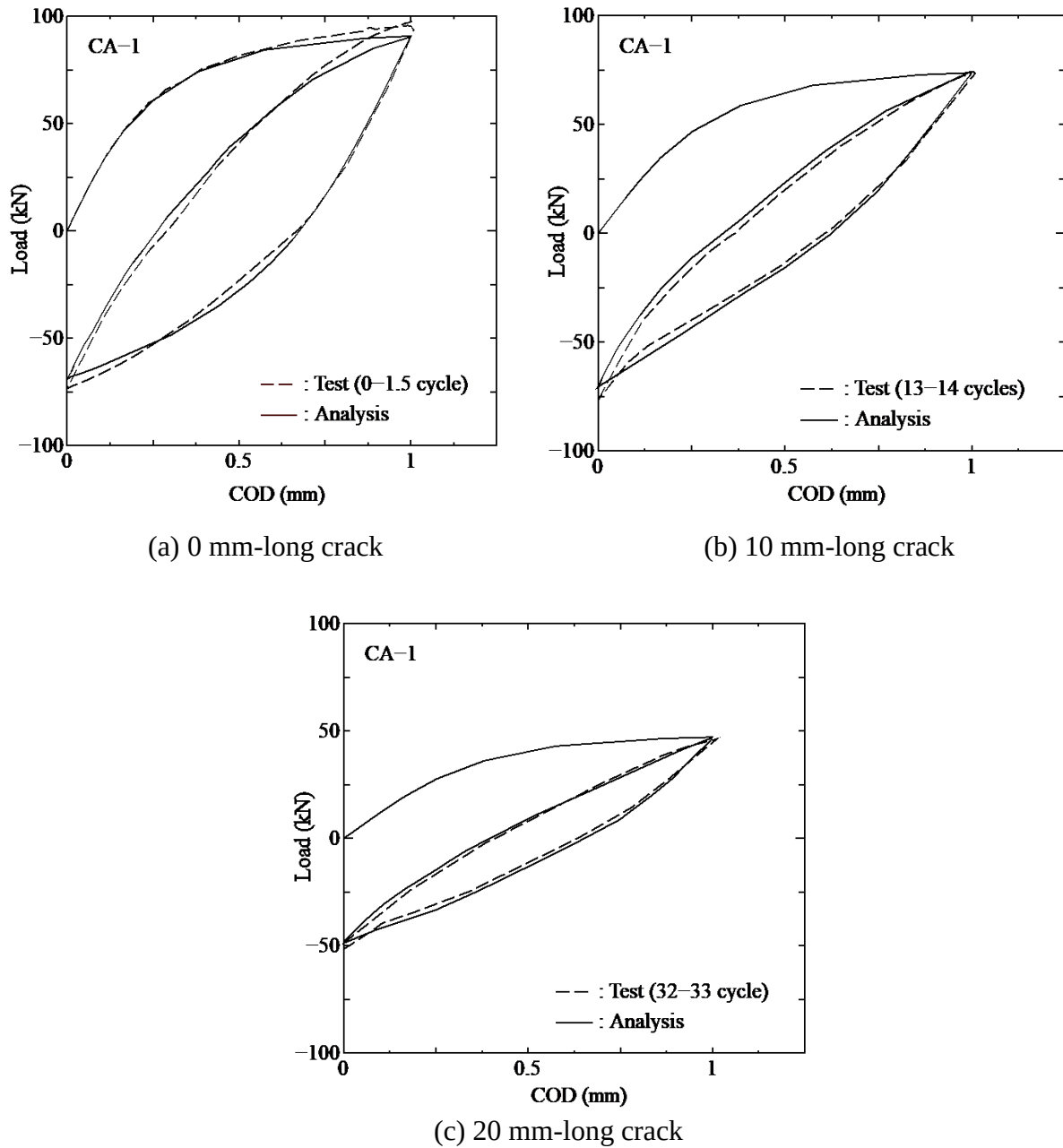


Fig. 3.4 Load-COD verification

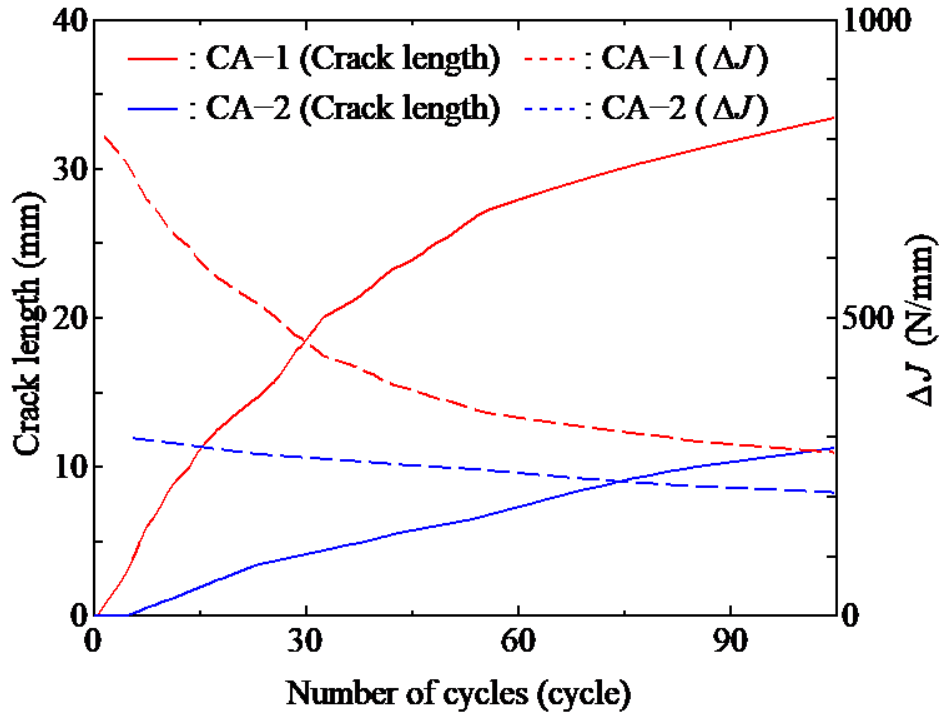


Fig. 3.5 Crack propagation behavior and ΔJ distribution under constant amplitude loadings

range increased, where CA-1 presents more clear strength degradation pattern than that under CA-2. Crack closure was also noticeably present, indicated by inflection points in unloading process where the crack surfaces have already experienced contact. This inflection point is identified where the hysteresis loop change its continuity on the unloading process.

Several crack lengths, i.e., 0, 10, 20 mm, applied under CA-1, for 1.5 cycles were selected for analysis-test verification since such crack lengths were also presented from experiment. The validation was done at the same crack lengths. As presented in **Fig. 3.4**, analysis agreed well with the experiment where the maximum and minimum loading point are almost same regardless the unidentified closure in analysis. Since for considered loading cycle, the ΔJ was principally computed from minimum to the maximum loading point, the closure effect at the unloading can be considered to be insignificant.

Fig. 3.5 exhibits the crack growth curves, the relationship between the number of

cycles and crack length. The crack length is defined by accumulating crack propagation rates from each cycle. In the figure, the fluctuation of ΔJ is also shown. Because of the displacement-controlled condition, the inclination of the crack growth curves gradually decreases. The change of curve inclination indicates that the crack propagation rates at the earlier cycles gradually decreased along with the crack extension. The same trend is also identified in ΔJ , where the ΔJ inclination curve is reduced with the crack length

Fig. 3.6 expresses the crack growth rates from both studies, this present and past study (Hanji et al., 2017), which are distributed in the same region within a narrow band. This tendency indicates that ΔJ correlates with crack growth rates regardless of the material and specimen size scale. Based on crack growth rate- ΔJ pattern, the regression curve for constant amplitude loading is reconstructed as follows:

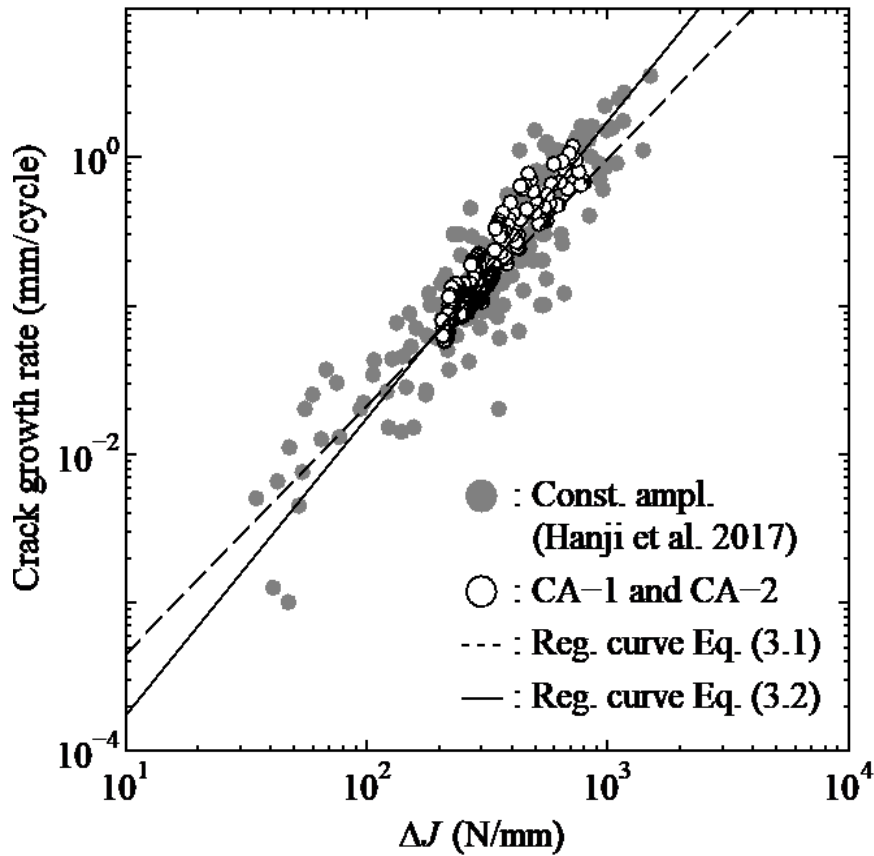
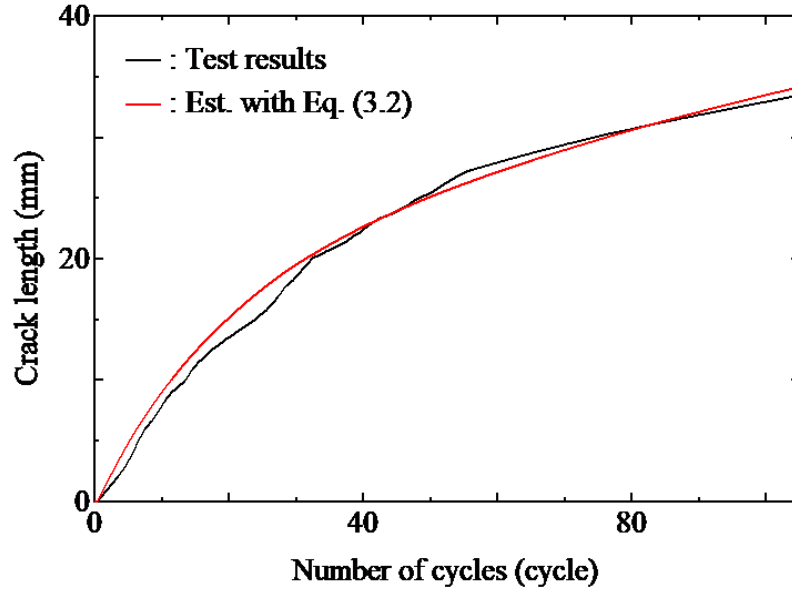


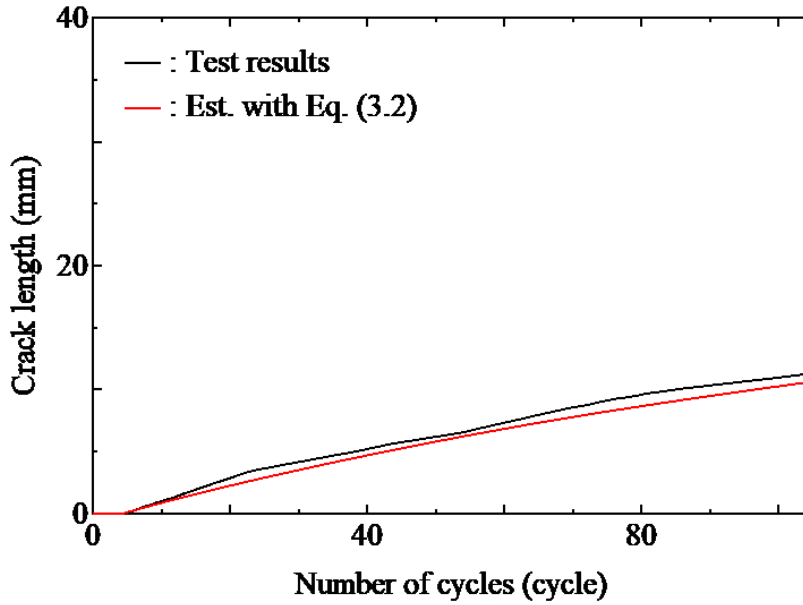
Fig. 3.6 ΔJ -crack growth rate relationships under constant amplitude loadings

$$\frac{da}{dN} = 1.74 \times 10^{-6} \cdot \Delta J^{2.00} \quad (3.2)$$

The regression lines by **Eq. (3.1)** and **(3.2)** are also shown in **Fig. 3.6**. A small difference is observed in their slopes, but it is not critical. Since **Eq. (3.2)** is established



(a) CA-1



(b) CA-2

Fig. 3.7 Crack length-number of cycles relationships under constant amplitude loadings

on a larger number of test data than **Eq. (3.1)** and considered to be reliable, **Eq. (3.2)** is used to provide the crack growth rate under constant amplitude loading in this study.

Crack propagation analysis and validation were conducted for constant amplitude loadings, and the crack growth rates were defined by **Eq. (3.2)**. Total crack length are obtained by accumulating crack growth rates during each cycle. The crack propagation analysis results are expressed in **Fig. 3.7**. From the figure, it can be shown that estimation model by **Eq. (3.2)** perform similar estimated crack lengths to those observed from the experiment and confirm the applicability of **Eq. (3.2)**.

3.4. Summary

This chapter discloses crack growth rate behavior under constant amplitude loadings. Even though at the past research work (Hanji et al. 2017), the crack propagation was observed under the same typical of loading, this present research attempted to employ constant amplitude loading to test the specimen with different material and specimen size from the past study to result rigorous estimation model for the crack propagation. The results from both studies, past and this current study, figured out that crack growth rates were distributes on the same region within narrow band and yield the estimation model. Crack propagation analysis presented acceptable results shown by identical manner of the crack length resulted from the analysis and experiment.

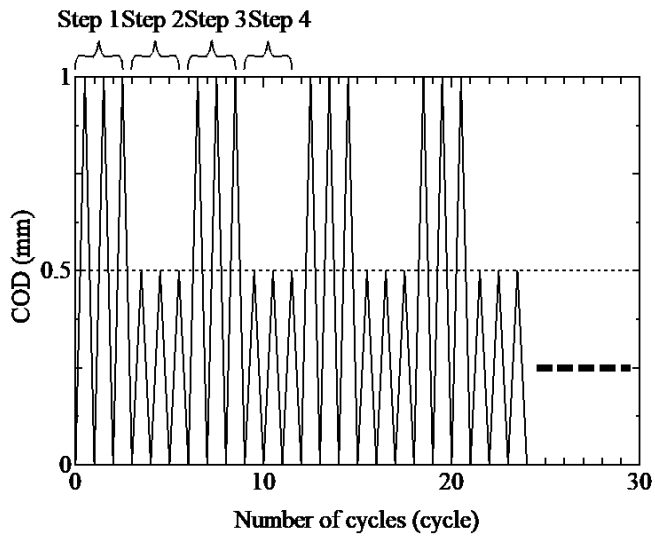
4. Two-steps Variable Amplitude Loading (VA)

4.1. Introduction

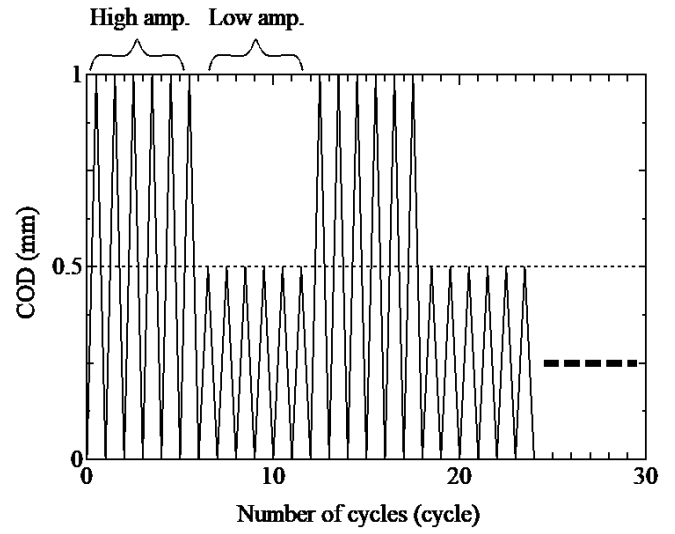
This chapter aims to evaluate crack propagation rate under variable amplitude loadings, particularly when the amplitudes was changed. Due to its simple loading attributes, two-steps variable was selected to examine amplitude alteration influence to the crack extension. The study was firstly done with evaluating crack growth behavior among test cases and investigating the crack growth patterns when the loading amplitudes were altered. Accordingly, collective plots of the crack growth under constant and variable amplitude loading at any crack length were outlined. After the crack growth patterns of two-steps variable cases were characterized, EPFM parameter computed from FE analysis, i.e., the ΔJ , was employed to specify crack growth manners. At the following step, estimation model was established and validated to the test result.

4.2. Load cases

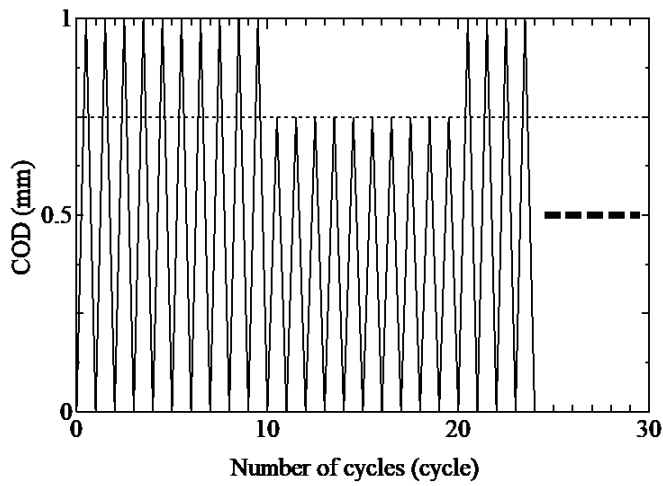
In this load cases, two different constant COD amplitudes were mixed for each specimen. **Fig. 4.1** shows the loading patterns where each loading history consists of high amplitude and low amplitude loading cycles. The magnitude of the high amplitude was 1.0 mm through VA-1 to VA-4. The magnitude of the low amplitude was variably employed in order to examine the effect of different amplitude loading. Besides the steps number at high and low amplitude loading was also considered to identify its influence



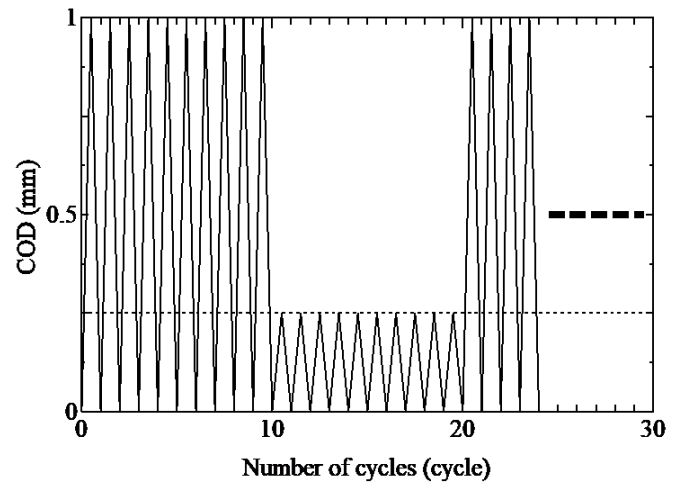
(a) VA-1



(b) VA-2



(c) VA-3



(d) VA-4

Fig. 4.1 Loading patterns of two-steps variable amplitude loadings

to the crack propagation rate. This arrangement is exemplified at the VA-1 and VA-2 where VA-1 and VA-2 have the same loading amplitude but have the difference in the number of steps at high and low amplitude loading.

4.3. Load-COD relationship

The behavior of the specimen under two-steps variable loadings was exemplified by

evaluating VA-1 test case until 6 cycles. The distinction was made for high amplitude in red line and low amplitude in blue line. As expressed on the **Fig. 4.2**, maximum loads of each cycle experience exhibited declining manner because of the crack growth. Due to the higher crack propagation rate, such tendency is more clearly shown at the higher amplitude steps than that under low amplitude loading employment.

In high and low amplitude loading, the crack closure was also figured out. As shown in figure, at the unloading process, the inflection points were indicated due to crack surfaces contact. In such process, when the compressive load was acted to the specimen, the crack faces are contacted together and cause un-continuity of curve where the compressive load significantly increases before it reaches 0-COD.

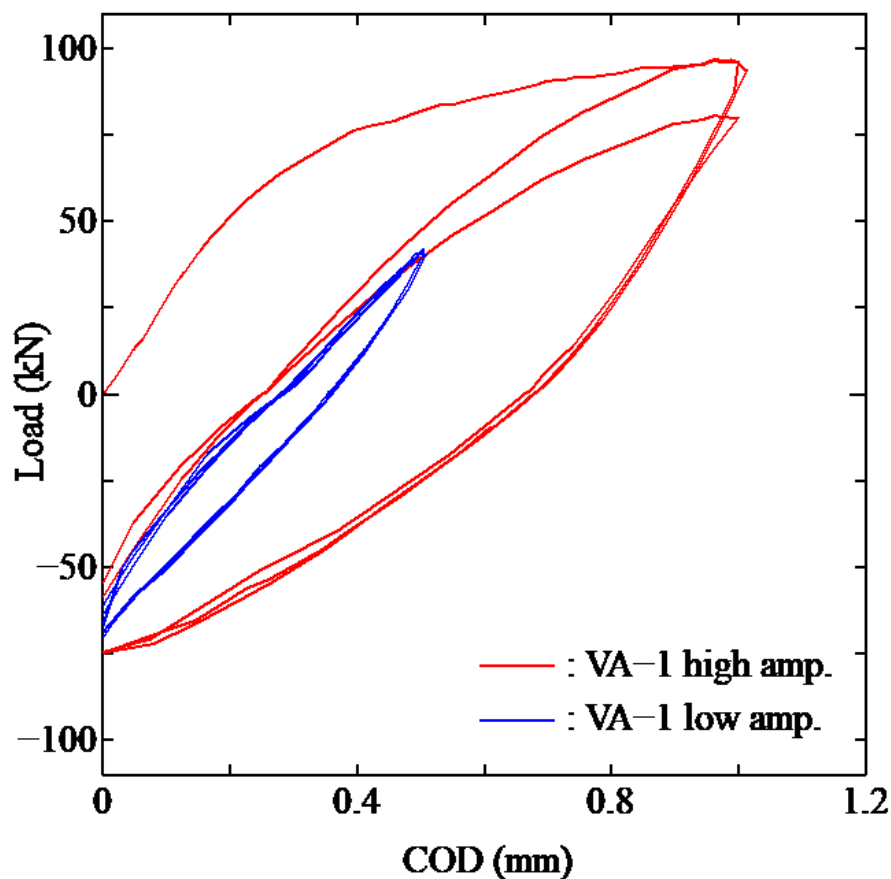


Fig. 4.2 Load-COD hysteresis loop under VA-1

4.4. Crack propagation behavior

Crack propagation behavior is presented with providing crack length and number of cycles as presented in **Fig 4.3**. As expressed in constant amplitude loading, the curve

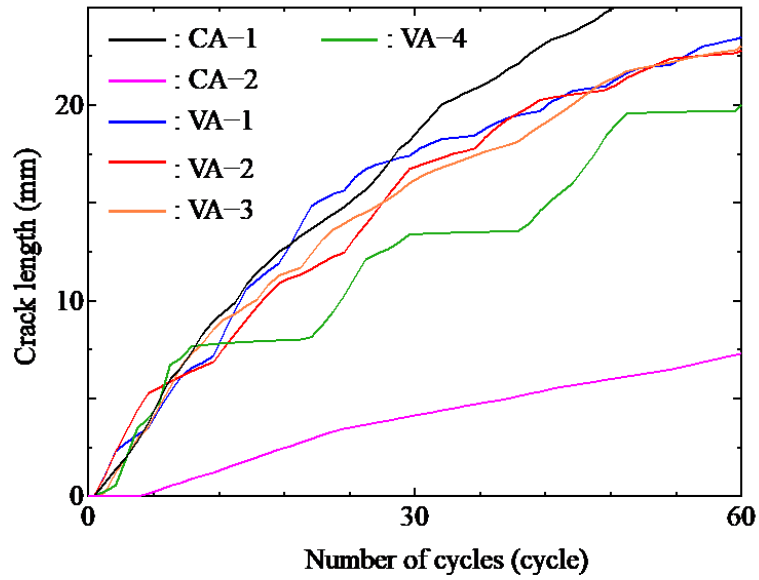


Fig. 4.3 Crack growth rate behavior

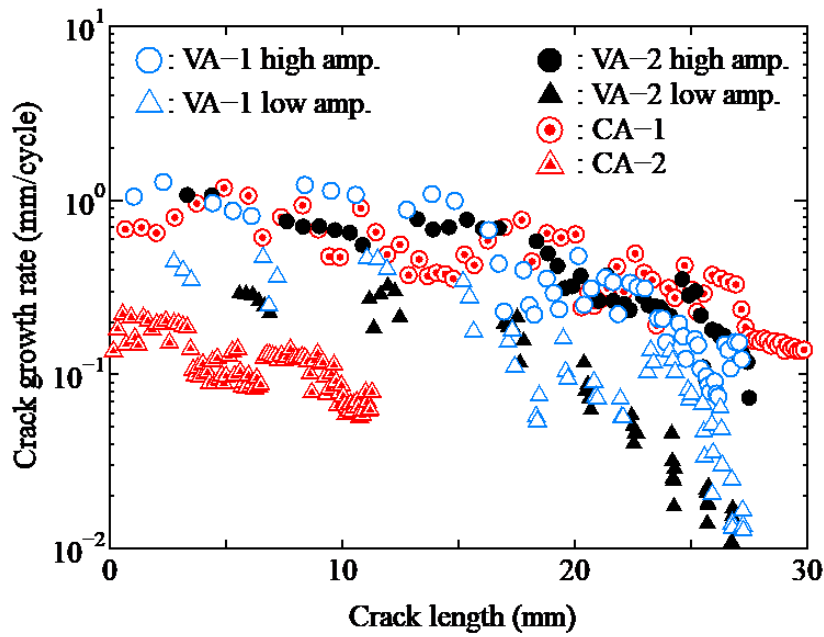


Fig. 4.4 Crack growth rate-crack length relationships

inclination gradually decreased in tandem with the crack extension due to the displacement-controlled conditions. Due to the lowest low amplitude among other two-steps variable cases, VA-4 experienced the most significant reduction in crack growth when the amplitude was dropped. VA-1 and VA-2 performed almost similar crack growth behavior even though their cumulative cycle number in each step are different. VA-3 performed the most identical crack growth behavior to the constant amplitude due to the close magnitude of high and low amplitude steps.

An example of the crack growth rate at any crack length is shown in **Fig. 4.4**. For the comparison, the crack growth rates under constant amplitude loading are also plotted. Comparing high amplitude loading in VA-1, VA-2 with CA-1, the difference identified in the crack growth rate is negligible, meaning the effect of variable amplitude loading can be neglected in high amplitude loading. Meanwhile, low amplitude loading in VA-1, VA-2 represents higher crack propagation rates from those under CA-2 even though COD amplitude is equivalent. This behavior implies that crack propagation rates are accelerated than those under constant amplitude loading when the amplitudes are decreased.

The accelerated crack propagation rates are highly expected by the remained plasticity area at the end of high amplitude loading employment. In such process, a drop of the amplitude is supposed to leave wider plastic zone than that under low amplitude. Accordingly, that wider plasticity area is possible to accelerate the crack growth. Those mechanism are hypothetically provided and need to be clarified for the future research work.

Fig. 4.5 shows the relationships between the crack growth rate and ΔJ under two-steps variable loadings. For high amplitude loading cycles, such relationships are shown in **Fig. 4.5(a)**. The equivalent trend of crack growths to those under constant amplitude loading is exhibited in this figure. Thus, such crack propagation can be predicted from the regression curve for constant amplitude loading. In contrast, the crack growth rates for low amplitude loading cycles, shown in **Fig. 4.5(b)**, are plotted above the regression

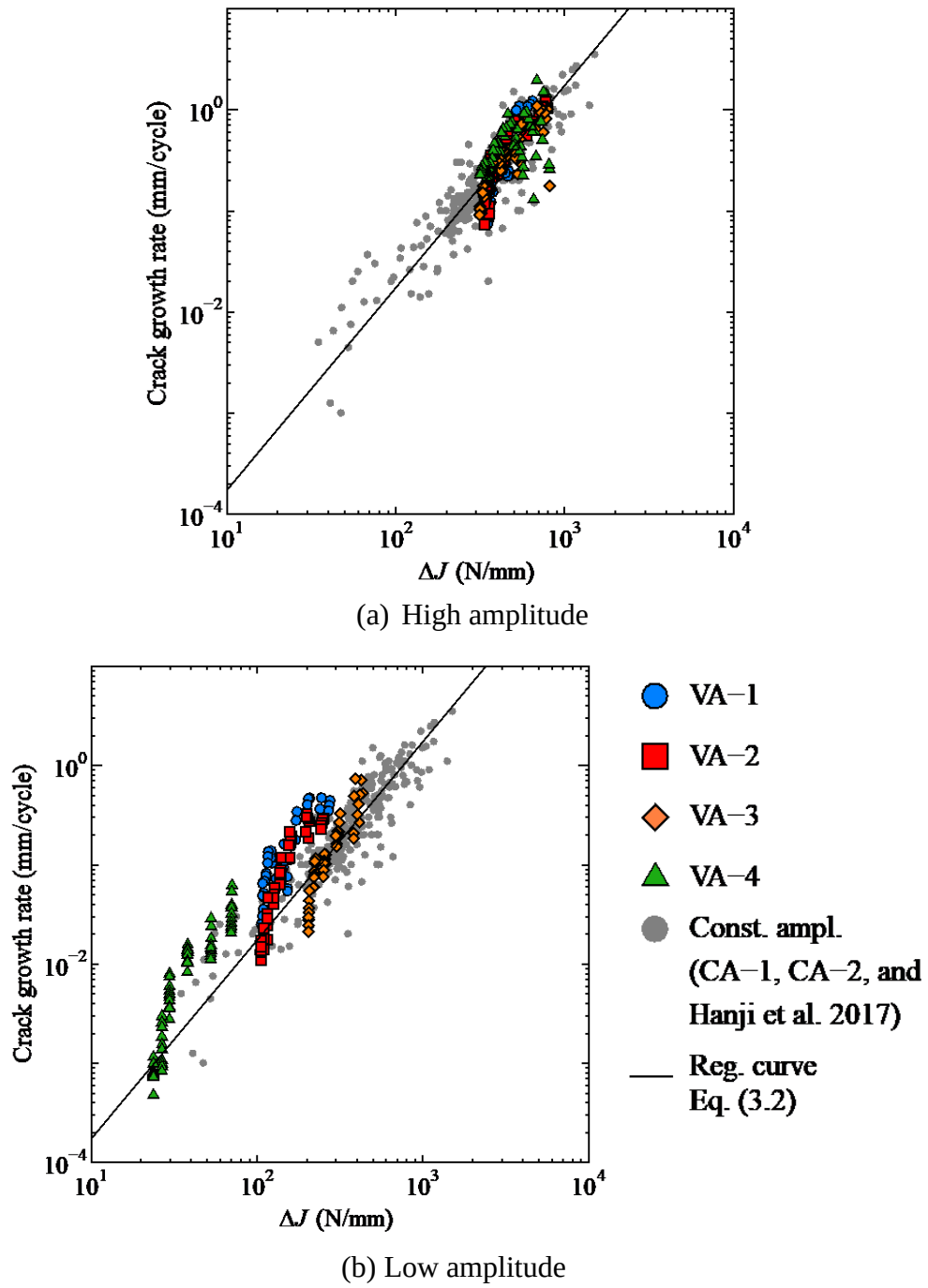


Fig. 4.5 ΔJ -crack growth rate relationships under constant and two-steps variable amplitude loadings

curve especially in relatively high ΔJ region. Hence, the constant amplitude regression curve provides the crack growth rate in non-conservative side. According to the findings,

the crack growth rate in low amplitude loading should be carefully investigated.

4.5. Amplification factor (AF)

To simulate crack growth under variable amplitude conditions, accelerated crack growth is expressed as the amplification factor (AF) that denotes the ratio of crack growth rate under variable amplitude loading to constant amplitude loading. Here, AF is expressed as:

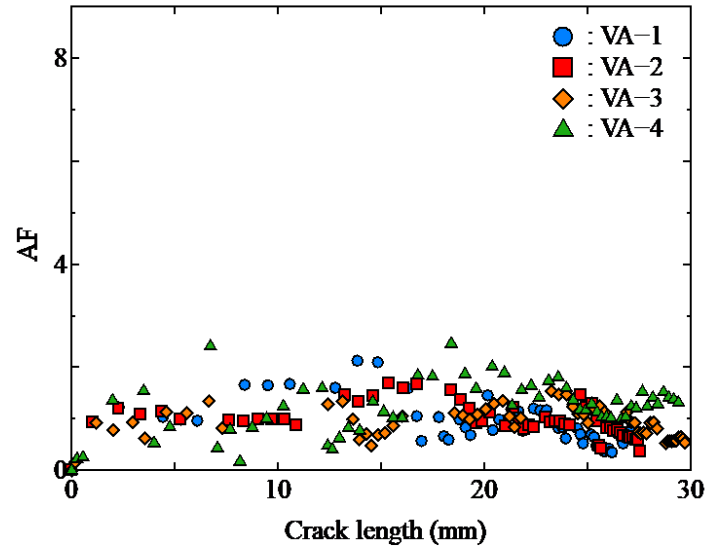
$$AF = \frac{\left(\frac{da}{dN}\right)_{var}}{\left(\frac{da}{dN}\right)_{const}} \quad (4.1)$$

where $(da/dN)_{var}$ is the crack growth rate under variable amplitude loading, $(da/dN)_{const}$ is the crack growth rate obtained from the regression curve presented in **Eq. (3.2)**.

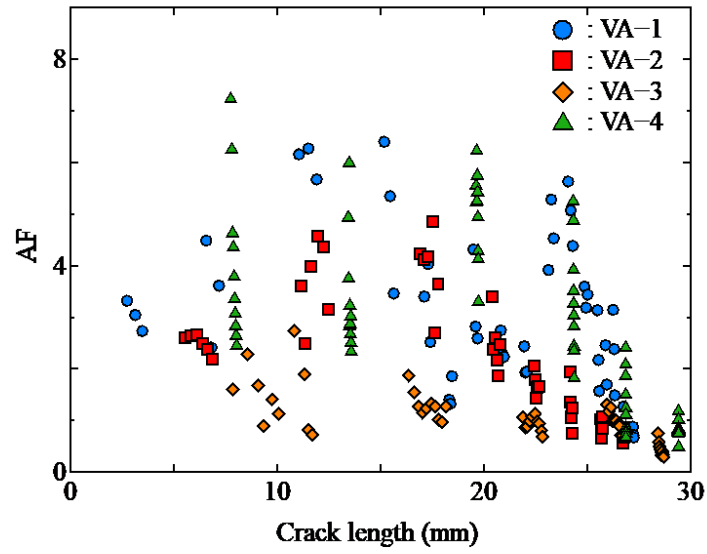
To obtain crack propagation rate under constant amplitude in **Eq. (4.1)**, the ΔJ is substituted to the **Eq. (3.2)** in which, this ΔJ is defined from the equivalent COD and equivalent crack length to those under variable amplitude.

Fig. 4.6 shows a scatter of AF during each loading history. The test data for high and low amplitude loading are individually presented. In both figures, between the results for VA-1 and VA-2, no obvious difference can be observed, meaning that the number of constant loading cycles has little effect on the AF . As shown in **Fig. 4.6(a)**, during high amplitude loading, the AF is scattered below three. In contrast, a high AF can be seen under low amplitude loading conditions, as shown in **Fig. 4.6(b)**. Generally, these high AF s are noticeably decreased with crack extension, and become 1.0 or less when the crack length reaches approximately 25 mm in each loading case, although AF values strongly depend on the loading case, i.e., the magnitude of low amplitude loading.

The AF is supposed to be influenced by a combination of the previous and the current loading amplitudes. Therefore, a new parameter of ΔJ ratio, $\Delta J_i/\Delta J_{i-1}$, which is



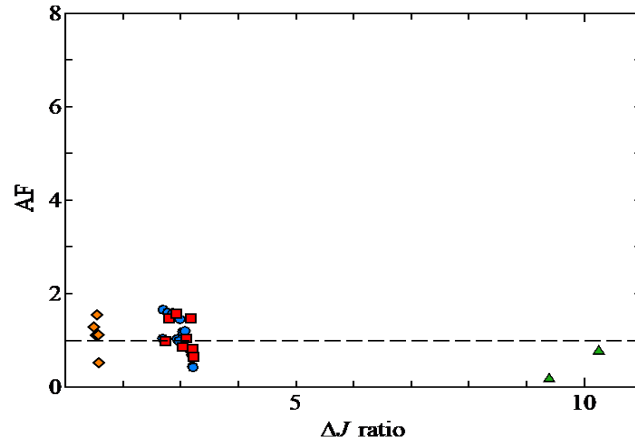
(a) High amplitude



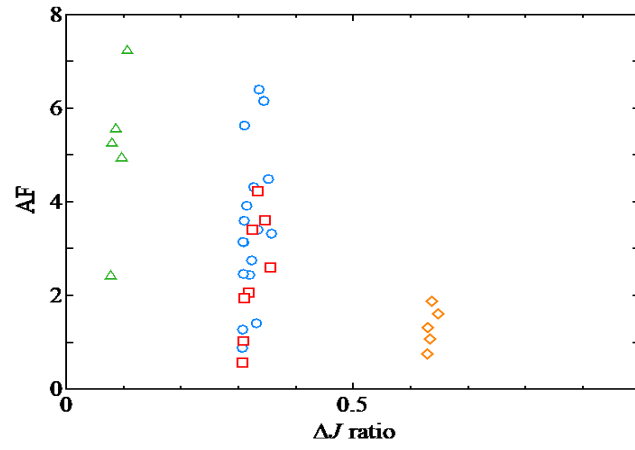
(b) Low amplitude

Fig. 4.6 *AF*-crack length relationships

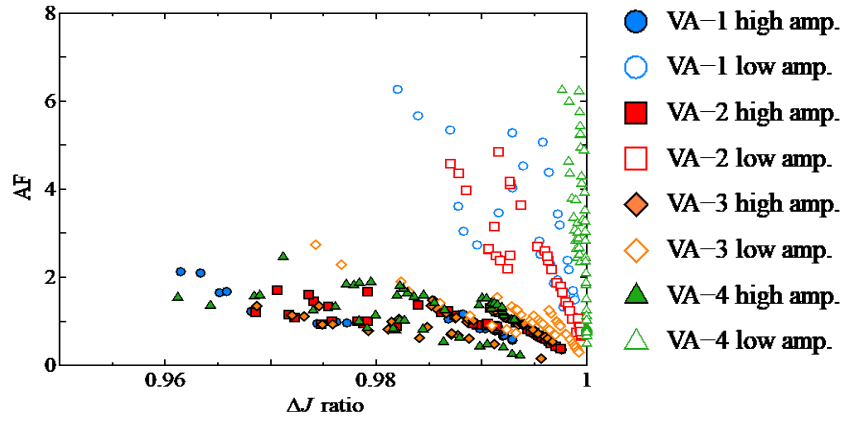
the ratio of ΔJ at an observed cycle to ΔJ at one cycle before the observed cycle, is examined. **Fig. 4.7** shows the relationships between the *AF* and the ΔJ ratio. The figures are separately indicated for the first loading cycle after amplitude alteration and others.



(a) First cycles, $\Delta J_i/\Delta J_{i-1} \geq 1$



(b) First cycles, $\Delta J_i/\Delta J_{i-1} < 0.95$



(c) Second and steady following cycles, $0.95 \leq \Delta J_i/\Delta J_{i-1} < 1$

Fig. 4.7 AF - ΔJ ratio relationships under two-steps variable amplitude loadings

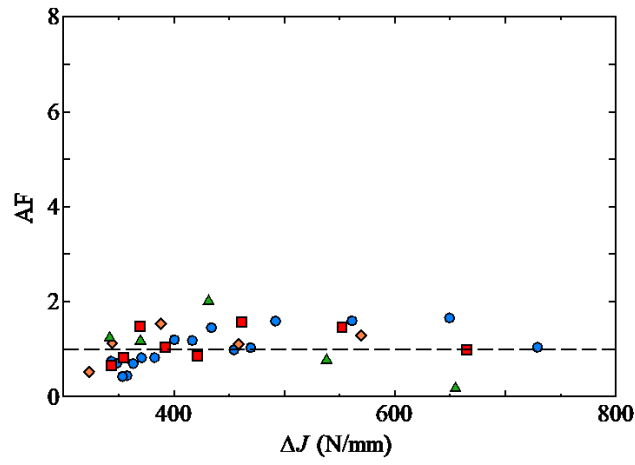
As shown in **Fig. 4.7(a)**, at the first cycle just after the amplitude increases from low to

high level, the AF is almost 1.0 constantly, which means that it can be disregarded when the ΔJ ratio is larger than 1.0. Meanwhile, as shown in **Fig. 4.7(b)**, at the first cycle when the amplitude is reduced from high to low level, the AF tends to increase with the decrease of the ΔJ ratio. At the second and steady following cycles, the ΔJ ratio was gradually decreased even though the COD amplitude was kept constant because of the displacement controlled condition, where the ΔJ ratios were distributed in the range from 0.95 to 1.0, and they have an approximately linear relation with the AF as shown in **Fig. 4.7(c)**. Relied on this figure, the inclination of each curve tends to be changed depending on the magnitude of the COD amplitude or ΔJ .

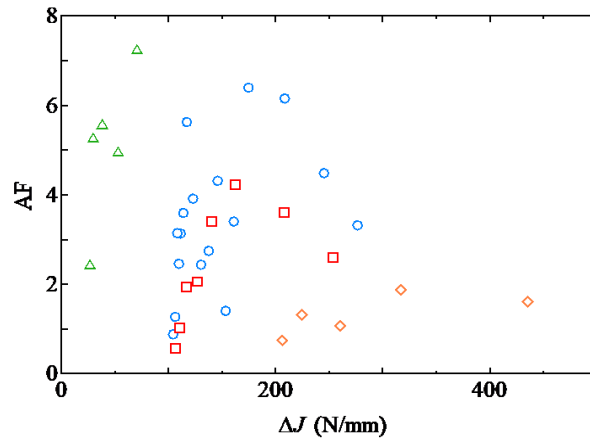
Fig. 4.8 presents the relationships between AF and ΔJ . **Fig. 4.8(a)** exhibits AF plots related to the ΔJ at the ΔJ ratio higher than 1.0, where the AF s lead to give the value of 1.0 constantly at the entire ΔJ values. **Fig. 4.8(b)** demonstrates the AF values at the ΔJ ratio less than 0.95, where the AF s perform elevating trend during reducing ΔJ values. **Fig. 4.8 (c)** shows the AF patterns of each test cases on the range within 0.95 to 1.0. The figure describes that AF s of high amplitude loadings are distributed within 300 to 800 N/mm- ΔJ and AF s of low amplitude loadings are scattered within, 0 to 300 N/mm- ΔJ . The figure exhibits that the low amplitude loading experience more narrow range of AF s scatter than those under high amplitude loading. This tendency is obtained since the crack rapidly propagates at the high amplitude and then dropped when the COD amplitude were decreased.

4.6. Estimation model on the AF

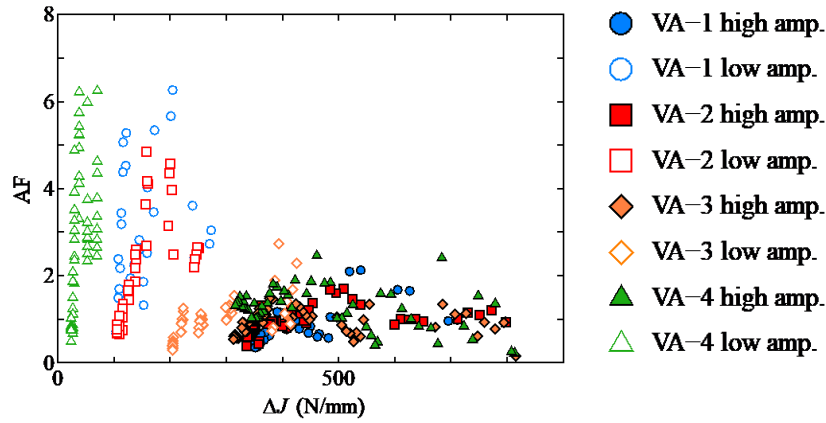
AF estimation formula was firstly established with considering a parameter related to the AF , ΔJ ratio. This parameter was selected since it is dimensionless and applicable to be used in any circumstances of ΔJ value. Accordingly, the estimation was attempted



(a) First cycles, $\Delta J_i/\Delta J_{i-1} \geq 1$



(b) First cycles, $\Delta J_i/\Delta J_{i-1} < 0.95$



(c) Second and steady following cycles, $0.95 \leq \Delta J_i/\Delta J_{i-1} < 1$

Fig. 4.8 AF - ΔJ relationships under two-steps variable amplitude loadings

to be constructed relied on **Fig 4.7**. Estimated AF for ΔJ ratio higher than 1.0 was

obtained from **Fig. 4.7(a)** where the AF s are clearly dispersed at the value equals to 1.0. **Fig. 4.7(b)** and **Fig. 4.7(c)** illustrate that even though the ΔJ ratios are nearly equal, AF s are widely scattered. Due to such sprinkled plots, additional descriptions comparing the test and estimation are required to clarify the accuracy of the estimation formula.

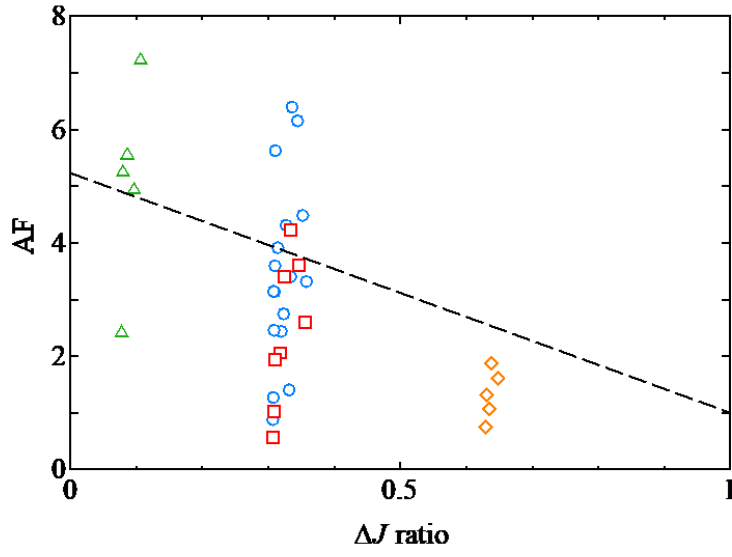
A linear regression is chosen to reflect the AF s plots since such regression line has the best coefficient determination (R squared) than other regression trend lines. However, as expressed on the **Fig 4.9(a)** and **Fig. 4.9(b)**, the linear regression is not able to satisfyingly represent overall dispersed AF s.

According to those findings, both parameters, ΔJ ratio and an absolute ΔJ are required to estimate the AF . Based on the scatter patterns of ΔJ and ΔJ ratio, a formula to estimate the AF is established and proposed as follows:

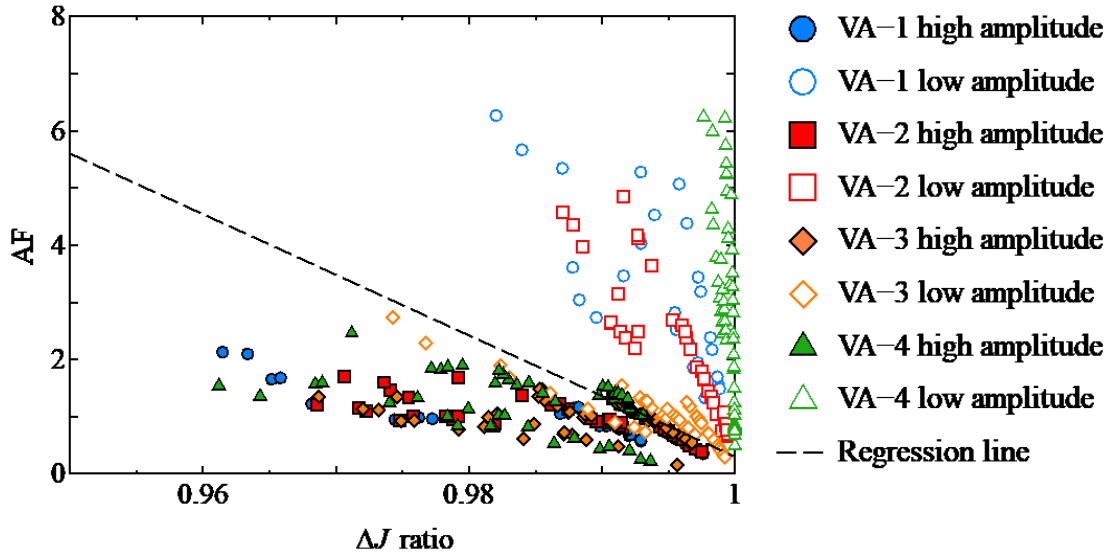
$$AF_{est} = \begin{cases} \left(\frac{787}{\Delta J_i}\right)^{0.4} + 1.8\left(1 - \frac{\Delta J_i}{\Delta J_{i-1}}\right)^{2.5} + 0.5 & \left(\frac{\Delta J_i}{\Delta J_{i-1}} < 0.95\right) \\ \left(\frac{5670}{\Delta J_i}\right)^{1.77} \left(1 - \frac{\Delta J_i}{\Delta J_{i-1}}\right) & \left(0.95 \leq \frac{\Delta J_i}{\Delta J_{i-1}} < 1\right) \\ 1.0 & \left(\frac{\Delta J_i}{\Delta J_{i-1}} \geq 1\right) \end{cases} \quad (4.2)$$

Fig. 4.7(b) and **Fig 4.8(b)** are utilized to obtain estimated AF on the ΔJ ratio range less than 0.95. For the accuracy, the coefficients for ΔJ ratio and ΔJ in **Eq. (4.2)** are obtained by Microsoft Excel's regression analysis relying on AF s scatter. Accordingly, the power for each parameter is firstly determined and by that feature the coefficients then are solved.

Estimated AF on the ΔJ ratio range within 0.95 to 1.0 is derived from the scatter plots of AF s presented in **Fig. 4.7(c)** and **Fig. 4.8(c)**. The equation also considers ΔJ ratio and ΔJ to determine estimated AF where it is resulted with assigning relationship between $AF/(1-\Delta J \text{ ratio})$ as the vertical axis and ΔJ as the horizontal axis. Thus, relying on the



(a) First cycles, $\Delta J_i/\Delta J_{i-1} \geq 1$



(b) Second and steady following cycles, $0.95 \leq \Delta J_i/\Delta J_{i-1} < 1$

Fig. 4.9 Cumulative plots of estimation and test results on AF - ΔJ ratio relationships

plots of $AF/(1-\Delta J \text{ ratio})$, the regression equation to obtain $AF/(1-\Delta J \text{ ratio})$ is stated as the function of ΔJ as shown in **Fig. 4.10**. For the validation of **Eq. (4.2)**, estimated AF by the above equation was utilized to yield estimated crack growth rate which is acceptable for fatigue life estimation as presented in later section.

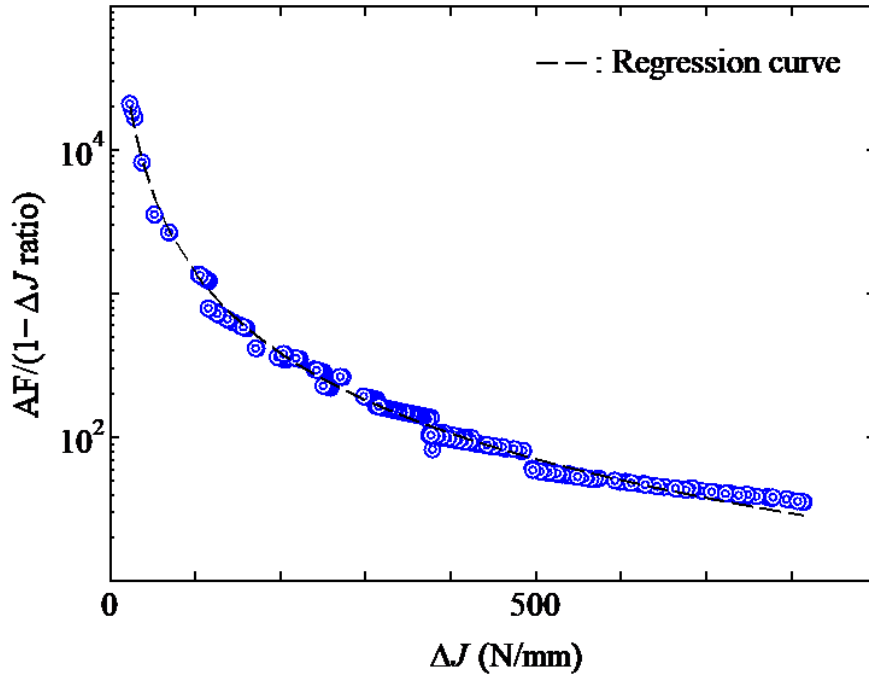


Fig. 4.10 Regression process to obtain estimated AF at $0.95 \leq \Delta J_i / \Delta J_{i-1} < 1$

To illustrate the pattern of estimated AF at the ΔJ ratio less than 1.0, the plots then was established within range 0 to 1.0 relied on two-steps variable amplitude loading as shown in **Fig. 4.11**. In general, this figure indicates that the estimated AF s perform the discontinuity in the plots and specific patterns depending on the ΔJ ratio range. At the range within 0.95 to 1.0, the detail pattern of AF s is given in **Fig. 4.12**. The **Eq. (4.2)** for the range within 0.95 to 1.0 is derived from two-steps cases, so that this formula is only valid for variable amplitude loading at the low cycle fatigue condition.

4.7. Validation on the estimation to the test results crack growth rate

In this research, the estimated crack growth rate $(da/dN)_{est}$ is expressed as follows:

$$\left(\frac{da}{dN}\right)_{est} = AF_{est} \cdot \left(\frac{da}{dN}\right)_{const} \quad (4.3)$$

The estimated crack growth rate for constant amplitude loading presented in **Eq.**

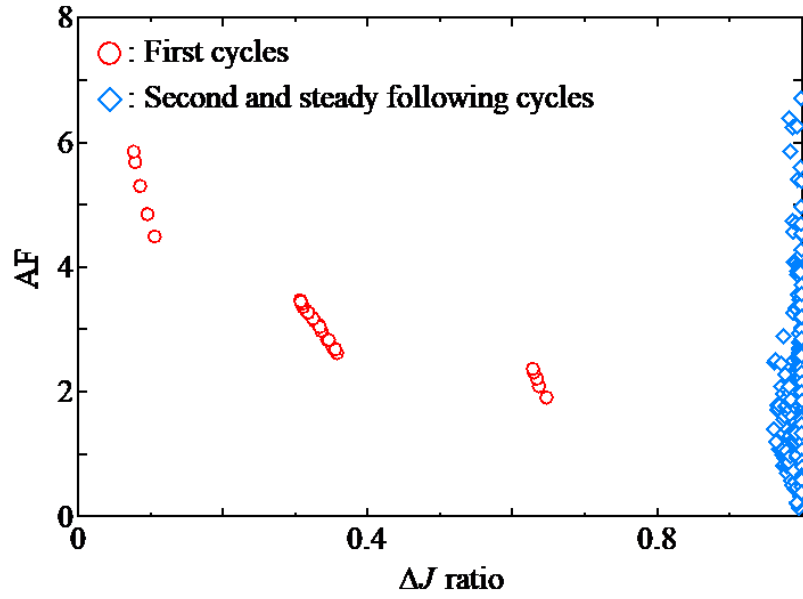


Fig. 4.11 Estimated AF at the $\Delta J_i/\Delta J_{i-1} < 1$

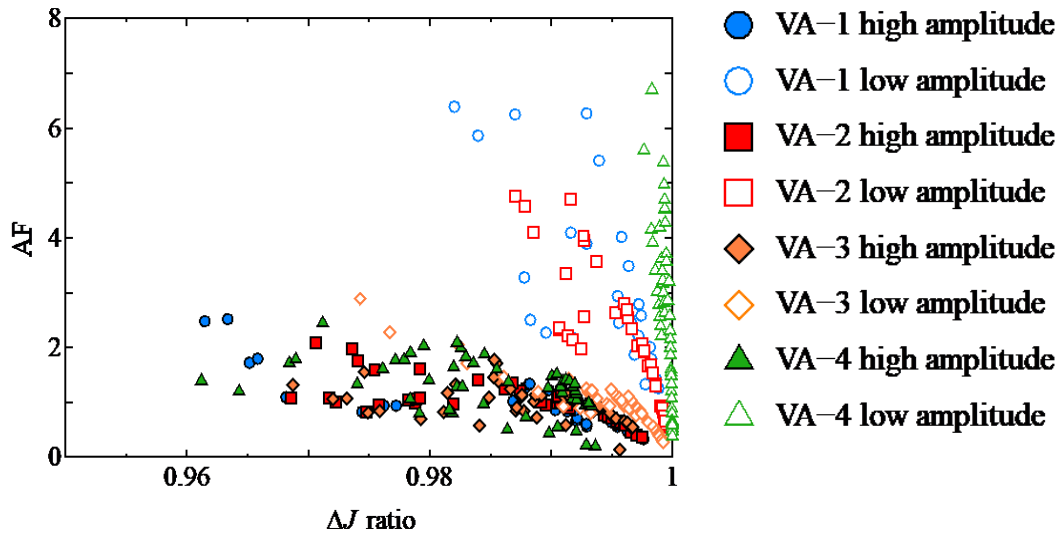


Fig. 4.12 Estimated AF at the $0.95 \leq \Delta J_i/\Delta J_{i-1} < 1$

(4.3), $(da/dN)_{const}$, was calculated from Eq. (3.2) by substituting the ΔJ based on constant amplitude, as explained in Section 3. As shown in Fig. 4.13, estimated and test result crack growth rate were distributed in the same region within narrow band and confirmed applicability of Eq. (3.2) corrected by Eq. (4.2) to estimate crack propagation.

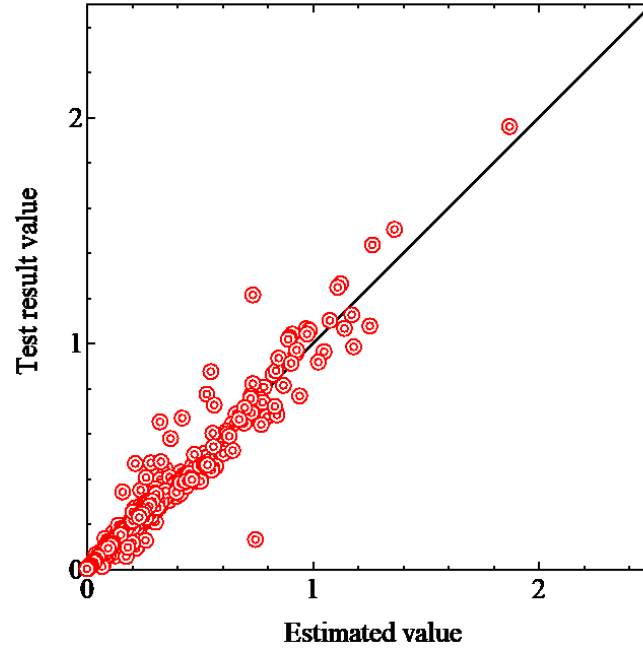


Fig. 4.13 Verification on the estimation model to the test result crack growth rate (unit: mm/cycle)

4.8. Crack propagation analysis

Crack propagation analysis and validation were conducted for all test cases, and the crack growth rates were defined under the two-steps variable amplitude loadings in **Eq. (4.3)**. In this equation, the crack growth rate under constant amplitude loading is calculated from **Eq. (3.2)**, while the estimated AF is calculated from **Eq. (4.2)**. Total crack lengths are obtained by accumulating crack growth rates during each cycle. The crack growth curves, the relationships between the crack length and the number of cycles, are shown in **Fig. 4.14**. For comparison, the prediction curve neglecting the AF is also presented. The figures indicate that disregarding the AF estimates the crack length shorter than the actual one obtained from the experiment, and their gaps are significant. Upon consideration of the AF , the estimated crack length presents close pattern to the test results.

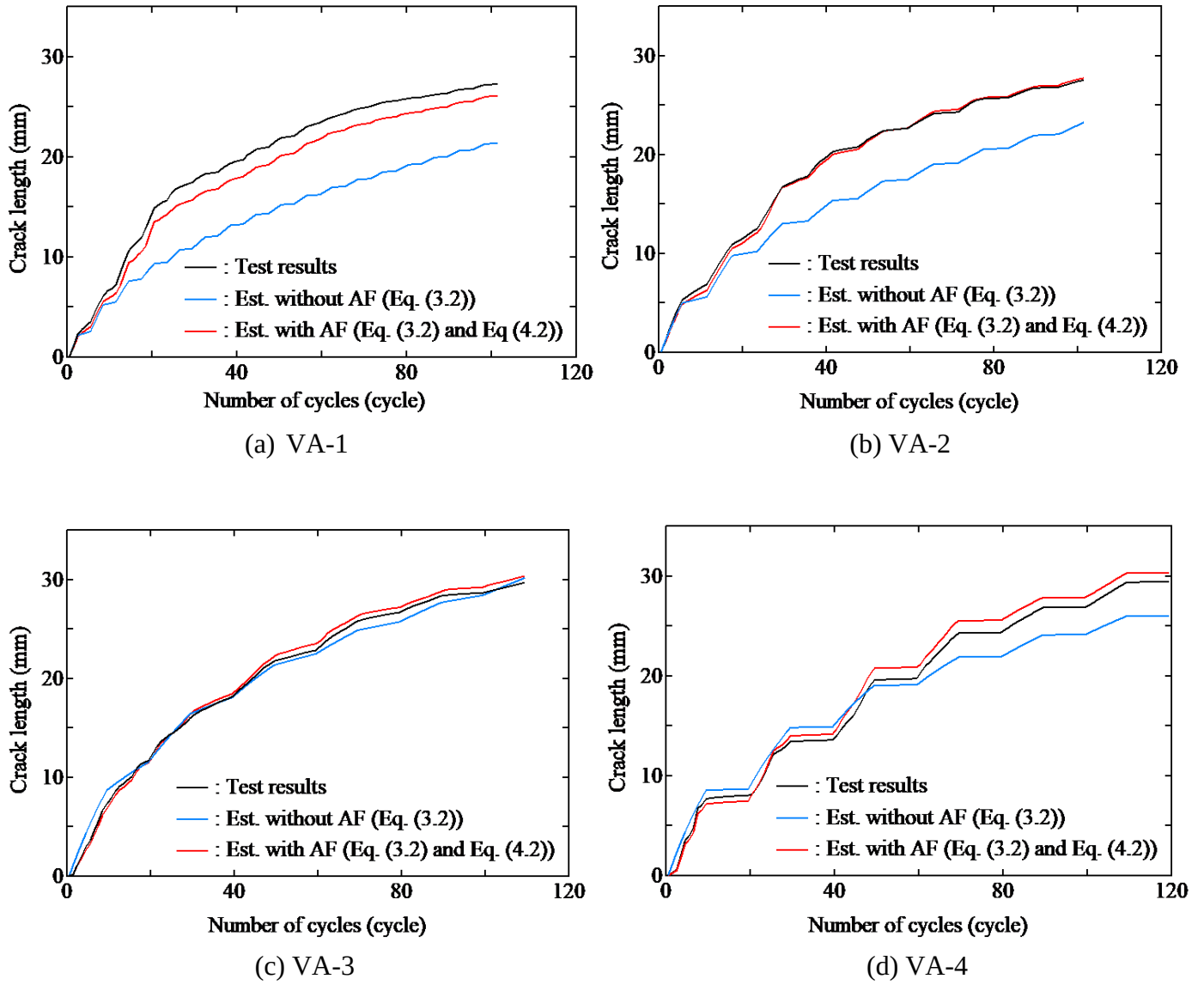


Fig. 4.14 Crack propagation analysis results

Therefore, the formula presented in **Eq. (3.2)** corrected by **Eq. (4.2)** is applicable to estimate the crack growth rate under two-steps variable amplitude loading.

4.9. Summary

This chapter elaborates crack propagation behavior under two-steps variable amplitude loading. The observation was emphasized on the loading amplitude alteration

from high to low levels and from low to high levels. When the amplitudes were dropped from high to low levels, it was found that the crack growth rates increase than those observed under constant amplitude loadings. Meanwhile, when amplitude loading increased from low to high levels, the crack growth rates were similar to those observed under constant amplitude conditions. Furthermore, during the second and steady subsequent cycles, the crack growth rate decreased continuously in the same manner as those under constant amplitude loading.

The crack growth rates under two-steps variable amplitude loading conditions can also be simulated by AF , which is the ratio of the crack growth rate under two-steps variable amplitude loading conditions to that under constant amplitude loading conditions. Estimation model are also presented from this study and yields the formula which is considered acceptable, as shown by the close pattern between the estimation and experimental results.

5. Random Variable Amplitude Loading (RA)

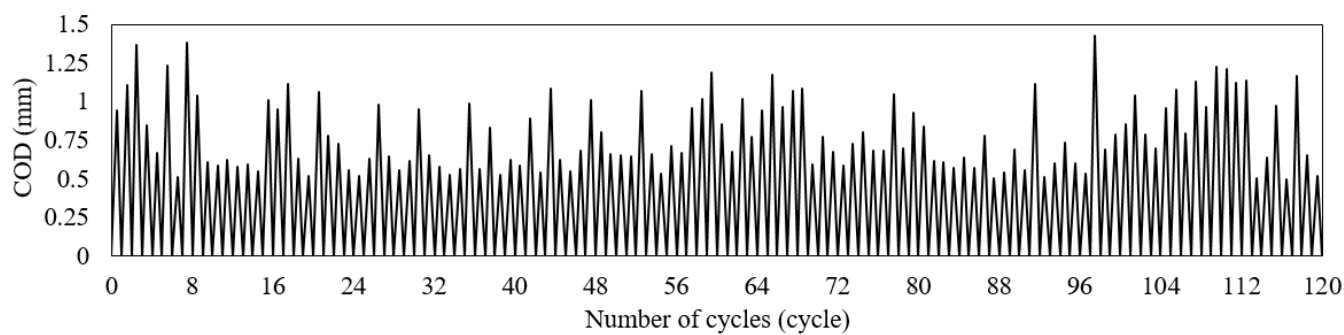
5.1. Introduction

Crack propagation behavior under random amplitude loading was discussed in this chapter. This random amplitude loading considered different amplitude loading ranges and the average of COD. Firstly, the amplitudes of random loading was determined where they were un-repetitive, to obtain value of fully random. Next, to identify the behavior of random and constant amplitude conditions, crack growth rates from both typical loadings were togetherly plotted. Since at two-steps variable amplitude loading, crack growth rates were governed by ΔJ ratio, the growth rate under random loading was also distinguished based on such parameter. At the final work, estimation model from two-steps variable was employed to the random amplitude loading and then validated.

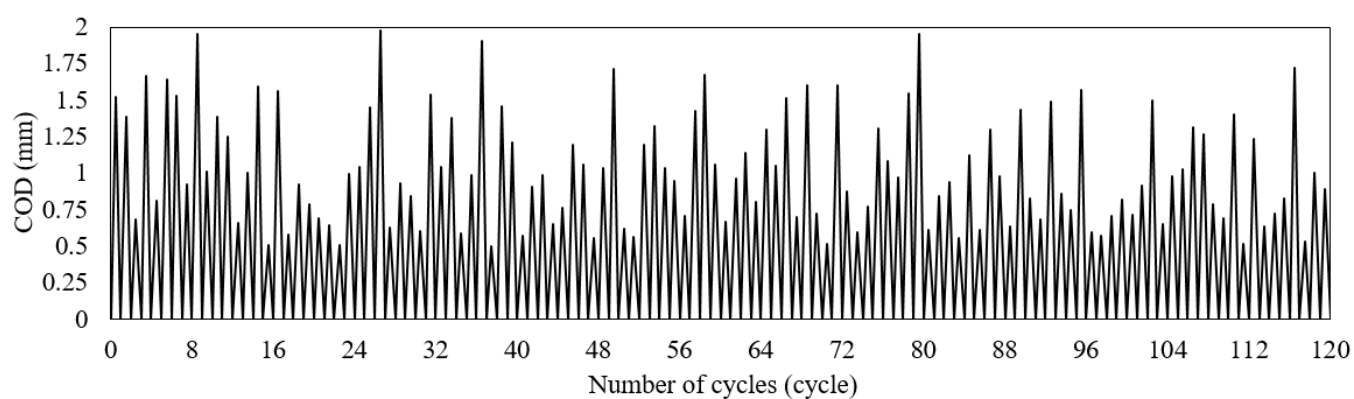
5.2. Load cases

The COD amplitudes were obtained by arranging each peak COD in range within 0 to 300-level, in which 0-levels refers to minimum COD and 300-level is addressed to maximum COD.

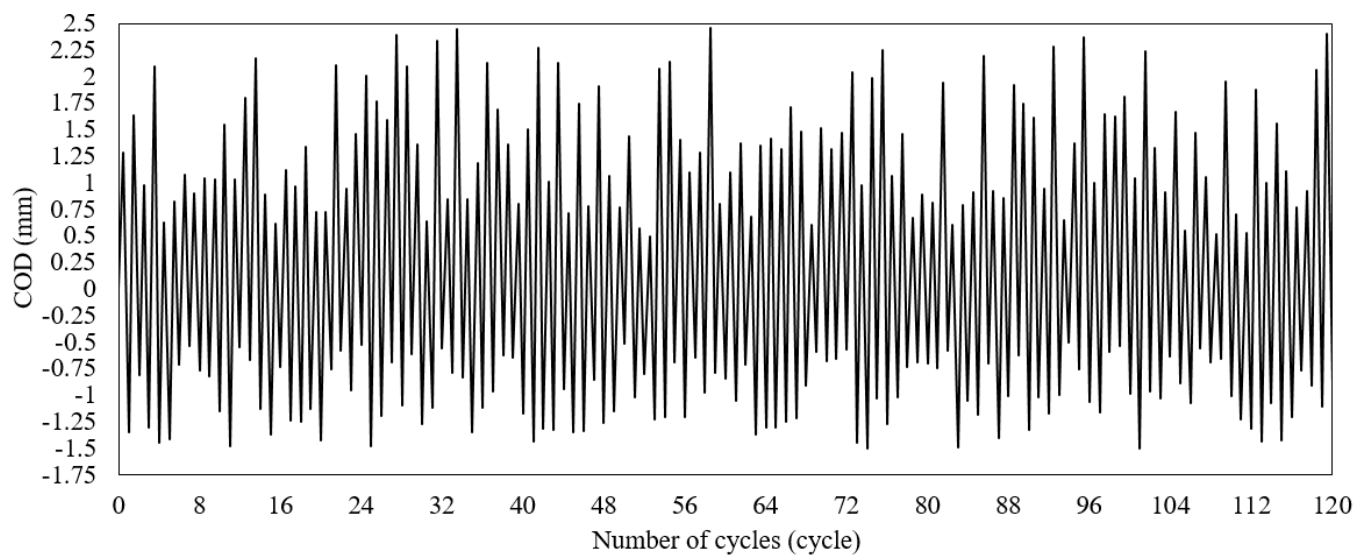
Three different ranges of COD were employed: RA-1, RA-2, and RA-3, with range values of 0.5 to 1.5 mm, 0.5 to 2.0 mm, and -1.5 to 2.5 mm, respectively for random amplitude loadings, as shown in **Fig. 5.1**. All CODs are un-repetitive and attributed to be



(a) RA-1



(b) RA-2



(c) RA-3

Fig. 5.1 Loading patterns of random variable amplitude loading

random entirely. The RA-1 and RA-2 are employed to examine the effect of amplitude

range differences to the crack extension. Both have minimum COD set to be 0 mm. In the RA-3, the minimum COD are set to be -1.5 mm to evaluate effect of the average COD on the crack propagation. As for the RA-3, the minimum COD value more than -1.5 mm is not recommended since it is required very high compressive load which is difficult to be reached on the test.

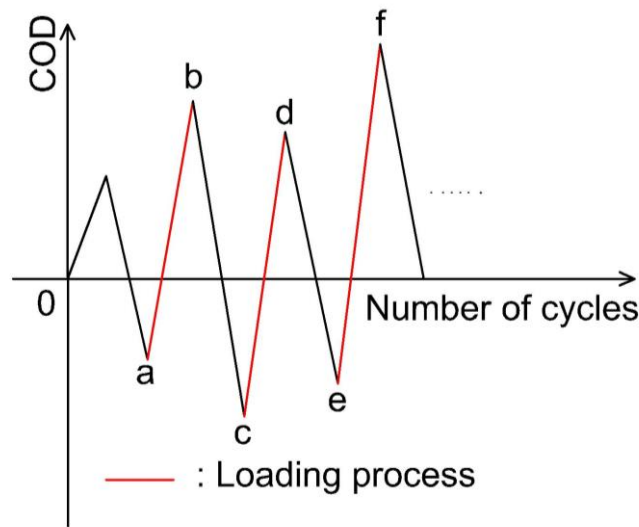


Fig. 5.2 Loading process used to define the ΔJ for RA-3

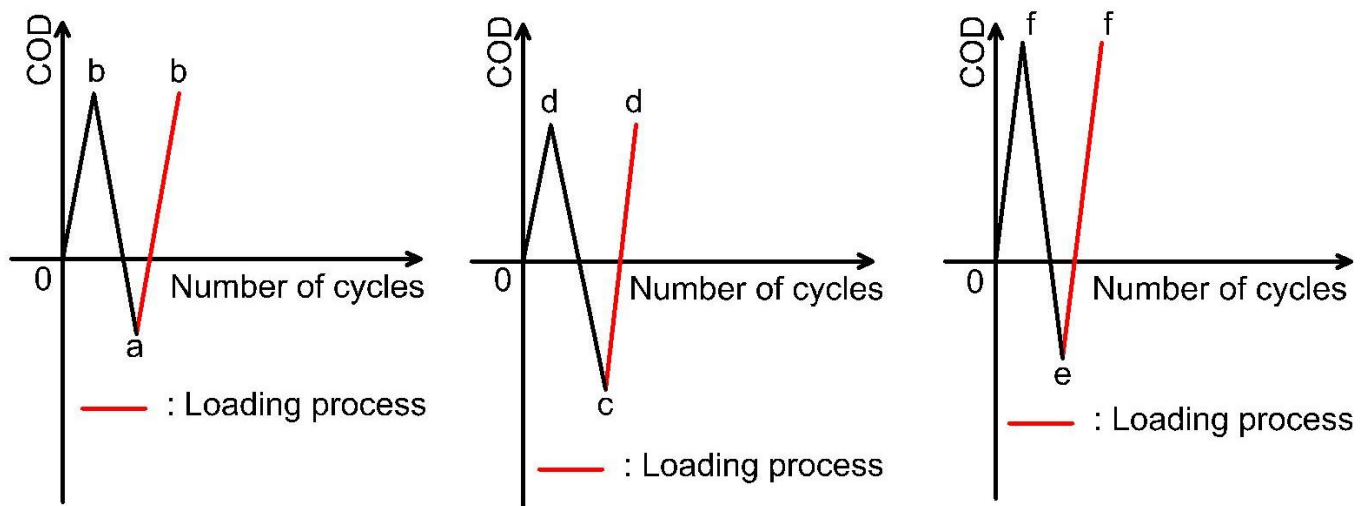


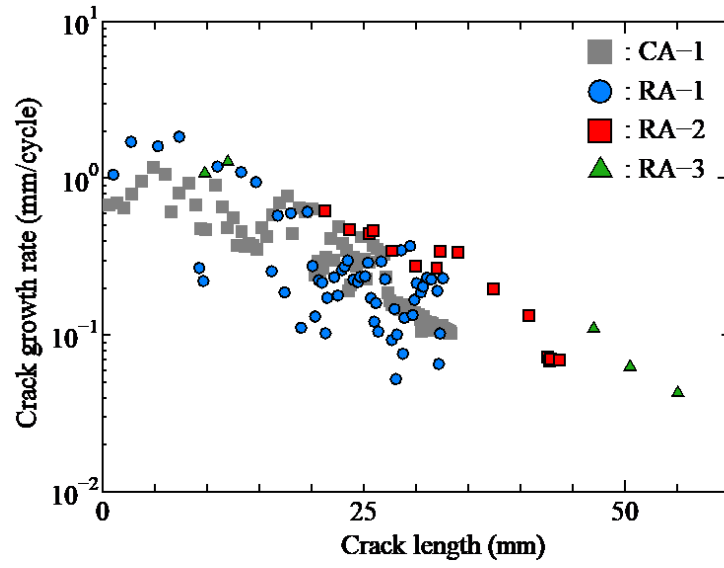
Fig. 5.3 The ΔJ for RA-3 defined based on constant amplitude principle

5.3. Test result arranged with ΔJ

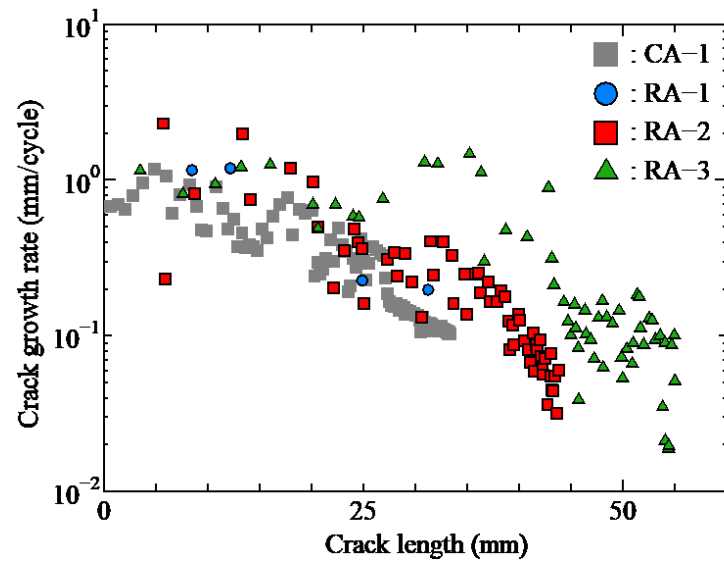
In random loading, the ΔJ is calculated in each cycle in the loading process as previously illustrated in **Fig. 2.5**. Loading simulation on the model is applied until 1.5 cycles. For RA-1 and RA-2, the ΔJ is defined with the same principle as constant amplitude with minimum COD equals to 0. Meanwhile, for RA-3, the ΔJ is computed and refers to an illustration in **Fig. 5.2**. The figure outlines several cycles to define the ΔJ in a loading history. Line a-b, c-d, e-f are the loading processes that need to be defined for the ΔJ . **Fig. 5.3** outlines each loading process in which ΔJ is computed from the minimum COD to maximum COD.

It is obtained from two-steps variable amplitude cases, that the crack growth rate is governed by a combination between the previous and the current loading amplitudes. Therefore, the ΔJ ratio, $\Delta J_i/\Delta J_{i-1}$, which is the ratio of ΔJ for an observed cycle to ΔJ at one cycle before the observed cycle, was also employed in these typical loadings.

Fig. 5.4 shows an example of the crack growth rate at any crack length which provides comparison between random amplitude loadings and CA-1. The figures are individually presented for the ΔJ ratio equal to or larger than 1.0 and for the ΔJ ratio less than 1.0. To obtain proportional comparison on the crack growth rates, 0.9 to 1.1 mm-COD range is selected for random loading and then compared to the CA-1. Since the COD of both types of loadings are almost equivalent, the crack growth comparison can be considered reliable. In figure, the differences in crack growth rates between CA-1 and random amplitude conditions at ΔJ ratio higher than 1.0 are indistinguishable as shown in **Fig. 5.4(a)**. Hence, the influence of random amplitude loadings can be neglected when the amplitudes increase from low to high levels. In contrast, as expressed in **Fig. 5.4(b)**, the crack growth rate observed during ΔJ ratio less than 1.0 shows considerable distinctions from those observed during CA-1, even though the COD amplitude is equivalent. This implies that accelerated crack growth should be considered when the



(a) ΔJ ratio ≥ 1



(b) ΔJ ratio < 1

Fig. 5.4 Crack growth rate-crack length relationships under CA-1 and random amplitude loadings

amplitude was dropped. Additionally, the graph also figured out that the random amplitude range affects the crack growth rates, i.e., the RA-3 provides higher crack growth rate than those observed in the RA-1 and RA-2.

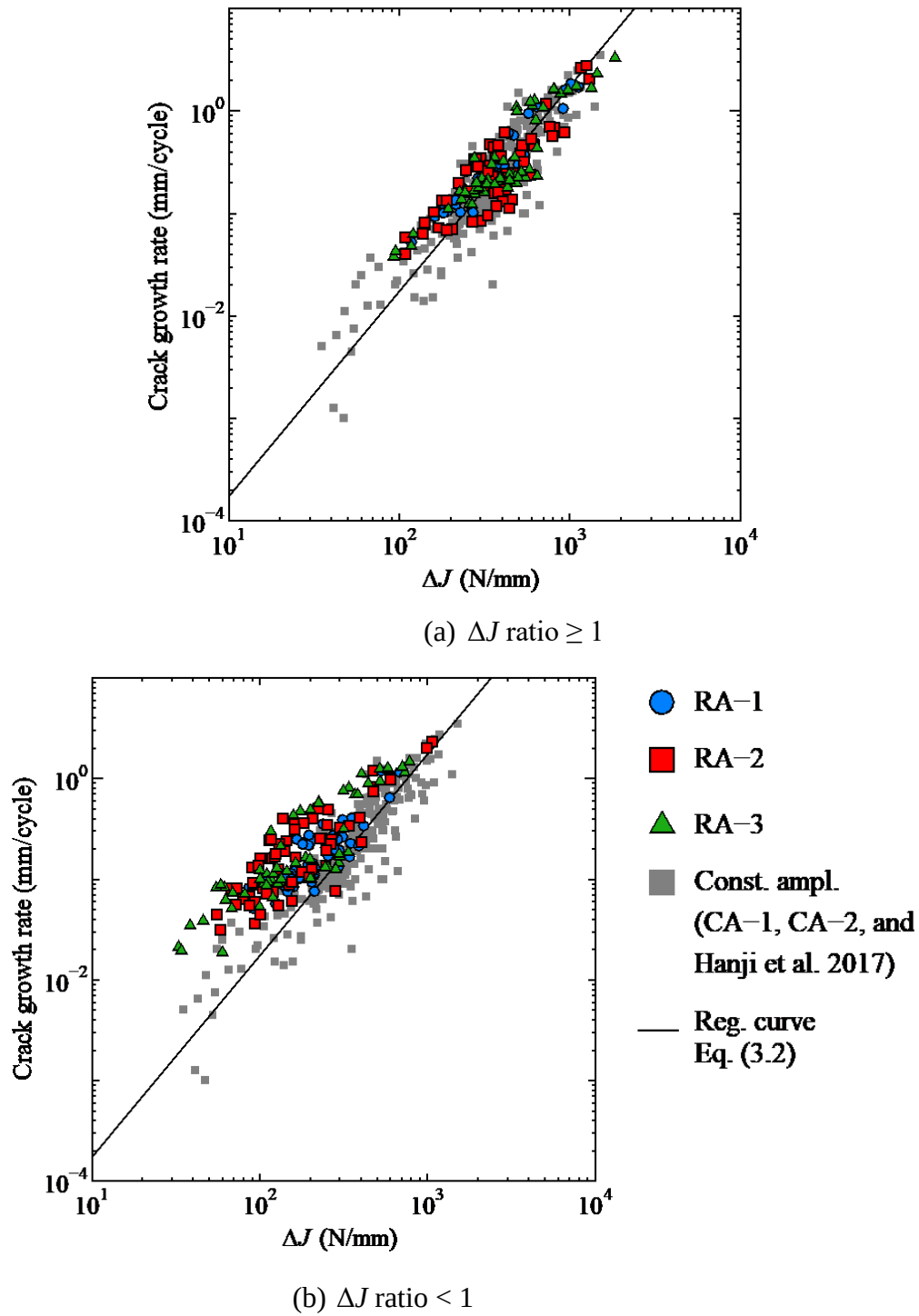
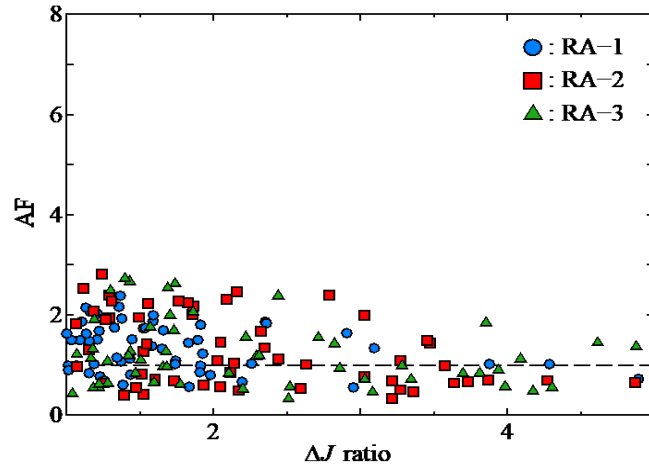
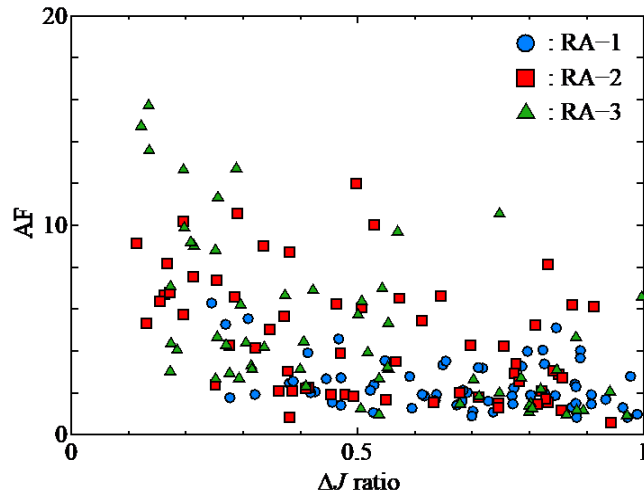


Fig. 5.5 ΔJ -crack growth rate relationships under constant and random amplitude loadings

Fig. 5.5 shows relationships between the crack growth rate and the ΔJ under random variable amplitude loading where the relationships are distinguished based on ΔJ



(a) ΔJ ratio ≥ 1



(b) ΔJ ratio < 1

Fig. 5.6 AF - ΔJ ratio relationships under random amplitude loadings

ratio range, as previously explained. Similar pattern for all cases of random amplitudes loading to the constant amplitude conditions was indicated at the ΔJ ratio higher than 1.0 as shown in **Fig. 5.5(a)**. This behavior indicated that crack growth rate for ΔJ ratio higher than 1.0 can be interpreted by utilizing regression curve originated from constant amplitude loadings. On the contrary, as shown in **Fig. 5.5 (b)**, at the ΔJ ratio less than 1.0, a trend of the crack growth rate which is obviously different than that in constant amplitude loadings was interpreted where most of the plots are scattered above the regression curve. Therefore, the crack growth rates have to be corrected by the AF for the case of the ΔJ ratios less than 1.0.

In random amplitude cases, the AF was also considered since such parameter was noticeably to be influenced by the ΔJ ratio. Hence, relationship between AF and ΔJ ratio are presented in **Fig. 5.6**. When the amplitude increases from low to high level, the constant value of 1.0 in AF is exhibited as shown in **Fig. 5.6(a)**. This means that the contribution of the AF is meaningless when the ΔJ ratio is larger than 1.0. On the contrary, as shown in **Fig. 5.6(b)**, when the amplitude is reduced from high to low level, AF tends to increase with the decrease of the ΔJ ratio.

Fig. 5.7 expressed relationships between AF and ΔJ . The AF s lead to give the value of 1.0 constantly during entire ΔJ values at the ΔJ ratio higher than 1.0, as presented

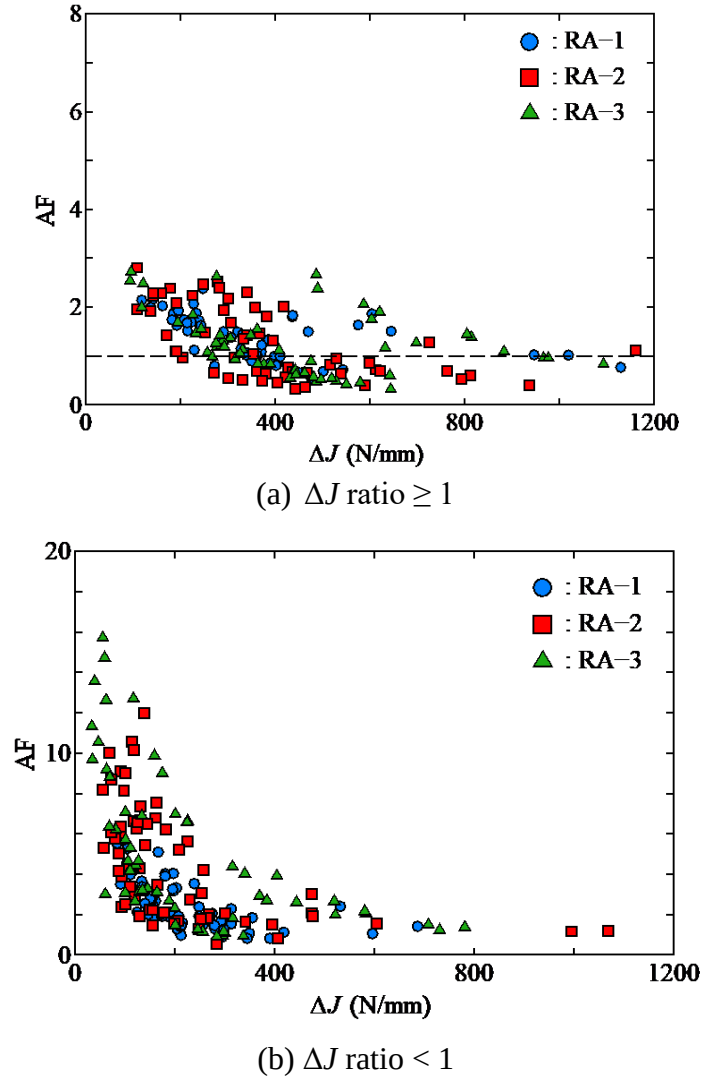


Fig. 5.7 AF - ΔJ relationships under random amplitude loadings

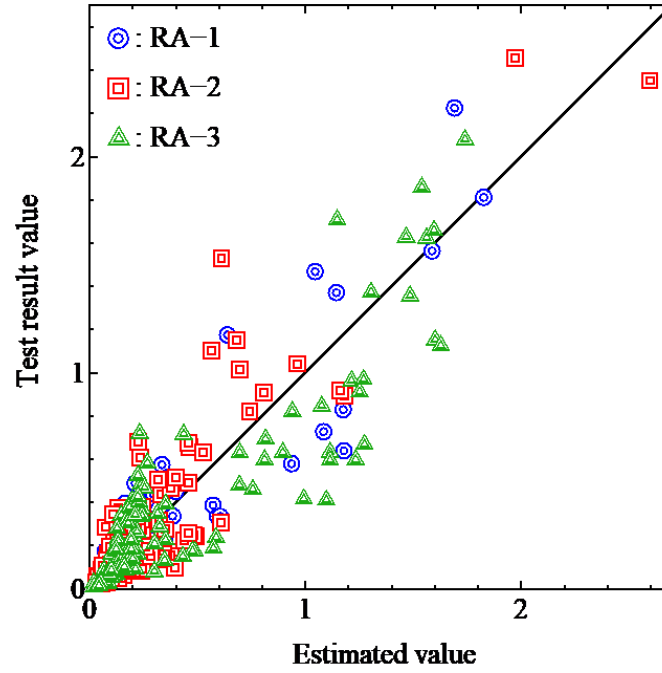


Fig. 5.8 Verification of estimated crack growth rate to the test result (unit: mm/cycle)

in **Fig 5.7(a)**. On the contrary, as exhibited in **Fig 5.7(b)**, the *AF*s performs elevating manner together with decreasing value of ΔJ at the ΔJ ratio less than 1.0.

As an estimation of the *AF*, the same equation to **Eq. (4.2)**, derived for two-steps variable amplitude loading, is adopted even for random variable amplitude loading. Since the random loading has un-repetitive COD, estimated *AF* only adopts two formulas from two-steps cases. The estimation formula for ΔJ ratio less than 0.95 and the estimation formula for ΔJ ratio higher than 1.0. Here, the estimation formula for ΔJ ratio less than 0.95 for two-steps is subjected to estimation formula for ΔJ ratio less than 1.0 in random loading. The comparison of the crack growth rate under random amplitude loading, shown in **Fig. 5.8**, elucidates the crack growth rate obtained by **Eq. (3.2)** and corrected by the *AF* in **Eq. (4.2)** is close to the test result, even though there is the scatter on the plots. This condition is identified since the regression line is established mostly from the crack propagation rate less than 1.0 mm/cycle. Hence, it may result a more wide range plots for crack propagation rate higher than 1.0 mm/cycle. The relationships

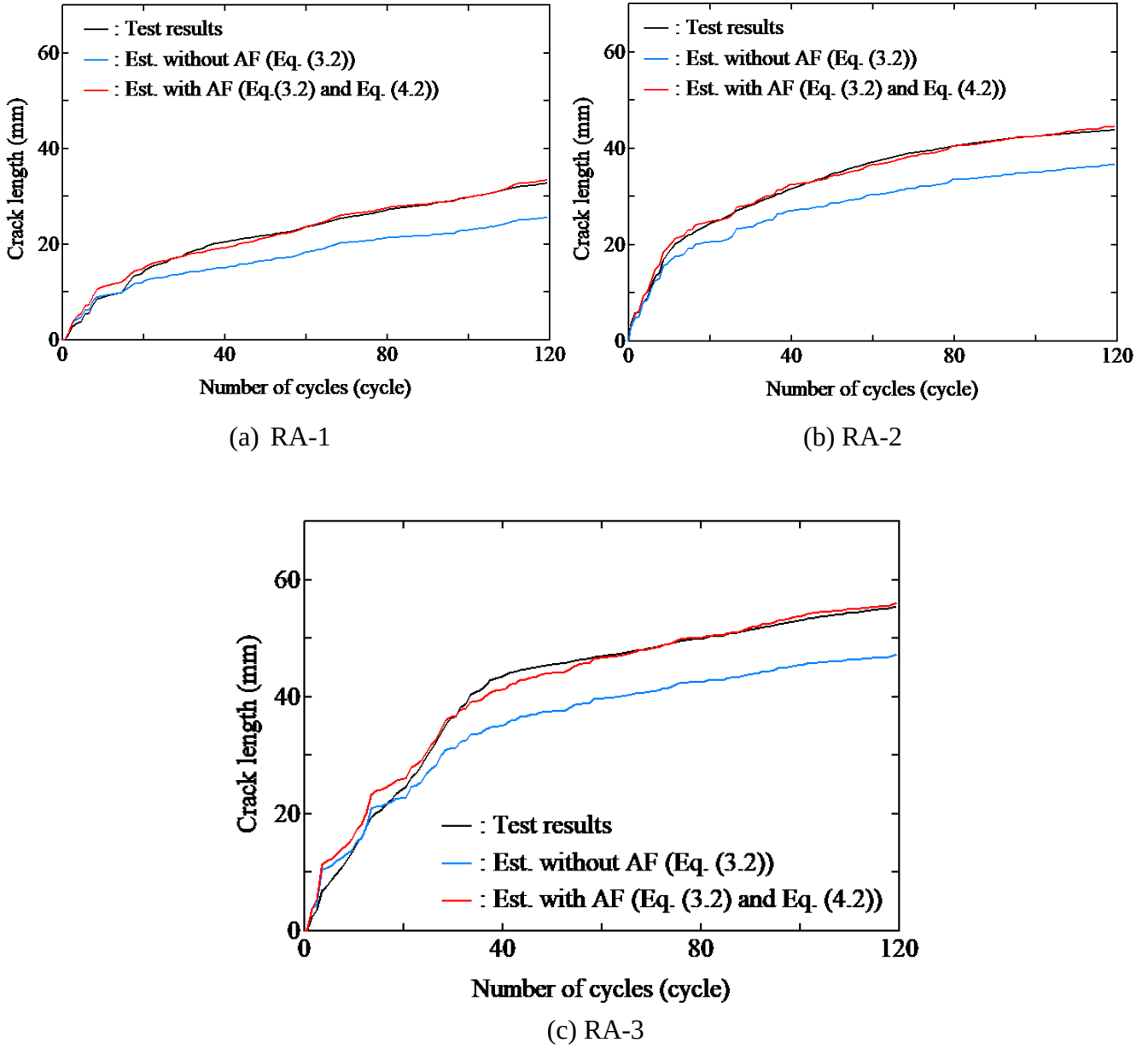


Fig. 5.9 Crack propagation analysis results

between the crack length and the number of cycles for random variable amplitude loading are shown in **Fig. 5.9**. In the figure, a prediction curve neglecting the *AF* is drawn together. It is confirmed that the neglecting presents shorter crack length than the test results. The consideration of the *AF* can improve the accuracy so that the crack growth curve expresses the pattern similar to the test results. Consequently, it is confirmed that the *AF* by **Eq.**

(4.2), derived from two-steps variable amplitude loading, is also applicable for random variable amplitude loading regardless the magnitude of COD and average of COD.

5.4. Summary

On the purpose to investigate crack propagation behavior under random amplitude loading, an experimental work and analytical study were conducted. As identified from two-steps variable loadings, crack growth rates were governed by ΔJ ratio. Accordingly, ΔJ ratio which is less than 1.0 and higher than 1.0 were considered and such distinction manners were employed to the random variable loadings. Relied on the experiment, crack growth rates increased when the amplitudes were reduced from high to low levels or at ΔJ ratio less than 1.0, than those observed under constant amplitude loading. Contrarily, the equivalent tendency of the crack growth rates to that under constant amplitude conditions were identified when the amplitudes were elevated from low to high levels or at the ΔJ ratio higher than 1.0. Estimation formula yielded from two-steps variable cases was applied to the random variable loading expressed that the estimation model can be accepted in domain of random loading regardless the COD magnitude and COD mean.

6. Summary and Conclusions

6.1. Conclusion

In this study, the author employed ΔJ as a parameter to evaluate crack growth rate. Under variable amplitude loading, LCF crack propagation was investigated through experiments and analysis. By those ways, the observations were emphasized on the crack growth behavior when the loading amplitudes were altered. The conclusions derived from this study are included as follows.

1. As a general summary, as discovered from the variable amplitude loading, there is a different tendency between the crack growth rates at the decreasing and at the increasing phases of amplitude when those two crack growth manners were evaluated to those under constant amplitude conditions.
2. Chapter 2 contemplates the different integration paths and loading cycles to examine the independency of ΔJ and relying on the analysis, the negligible difference on the ΔJ was verified. ΔJ which correlates to the crack length were also outlined in this chapter where the ΔJ s performed declining manner along with the extension of the crack due to displacement-controlled condition.
3. Chapter 3 reveals crack propagation pattern under constant amplitude loading. To ensure the accuracy of the estimation, study results from the past study (Hanji et al. 2017), also tested crack growth under constant amplitude conditions, were involved and collectively plotted with the present study. As for the validation, cumulative plots of load-COD response at the same crack lengths, resulted from

experiment and analysis, figure out that maximum and minimum loading points are identical which confirm that the analysis can interpret the experimental results. The present study employed constant amplitude with different material and specimen size from the previous research (Hanji et al., 2017). However, from both studies, the crack growth rates were distributed within narrow band regardless material grades, specimen dimensions and COD averages.

4. In chapter 4, two-steps variable amplitude loadings were employed to the specimen and crack growth rates were evaluated. At the amplitude change from high to low levels, escalation rates of the crack growth than those observed under constant amplitude condition were confirmed. In contrast, the close pattern of crack growth rates to those under constant amplitude loadings was indicated when the amplitudes were elevated from low to high levels. The second and following cycles on the two-steps variable loading performed crack growth rates sustainability decreased in the same manner as constant amplitude. In this chapter, an amplification factor (AF) which is the ratio of crack growth rate under variable amplitude to that under constant amplitude loading was introduced, and confirmed to be useful parameter to describe the crack growth rate manners. At the final work, estimation model was resulted from two-steps variable amplitude and then validated to be acceptable.
5. In chapter 5, random variable loading was employed to investigate crack propagation rate. Since the crack growth rates were noticed to be correlated to the ΔJ ratio from two-steps variable study, the plots were also distinguished based on ΔJ ratio in these random variable cases. In such cases, at the ΔJ ratio less than 1.0, the higher rates of crack growth than those observed under constant amplitude loadings is clarified. Meanwhile, at the ΔJ ratio higher than 1.0, an equivalent manner of crack growth rates as shown by constant amplitude loading was exhibited. At the final discussion, the acceptable results due to crack propagation

estimation was yielded when the formula acquired from two-steps variable study was also employed to the random variable cases.

6.2. Recommendation for future study

In this study, at the discussion of two-steps variable and random amplitude, a higher crack growth rate than those under constant amplitude was obviously clarified when the amplitudes were reduced from high to low levels. This behavior is hypothetically expected caused by wide plasticity area formed at the end of the higher amplitude loading employment. However, the reason of those accelerated crack growth rates has not been discussed in this dissertation, yet. Correspondingly, at the future study, it is suggested to establish a modelling that can completely simulate the crack progress to reveal the contribution of the plasticity to the crack growth rate manners.

As expressed in Chapter 1, LCF failures were identified on the bridge pier particularly at the beam column connection and base joint of steel bridge piers. However, the number of LCF crack propagation tests on this bridge pier is very limited and as far the tests have mainly been conducted under constant amplitude loading. At the real condition, the bridge pier interacts with seismic loading which is attributed to the variability of the cyclic strains. Therefore, to discover crack propagation tendency under the real loading, it is suggested to test the steel bridge pier under variable amplitude conditions to characterize crack growth manner due to seismic loading employment.

References

- Agrawal, R., Uddanwadiker, R., Padole, P. (2014): Low Cycle Fatigue Life Prediction, International Journal of Emerging Engineering Research and Technology, Vol. 2(4), pp. 5-15.
- Anderson, T. L. (2011): Fracture Mechanics: Fundamentals and Applications (3rd ed.), Taylor and Francis Group.
- Bhandari V. B. (2008): Design of Machine Elements (2nd ed.), 3rd Reprint, Mc Graw Hill Education.
- ASTM E1820-031. (2008): Standard Test Method for Measurement of Fracture Toughness, pp. 1-48.
- Chen, T. and Tateishi, K. (2007): Extremely Low Cycle Fatigue Assessment of Thick Walled Steel Pier Using Local Strain Approach, Journal of Structural Engineering, Vol. 53A, pp. 485-492.
- Chung, Y. S and Abel, A. (1988): Low Cycle Fatigue of Some Aluminium Alloys, American Society for Testing and Materials STP 942, pp. 94–106.
- Coffin, L. F. (1954): A Study of the Effect of Cyclic Thermal Stresses on a Ductile Metal, Transactions of the American Society of Mechanical Engineers, Vol. 76, pp. 931–950.
- Coffin, L. F. (1971): A Note on Low Cycle Fatigue Laws, Journal of Material Science, Vol. 6(2), pp. 388–402.
- Corbly, D. M., Packman, P. F. (1973): On the influence of single and multiple peak overloads on fatigue crack propagation in 7075-T6511 aluminum, Engineering Fracture Mechanics, Vol. 5(2), pp. 479-497, 1973.

- Dahlin, P., and Olsson, M., (2007): Fatigue crack growth – Mode I cycles with periodic Mode II loading, *International Journal of Fatigue*, Vol. 30(5), pp. 931-941.
- Dong, Q., Yang, P., Xu, G. and Deng, J. (2016): Mechanisms and Modeling of Low Cycle Fatigue Crack Propagation in A Pressure Vessel Steel Q345, *International Journal of Fatigue*, Vol. 89, pp. 2-10.
- Dowling, N. E., Begley, J. A. (1976): Fatigue crack growth during gross plasticity and the J-integral, *ASTM STP 590*, pp. 82-103.
- Gardner, L. and Davies, C.M., (2010): Extremely Low Cycle Fatigue Tests on Structural Carbon Steel and Stainless Steel, *Journal of Constructional Steel Research*, Vol. 66(1), pp. 96-110.
- Gonzalez-Herrera, A., and Zapatero, J. (2008): Tri-dimensional numerical modelling of plasticity induced fatigue crack closure, *Engineering Fracture Mechanics*, Vol. 75(15), pp.4513–4528, 2008.
- Hanji, T., Tateishi, K., Terao, N. and Shimizu, M. (2017): Fatigue Crack Growth Prediction of Welded Joints in Low-Cycle Fatigue Region. *International Journal of Material Joining*, Vol. 61(6R), pp. 1189-1197.
- Hatanaka, K., and Fujimitsu, T. (1984): Some considerations on cyclic stress–strain relation and low cycle fatigue life, *Transactions of the Japan Society of Mechanical Engineers A*, Vol. 50(451), pp. 291–300.
- Gray G. T., Williams J. C., and Thompson A. W. (1983): Roughness-Induced Crack Closure: An Explanation for Microstructurally Sensitive Fatigue Crack Growth *Metallurgical Transactions A*, Vol. 14, pp. 421-433.
- Gonzalez-Herrera, A. and Zapatero, J. (2008): Tri-dimensional numerical modelling of plasticity induced fatigue crack closure, *Engineering Fracture Mechanics*, Vol. 75(15), pp.4513–4528.
- Itoh, T. and Miyazaki, T. (2003): A Damage Model for Estimating Low Cycle Fatigue Lives under Non proportional Multiaxial Loading, *European Structural Integrity*

- Society Journal, Vol. 31, pp. 423-439.
- James, M. N. and Knott, J. F. (1985): Critical aspects of the characterization of crack tip closure by compliance techniques, *Materials Science and Engineering*, Volume 72(1), 1985, pp. L1-L4.
- Jiang, Y., Feng, M., and Ding, F. (2005): A reexamination of plasticity-induced crack closure in fatigue crack propagation. *International Journal of Plasticity*, Vol. 21(9), 1720–1740.
- Jono, M., Sugeta, A. and Ohta, S. (1993): Prediction Method of Elastic-Plastic Fatigue Crack Growth Rate under Variable Amplitude Loadings, *The Japan Society of Mechanical Engineers*, Vol. 59, No. 560, pp. 939-945 (in Japanese).
- Kanvinde, A. M. and Deierlein, G. G. (2005): Continuum based micro-models for ultra low cycle fatigue crack initiation in steel structures, *Forensic Engineering Symposium*, Vol. 19(171), pp. 20–24.
- Kim, N., (2015): *Introduction to Nonlinear Finite Element Analysis*, Springer Book.
- Komotori, J. and Shimizu, M., (1998): Fracture mechanism of ferritic ductile cast iron in extremely low cycle fatigue, *Elsevier Science Ltd*, Vol. 39, pp. 39–44.
- Kuroda M. (2002): Extremely low cycle fatigue life prediction based on a new cumulative fatigue damage model, *International Journal of Fatigue*, Vol. 24(6), pp.699–703.
- Lamba, H. S. (1975): The J-integral applied to cyclic loading, *Engineering Fracture Mechanics*, Vol. 7(4), pp. 693-703.
- Lambert, Y., Saliard, P., Bathias, C. (1988): Application of the J concept to fatigue crack growth rate in large-scale yielding, *ASTM STP 969*, pp. 318-329.
- Ling, M. R. and Schijve, J. (1992). The Effect of Intermediate Heat Treatments on Overload Induced Retardation during Fatigue Crack Growth in An Al-Alloy, *Fatigue and Fracture Engineering Materials and Structures*, Vol. 15, pp. 421–430.
- Manson, S. S. (1965): Fatigue: A complex Subject – Some Simple Approximation,

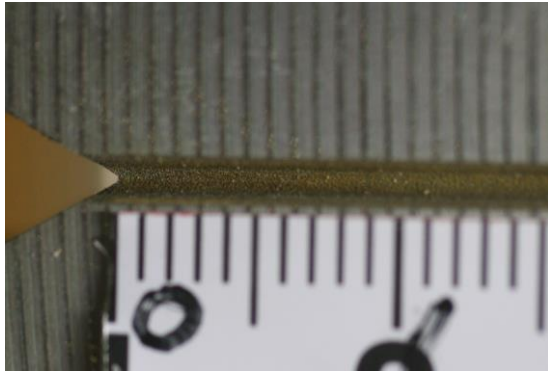
- Experiment Mechanical, Vol. 5, pp.193–226.
- Masatoshi, K. (2002): Extremely Low Cycle Fatigue Life Prediction based on a new cumulative fatigue damage model. *International Journal of Fatigue*, 24(6), 699-703.
- Matthews, W. T., Barratta, F. I., Driscoll, G. W. (1971). Experimental observations of a stress intensity history effect upon fatigue crack growth rate, *International Journal of Fracture Mechanics*, Vol. 7 (2), pp. 224-228.
- McClung, R. C. and Sehitoglu. (1989): On the finite element analysis of fatigue crack closure-2. Numerical results, *Engineering Fracture Mechanics*, Vol. 33, pp. 253–272.
- McMillan, J. C. and Pelloux, R, M. N. (1967): Fatigue Crack propagation under program and random loads, *Fatigue Crack Propagation*, ASTM STP 415, Vol. 505.
- Miki, C. (1997): Fractures in Seismically Loaded Bridges. *Progress in Structural Engineering and Materials*, Vol. 1(1), pp. 115-121.
- Murakami, Y., Harada, S., Tani-ishi, H., Fukushima, Y., and Endo, T. (1983): Correlations among Propagation Law of Small Cracks, Manson-Coffin Law and Miner Rule under Low-Cycle Fatigue, *Japan Society of Mechanical Engineers*, Vol. A49, pp. 1411-1419. (in Japanese).
- Ljustel, P. (2013): Fatigue crack growth experiments on specimens subjected to monotonic large scale yielding, *Engineering Fracture Mechanics*, Vol. 110, pp. 138-165.
- Paris, P. and Erdogan, F. (1963): A Critical Analysis of Crack Propagation Laws, *Transactions of the American Society of Mechanical Engineers Journal of Basic Engineering*, Vol. 85(4), pp.528–533.
- Park, J. E., Hanji, T., and Tateishi, K. (2015): Extremely Low Cycle Fatigue Assessment of Corner Crack in Concrete-Filled Steel Piers based on Effective Notch Strain Approach, *Journal of Structural Engineering*, Vol. 61A, pp. 662-671.
- Pommier, S. N. (2001): A study of the relationship between variable level fatigue crack growth and the cyclic constitutive behaviour of steel, *International Journal of Fatigue*,

Vol. 23, pp.111-118

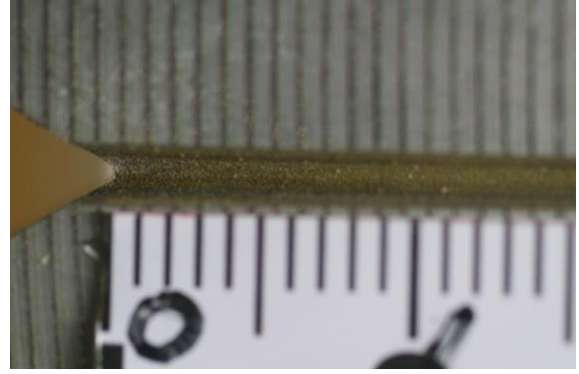
- Pommier, S. N., Bompard, P. (2000): Bauschinger effect of alloys and plasticity-induced crack closure: A finite element analysis, *Fatigue and Fracture of Engineering Materials and Structures*, Vol. 23(2), pp.129-139.
- Raghuvir K., Arbind K., Sunil, K. (1996): Delay effects in fatigue crack propagation, *International Journal of Pressure Vessels and Piping*, Vol. 67(1), pp. 1-5.
- Rice, P. (1968): A Path Integral and the Approximate Analysis of Strain Concentration by Notches and Cracks, *Journal of Applied Mechanics*, Vol. 35(2), pp. 379-386.
- Sakano, M., Kishigami, N., Ono, T., and Mikami, I. (1998): Super-Low-Cycle Fatigue Behavior of Steel Pier Base Joint with Triangular Ribs, *Journal of Structural Engineering*, Vol. 44A, pp. 1281-1288 (in Japanese).
- Sakano, M. and Wahab, M. A. (2001): Extremely Low Cycle (ELC) Fatigue Cracking Behavior in Steel Bridge Rigid Frame Piers, *Journal of Materials Processing Technology*, Vol. 118, pp. 36-39.
- Shahani, A. R., Kashani, H. M., Rastegar, M., and Dehkordi, M. B. (2009): A unified model for the fatigue crack growth rate in variable stress ratio, *Fatigue and Fracture Engineering Materials and Structures*, Vol.32(2), pp.105–118.
- Shercliff, H. R. and Fleck N. A. (1990): Effect of Specimen Geometry on Fatigue Crack Growth in Plane Strain-II. Overload Response, *Fatigue and Fracture Engineering Materials and Structures*, Vol. 13, No. 3, pp. 297–310.
- Shimada, K., Komotori, J., and Shimizu, M. (1987): The applicability of the Manson–Coffin law and Miner’s law to extremely low cycle fatigue, *Transactions of the Japan Society of Mechanical Engineers A.*, Vol. 53(491), pp. 1178–85.
- Solanki, K., Daniewicz, S. R., and Newman, J. C. (2004): Finite Element Analysis of Plasticity-Induced Fatigue Crack Closure: An Overview, *Engineering Fracture Mechanics*, Vol. 71, pp. 149–171.
- Tanaka, K., Takahashi, H. and Akiniwa, Y. (2006): Fatigue Crack Propagation from A

- Hole in Tubular Specimens under Axial and Torsional Loading, *International Journal of Fatigue*, Vol. 28, No. 4, pp. 324-334.
- Tanaka, K., Akiniwa, Y., Takahashi, A., Mikuriya, T. (2003): Fatigue Crack Propagation from A Hole in Thin-Walled Tubular Steel Specimens under Torsional-Axial Loading, *Transactions of the Japan Society of Mechanical Engineers*, Vol. 69, pp.1001–1008.
- Tanuma, Y. and Kobayashi, H. (2002): Study on Ultra Low Cycle Fatigue Crack Growth Characteristics of High Strength Steel, *Journal of Structural and Construction Engineering*, Vol. 67, No. 553, pp. 105–112. (in Japanese).
- Tateishi, K., Hanji, T., Minami, K. (2007): A prediction model for extremely low cycle fatigue strength of structural steel, *International Journal of Fatigue*, Vol. 29(5), pp. 887–896.
- Terao, N., Hanji, T. and Tateishi, K. (2015): Crack Propagation Behavior of Structural Steel in Extremely Low Cycle Fatigue Region, *IABSE Conference NARA*.
- Terao, N., Hanji, T., Tateishi, K. and Shimizu, M. (2015): A Prediction Method for Extremely Low Cycle Fatigue Crack Propagation of Structural Steel, *Proceedings of the 8th International Symposium on Steel Structures*, pp. 387-388.
- Trebules, V. W., Roberts, R., Hertzberg, R. W. (1973): Effects of multiple overloads on fatigue crack propagation in 2024-T3 aluminum alloy, *Progress in Flaw Growth and Fracture Toughness Testing*, ASTM STP 536, Vol. 113.
- Usami, T., and Kumar, S. (1996): Damage Evaluation in Steel Box Columns by Pseudodynamic Tests. *Journal of Structural Engineering*, Vol. 122, No. 6, pp. 635-642.

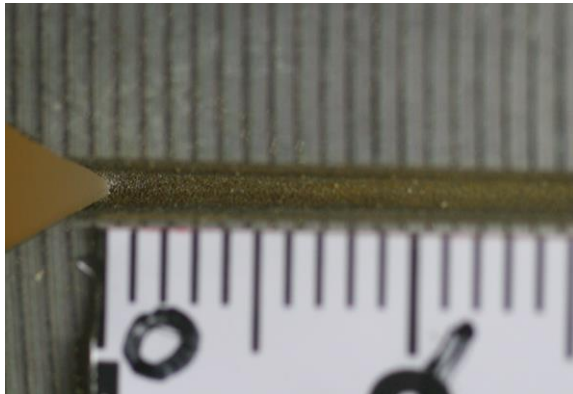
APPENDIX A: Digital image of the crack



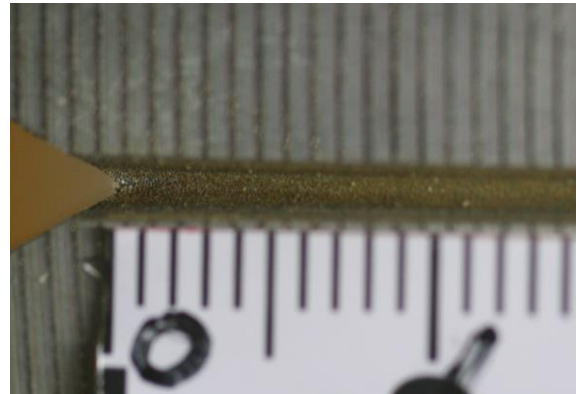
(a) 0-cycle number



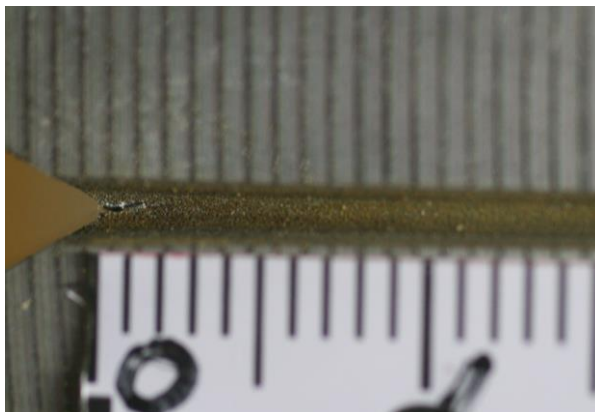
(b) 0.5-cycle number



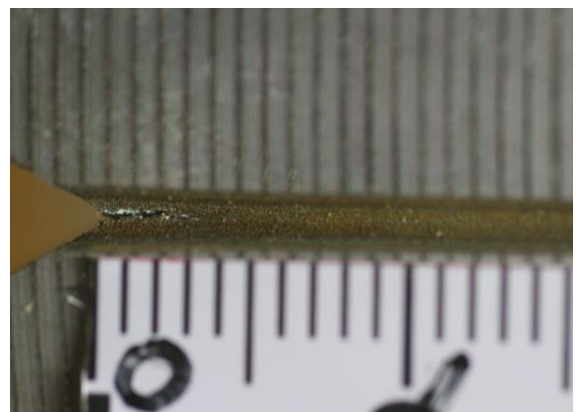
(c) 1.5-cycle number



(d) 2.5-cycle number



(e) 3.5-cycle number



(f) 4.5-cycle number

Fig. A1 Digital image of CA-1 crack

APPENDIX B: Crack propagation analysis

Table A1 Crack propagation analysis of CA-1

Cycle Number	Test result		Estimation		
	Crack growth rate	Crack length	ΔJ	Crack growth rate	Crack length
	(mm/cycle)	(mm)	(N/mm)	(mm/cycle)	(mm)
0	0	0	0	0	0
0.5	0	0	0	0	0
1.5	0.68	0.68	805.11	1.13	1.13
2.5	0.69	1.37	791.20	1.09	2.22
3.5	0.64	2.01	778.27	1.05	3.27
4.5	0.79	2.80	762.40	1.01	4.28
5.5	0.95	3.75	743.29	0.96	5.24
6.5	1.17	4.92	719.80	0.90	6.15
7.5	1.05	5.97	698.62	0.85	6.99
8.5	0.61	6.58	686.43	0.82	7.81
9.5	0.80	7.37	670.44	0.78	8.60
10.5	0.93	8.30	651.80	0.74	9.34
11.5	0.67	8.97	638.25	0.71	10.04
12.5	0.47	9.45	628.75	0.69	10.73
13.5	0.47	9.91	619.34	0.67	11.40
14.5	0.89	10.81	601.42	0.63	12.03
15.5	0.65	11.46	588.33	0.60	12.63
16.5	0.48	11.94	578.60	0.58	13.21
17.5	0.55	12.49	567.49	0.56	13.77
18.5	0.37	12.86	560.07	0.55	14.32
19.5	0.45	13.32	550.93	0.53	14.85
20.5	0.36	13.68	543.61	0.51	15.36
21.5	0.38	14.06	535.96	0.50	15.86
22.5	0.37	14.44	528.45	0.49	16.35
23.5	0.35	14.79	521.39	0.47	16.82
24.5	0.48	15.27	511.66	0.46	17.28
25.5	0.42	15.69	503.19	0.44	17.72
26.5	0.58	16.28	492.85	0.42	18.14
27.5	0.70	16.97	482.40	0.40	18.54
28.5	0.76	17.74	470.90	0.39	18.93
...	...	...	...	...	...
104.5	0.10	33.40	274.34	0.13	34.09

Table A2 Crack propagation analysis of CA-2

Cycle Number	Test result		Estimation		
	Crack growth rate	Crack length	ΔJ	Crack growth rate	Crack length
	(mm/cycle)	(mm)	(N/mm)	(mm/cycle)	(mm)
0	0	0	0	0	0
0.5	0	0	0	0	0
1.5	0.68	0.68	805.11	1.13	1.13
2.5	0.69	1.37	791.20	1.09	2.22
3.5	0.64	2.01	778.27	1.05	3.27
4.5	0.79	2.80	762.40	1.01	4.28
5.5	0.95	3.75	743.29	0.96	5.24
6.5	1.17	4.92	719.80	0.90	6.15
7.5	1.05	5.97	698.62	0.85	6.99
8.5	0.61	6.58	686.43	0.82	7.81
9.5	0.80	7.37	670.44	0.78	8.60
10.5	0.93	8.30	651.80	0.74	9.34
11.5	0.67	8.97	638.25	0.71	10.04
12.5	0.47	9.45	628.75	0.69	10.73
13.5	0.47	9.91	619.34	0.67	11.40
14.5	0.89	10.81	601.42	0.63	12.03
15.5	0.65	11.46	588.33	0.60	12.63
16.5	0.48	11.94	578.60	0.58	13.21
17.5	0.55	12.49	567.49	0.56	13.77
18.5	0.37	12.86	560.07	0.55	14.32
19.5	0.45	13.32	550.93	0.53	14.85
20.5	0.36	13.68	543.61	0.51	15.36
21.5	0.38	14.06	535.96	0.50	15.86
22.5	0.37	14.44	528.45	0.49	16.35
23.5	0.35	14.79	521.39	0.47	16.82
24.5	0.48	15.27	511.66	0.46	17.28
25.5	0.42	15.69	503.19	0.44	17.72
26.5	0.58	16.28	492.85	0.42	18.14
27.5	0.70	16.97	482.40	0.40	18.54
28.5	0.76	17.74	470.90	0.39	18.93
29.5	0.44	18.18	464.26	0.38	19.31
30.5	0.64	18.82	454.61	0.36	19.67
...	...	...	...	...	...
104.5	0.10	33.40	274.34	0.13	34.09

Table A3 Crack propagation analysis of VA-1

Cycle Numb.	Test result		Estimation						
	Crack growth rate	Crack length	ΔJ	ΔJ ratio	Crack growth rate (const.ampl) Eq. (3.2)	AF Eq. (4.2)	Crack growth rate (var. ampl.) Eq. (4.3)	Crack length (Without AF)	Crack length (With AF)
	(mm/cycle)	(mm)	(N/mm)	-	(mm/cycle)	-	(mm/cycle)	(mm)	(mm)
0	0	0	0	0	0	0	0	0	0
0.5	0	0	818.69	0	0	0	0	0	0
1.5	1.04	1.04	797.77	0.97	1.11	0.82	0.91	1.11	0.91
2.5	1.26	2.30	772.41	0.97	1.04	1.08	1.12	2.15	2.03
3.5	0.44	2.74	276.88	0.36	0.13	2.61	0.35	2.28	2.38
4.5	0.40	3.14	273.64	0.99	0.13	2.50	0.33	2.41	2.71
5.5	0.35	3.49	270.80	0.99	0.13	2.26	0.29	2.54	3.00
6.5	0.95	4.44	729.41	2.69	0.93	1	0.93	3.46	3.92
7.5	0.86	5.30	712.09	0.98	0.88	0.93	0.82	4.34	4.74
8.5	0.80	6.11	695.91	0.98	0.84	0.93	0.78	5.19	5.53
9.5	0.47	6.58	245.53	0.35	0.10	2.70	0.28	5.29	5.81
10.5	0.25	6.82	243.50	0.99	0.10	2.17	0.22	5.40	6.04
11.5	0.36	7.19	240.53	0.99	0.10	3.27	0.33	5.50	6.36
12.5	1.21	8.40	649.79	2.70	0.73	1	0.73	6.23	7.10
13.5	1.13	9.53	627.17	0.97	0.68	1.71	1.17	6.92	8.27
14.5	1.06	10.59	605.77	0.97	0.64	1.79	1.14	7.55	9.41
15.5	0.47	11.06	208.89	0.34	0.08	2.83	0.21	7.63	9.63
16.5	0.46	11.51	205.14	0.98	0.07	6.38	0.47	7.70	10.09
17.5	0.40	11.92	201.86	0.98	0.07	5.85	0.42	7.77	10.51
18.5	0.87	12.79	561.55	2.78	0.55	1	0.55	8.32	11.06
19.5	1.08	13.86	539.93	0.96	0.51	2.47	1.25	8.83	12.31
20.5	0.98	14.85	520.16	0.96	0.47	2.51	1.18	9.30	13.49
21.5	0.34	15.19	175.10	0.34	0.05	2.97	0.16	9.35	13.65
22.5	0.28	15.47	172.83	0.99	0.05	6.24	0.32	9.41	13.97
23.5	0.18	15.64	171.39	0.99	0.05	4.08	0.21	9.46	14.18
24.5	0.67	16.31	492.33	2.87	0.42	1	0.42	9.88	14.60
25.5	0.43	16.74	485.88	0.99	0.41	1.01	0.42	10.29	15.02
26.5	0.23	16.97	482.46	0.99	0.41	0.55	0.22	10.69	15.24
27.5	0.15	17.12	161.36	0.33	0.05	3.04	0.14	10.74	15.38
28.5	0.18	17.30	160.23	0.99	0.04	3.88	0.17	10.78	15.56
29.5	0.11	17.41	159.52	1.00	0.04	2.44	0.11	10.83	15.66
...	...	...	...	...	...	...	...	...	...
101.5	0.01	27.24	104.61	1.00	0.02	0.58	0.01	21.35	26.06

Table A4 Crack propagation analysis of VA-2

Cycle Numb.	Test result		Estimation						
	Crack growth rate	Crack length	ΔJ	ΔJ ratio	Crack growth rate (const.ampl) Eq. (3.2)	AF Eq. (4.2)	Crack growth rate (var. ampl.) Eq. (4.3)	Crack length (Without AF)	Crack length (With AF)
	(mm/cycle)	(mm)	(N/mm)	-	(mm/cycle)	-	(mm/cycle)	(mm)	(mm)
0	0	0	0	0	0	0	0	0	0
0.5	0	0	818.69	0	0	0	0	0	0
1.5	1.03	1.03	798.07	0.97	1.11	0.81	0.90	1.11	0.90
2.5	1.25	2.27	773.00	0.97	1.04	1.07	1.11	2.15	2.01
3.5	1.06	3.34	751.61	0.97	0.98	0.99	0.97	3.13	2.98
4.5	1.06	4.39	730.38	0.97	0.93	1.06	0.99	4.06	3.96
5.5	0.88	5.27	712.74	0.98	0.88	0.95	0.84	4.94	4.80
6.5	0.29	5.56	253.84	0.36	0.11	2.67	0.30	5.06	5.10
7.5	0.29	5.85	251.48	0.99	0.11	2.31	0.25	5.17	5.36
8.5	0.29	6.13	249.14	0.99	0.11	2.34	0.25	5.27	5.61
9.5	0.26	6.40	246.99	0.99	0.11	2.21	0.23	5.38	5.84
10.5	0.25	6.64	244.97	0.99	0.10	2.12	0.22	5.48	6.07
11.5	0.22	6.87	243.13	0.99	0.10	1.97	0.20	5.59	6.27
12.5	0.75	7.62	665.49	2.74	0.77	1	0.77	6.36	7.04
13.5	0.70	8.32	651.41	0.98	0.74	0.97	0.72	7.10	7.76
14.5	0.70	9.02	637.27	0.98	0.71	1.04	0.73	7.80	8.49
15.5	0.67	9.69	623.82	0.98	0.68	1.05	0.71	8.48	9.20
16.5	0.65	10.34	610.84	0.98	0.65	1.07	0.70	9.13	9.90
17.5	0.55	10.89	599.80	0.98	0.63	0.96	0.60	9.75	10.50
18.5	0.27	11.16	208.07	0.35	0.08	2.82	0.21	9.83	10.71
19.5	0.18	11.34	206.56	0.99	0.07	2.55	0.19	9.90	10.90
20.5	0.29	11.63	204.21	0.99	0.07	4.09	0.30	9.98	11.20
21.5	0.32	11.95	201.56	0.99	0.07	4.75	0.34	10.05	11.53
22.5	0.30	12.25	199.11	0.99	0.07	4.56	0.31	10.12	11.85
23.5	0.21	12.47	197.37	0.99	0.07	3.34	0.23	10.18	12.08
24.5	0.77	13.24	552.51	2.80	0.53	1	0.53	10.72	12.61
25.5	0.67	13.91	538.96	0.98	0.51	1.58	0.80	11.22	13.40
26.5	0.69	14.61	525.00	0.97	0.48	1.75	0.84	11.70	14.24
27.5	0.77	15.37	509.60	0.97	0.45	2.08	0.94	12.15	15.18
28.5	0.68	16.05	496.17	0.97	0.43	1.96	0.84	12.58	16.02
29.5	0.69	16.74	485.86	0.98	0.41	1.61	0.66	12.99	16.68
...	...	...	...	...	...	...	...	...	...
101.5	0.07	27.52	337.11	1.00	0.20	0.36	0.07	23.26	27.73

Table A5 Crack propagation analysis of VA-3

Cycle Numb.	Test result		Estimation						
	Crack growth rate	Crack length	ΔJ	ΔJ ratio	Crack growth rate (const.ampl) Eq. (3.2)	AF Eq. (4.2)	Crack growth rate (var. ampl.) Eq. (4.3)	Crack length (Without AF)	Crack length (With AF)
	(mm/cycle)	(mm)	(N/mm)	-	(mm/cycle)	-	(mm/cycle)	(mm)	(mm)
0	0	0	0	0	0	0	0	0	0
0.5	0	0	818.69	0	0	0	0	0	0
1.5	0.18	0.18	815.10	1.00	1.16	0.14	0.16	1.16	0.16
2.5	1.02	1.20	794.66	0.97	1.10	0.81	0.89	2.25	1.05
3.5	0.82	2.02	778.17	0.98	1.05	0.70	0.73	3.31	1.78
4.5	0.93	2.95	759.40	0.98	1.00	0.85	0.85	4.31	2.63
5.5	0.60	3.55	747.32	0.98	0.97	0.57	0.56	5.28	3.19
6.5	1.04	4.59	726.41	0.97	0.92	1.06	0.97	6.20	4.16
7.5	0.97	5.56	706.93	0.97	0.87	1.07	0.93	7.07	5.09
8.5	1.10	6.66	684.81	0.97	0.82	1.32	1.08	7.89	6.17
9.5	0.64	7.30	671.90	0.98	0.79	0.82	0.65	8.67	6.81
10.5	0.53	7.83	435.21	0.65	0.33	1.90	0.63	9.00	7.44
11.5	0.72	8.55	425.08	0.98	0.31	2.28	0.72	9.32	8.16
12.5	0.51	9.06	417.87	0.98	0.30	1.71	0.52	9.62	8.68
13.5	0.27	9.33	414.10	0.99	0.30	0.93	0.28	9.92	8.95
14.5	0.41	9.74	408.34	0.99	0.29	1.46	0.42	10.21	9.38
15.5	0.32	10.06	403.80	0.99	0.28	1.19	0.34	10.49	9.72
16.5	0.74	10.80	393.41	0.97	0.27	2.89	0.78	10.76	10.49
17.5	0.49	11.29	386.46	0.98	0.26	2.05	0.53	11.02	11.03
18.5	0.21	11.50	383.53	0.99	0.26	0.89	0.23	11.28	11.26
19.5	0.18	11.68	380.93	0.99	0.25	0.80	0.20	11.53	11.46
20.5	0.73	12.41	569.22	1.49	0.56	1	0.56	12.09	12.02
21.5	0.72	13.13	554.75	0.97	0.54	1.55	0.83	12.63	12.85
22.5	0.51	13.64	544.47	0.98	0.52	1.17	0.60	13.15	13.46
23.5	0.30	13.94	538.38	0.99	0.50	0.72	0.36	13.65	13.82
24.5	0.35	14.29	531.44	0.99	0.49	0.85	0.42	14.14	14.24
25.5	0.23	14.52	526.78	0.99	0.48	0.59	0.28	14.62	14.52
26.5	0.32	14.84	520.32	0.99	0.47	0.84	0.40	15.10	14.92
27.5	0.33	15.17	513.64	0.99	0.46	0.90	0.41	15.55	15.33
28.5	0.39	15.56	505.87	0.98	0.45	1.09	0.49	16.00	15.82
29.5	0.45	16.01	496.83	0.98	0.43	1.33	0.57	16.43	16.39
...	...	...	...	...	...	...	...	...	...
109.5	0.09	29.70	312.60	1.00	0.17	0.55	0.09	30.18	30.35

Table A6 Crack propagation analysis of VA-4

Cycle Numb.	Test result		Estimation						
	Crack growth rate	Crack length	ΔJ	ΔJ ratio	Crack growth rate (const.ampl) Eq. (3.2)	AF Eq. (4.2)	Crack growth rate (var. ampl.) Eq. (4.3)	Crack length (Without AF)	Crack length (With AF)
	(mm/cycle)	(mm)	(N/mm)	-	(mm/cycle)	-	(mm/cycle)	(mm)	(mm)
0	0	0	0	0	0	0	0	0	0
0.5	0	0	818.69	0	0	0	0	0	0
1.5	0.26	0.26	813.47	0.99	1.15	0.20	0.23	1.15	0.23
2.5	0.29	0.55	807.69	0.99	1.14	0.22	0.25	2.29	0.48
3.5	1.43	1.98	778.84	0.96	1.06	1.20	1.26	3.34	1.75
4.5	1.50	3.48	748.62	0.96	0.98	1.40	1.36	4.32	3.11
5.5	0.51	3.99	738.43	0.99	0.95	0.50	0.48	5.27	3.58
6.5	0.77	4.76	723.03	0.98	0.91	0.80	0.73	6.18	4.31
7.5	1.96	6.72	683.65	0.95	0.81	2.30	1.87	6.99	6.18
8.5	0.35	7.06	676.70	0.99	0.80	0.44	0.35	7.79	6.53
9.5	0.61	7.67	664.42	0.98	0.77	0.81	0.62	8.55	7.15
10.5	0.06	7.74	70.60	0.11	0.01	4.48	0.04	8.56	7.19
11.5	0.05	7.79	70.43	1.00	0.01	5.60	0.05	8.57	7.24
12.5	0.04	7.83	70.31	1.00	0.01	4.16	0.04	8.58	7.27
13.5	0.04	7.87	70.19	1.00	0.01	3.91	0.03	8.59	7.30
14.5	0.03	7.90	70.09	1.00	0.01	3.41	0.03	8.60	7.33
15.5	0.03	7.93	70.00	1.00	0.01	3.02	0.03	8.61	7.36
16.5	0.03	7.95	69.92	1.00	0.01	2.77	0.02	8.61	7.38
17.5	0.02	7.98	69.85	1.00	0.01	2.56	0.02	8.62	7.40
18.5	0.02	8.00	69.78	1.00	0.01	2.39	0.02	8.63	7.43
19.5	0.02	8.02	69.72	1.00	0.01	2.22	0.02	8.64	7.44
20.5	0.13	8.15	654.81	9.39	0.75	1	0.75	9.39	8.19
21.5	0.60	8.75	642.77	0.98	0.72	0.87	0.62	10.10	8.81
22.5	0.69	9.44	628.87	0.98	0.69	1.06	0.73	10.79	9.54
23.5	0.81	10.25	612.53	0.97	0.65	1.33	0.87	11.45	10.41
24.5	0.96	11.22	593.19	0.97	0.61	1.71	1.05	12.06	11.46
25.5	0.92	12.13	574.78	0.97	0.57	1.78	1.02	12.63	12.49
26.5	0.27	12.40	569.34	0.99	0.56	0.55	0.31	13.20	12.80
27.5	0.22	12.63	564.82	0.99	0.56	0.47	0.26	13.75	13.06
28.5	0.34	12.97	557.99	0.99	0.54	0.73	0.40	14.29	13.46
29.5	0.43	13.40	549.30	0.98	0.53	0.97	0.51	14.82	13.96
...	...	...	...	...	...	...	...	...	...
119.5	0.00	29.40	23.71	1.00	0.00	0.38	0.00	26.01	30.34

Table A7 Crack propagation analysis of RA-1

Cycle Numb.	Test result		Estimation						
	Crack growth rate	Crack length	ΔJ	ΔJ ratio	Crack growth rate (const.ampl) Eq. (3.2)	AF Eq. (4.2)	Crack growth rate (var. ampl.) Eq. (4.3)	Crack length (Without AF)	Crack length (With AF)
	(mm/cycle)	(mm)	(N/mm)	-	(mm/cycle)	-	(mm/cycle)	(mm)	(mm)
0	0	0	0	0	0	0	0	0	0
0.5	0	0	754.74	0	0	0	0	0	0
1.5	1.05	1.05	917.95	1.22	1.47	1	1.47	1.47	1.47
2.5	1.69	2.74	1130.35	1.23	2.22	1	2.22	3.69	3.69
3.5	0.64	3.38	596.81	0.53	0.62	1.89	1.17	4.31	4.86
4.5	0.34	3.72	418.89	0.70	0.31	1.87	0.57	4.61	5.43
5.5	1.59	5.31	947.28	2.26	1.56	1	1.56	6.18	7.00
6.5	0.21	5.52	262.77	0.28	0.12	2.85	0.34	6.30	7.34
7.5	1.83	7.34	1019.76	3.88	1.81	1	1.81	8.11	9.15
8.5	1.15	8.49	687.34	0.67	0.82	1.66	1.37	8.93	10.52
9.5	0.26	8.75	313.60	0.46	0.17	2.34	0.40	9.10	10.92
10.5	0.25	9.00	292.86	0.93	0.15	1.99	0.30	9.25	11.21
11.5	0.27	9.27	320.87	1.10	0.18	1	0.18	9.43	11.39
12.5	0.17	9.44	280.70	0.87	0.14	2.02	0.28	9.56	11.67
13.5	0.22	9.66	291.15	1.04	0.15	1	0.15	9.71	11.82
14.5	0.17	9.83	256.68	0.88	0.11	2.07	0.24	9.83	12.05
15.5	1.18	11.01	604.73	2.36	0.64	1	0.64	10.46	12.69
16.5	1.18	12.19	532.22	0.88	0.49	1.68	0.83	10.96	13.52
17.5	1.09	13.27	645.41	1.21	0.72	1	0.72	11.68	14.24
18.5	0.27	13.54	275.27	0.43	0.13	2.47	0.33	11.81	14.57
19.5	0.21	13.76	197.84	0.72	0.07	2.31	0.16	11.88	14.73
20.5	0.94	14.69	575.66	2.91	0.58	1	0.58	12.46	15.30
21.5	0.40	15.10	355.41	0.62	0.22	2.04	0.45	12.68	15.75
22.5	0.39	15.48	313.42	0.88	0.17	1.95	0.33	12.85	16.08
23.5	0.24	15.72	203.55	0.65	0.07	2.35	0.17	12.92	16.25
24.5	0.23	15.95	181.08	0.89	0.06	2.31	0.13	12.98	16.38
25.5	0.25	16.20	247.91	1.37	0.11	1	0.11	13.08	16.49
26.5	0.57	16.78	470.27	1.90	0.38	1	0.38	13.47	16.88
27.5	0.25	17.03	248.88	0.53	0.11	2.36	0.25	13.58	17.13
28.5	0.22	17.25	195.93	0.79	0.07	2.28	0.15	13.64	17.28
29.5	0.19	17.43	227.83	1.16	0.09	1	0.09	13.73	17.37
...	...	...	...	...	...	...	...	...	...
119.5	0.05	32.73	91.57	0.65	0.01	2.99	0.04	25.51	33.36

Table A8 Crack propagation analysis of RA-2

Cycle Numb.	Test result		Estimation						
	Crack growth rate	Crack length	ΔJ	ΔJ ratio	Crack growth rate (const.ampl) Eq. (3.2)	AF Eq. (4.2)	Crack growth rate (var. ampl.) Eq. (4.3)	Crack length (Without AF)	Crack length (With AF)
	(mm/cycle)	(mm)	(N/mm)	-	(mm/cycle)	-	(mm/cycle)	(mm)	(mm)
0	0	0	0	0	0	0	0	0	0
0.5	3.44	3.44	1254.65	0	2.74	1	2.74	2.74	2.74
1.5	2.28	5.72	1068.64	0.85	1.99	1.40	2.78	4.73	5.52
2.5	0.23	5.95	407.35	0.38	0.29	2.34	0.68	5.01	6.20
3.5	2.02	7.97	1310.48	3.22	2.99	1	2.99	8.00	9.19
4.5	0.81	8.78	475.59	0.36	0.39	2.31	0.91	8.40	10.09
5.5	2.60	11.38	1162.13	2.44	2.35	1	2.35	10.75	12.44
6.5	1.97	13.35	995.44	0.86	1.72	1.42	2.46	12.47	14.90
7.5	0.74	14.10	476.06	0.48	0.39	2.08	0.82	12.87	15.72
8.5	2.74	16.84	1254.42	2.63	2.74	1	2.74	15.60	18.46
9.5	1.18	18.02	475.21	0.38	0.39	2.27	0.89	16.00	19.35
10.5	1.16	19.18	726.08	1.53	0.92	1	0.92	16.91	20.27
11.5	0.97	20.15	605.07	0.83	0.64	1.63	1.04	17.55	21.30
12.5	0.49	20.65	225.04	0.37	0.09	2.71	0.24	17.64	21.54
13.5	0.61	21.26	419.06	1.86	0.31	1	0.31	17.94	21.85
14.5	0.68	21.94	812.02	1.94	1.15	1	1.15	19.09	23.00
15.5	0.20	22.14	132.14	0.16	0.03	3.70	0.11	19.12	23.11
16.5	0.70	22.84	762.94	5.77	1.01	1	1.01	20.13	24.12
17.5	0.35	23.19	162.84	0.21	0.05	3.37	0.16	20.18	24.28
18.5	0.46	23.65	340.91	2.09	0.20	1	0.20	20.38	24.48
19.5	0.48	24.14	257.67	0.76	0.12	2.12	0.24	20.50	24.72
20.5	0.40	24.53	209.04	0.81	0.08	2.23	0.17	20.57	24.89
21.5	0.36	24.89	182.92	0.88	0.06	2.30	0.13	20.63	25.03
22.5	0.16	25.05	118.25	0.65	0.02	2.77	0.07	20.66	25.09
23.5	0.44	25.49	358.16	3.03	0.22	1	0.22	20.88	25.32
24.5	0.46	25.95	381.77	1.07	0.25	1	0.25	21.13	25.57
25.5	0.47	26.42	612.91	1.61	0.65	1	0.65	21.79	26.22
26.5	0.61	27.03	936.69	1.53	1.53	1	1.53	23.31	27.75
27.5	0.31	27.34	161.54	0.17	0.05	3.51	0.16	23.36	27.91
28.5	0.34	27.68	301.35	1.87	0.16	1	0.16	23.52	28.07
29.5	0.34	28.02	253.64	0.84	0.11	2.09	0.23	23.63	28.30
...	...	...	...	...	...	...	...	...	...
119.5	0.06	43.81	155.77	0.81	0.04	2.44	0.10	36.64	44.56

Table A7 Crack propagation analysis of RA-3

Cycle Numb.	Test result		Estimation						
	Crack growth rate	Crack length	ΔJ	ΔJ ratio	Crack growth rate (const.ampl) Eq. (3.2)	AF Eq. (4.2)	Crack growth rate (var. ampl.) Eq. (4.3)	Crack length (Without AF)	Crack length (With AF)
	(mm/cycle)	(mm)	(N/mm)	-	(mm/cycle)	-	(mm/cycle)	(mm)	(mm)
0	0	0	0	0	0	0	0	0	0
0.5	0.00	0.00	1128.47	0	0.00	0	0.00	0.00	0.00
1.5	2.32	2.32	1445.70	1.28	3.64	1	3.64	3.64	3.64
2.5	1.15	3.46	730.49	0.51	0.93	1.84	1.71	4.57	5.35
3.5	3.30	6.77	1841.15	2.52	5.90	1	5.90	10.46	11.24
4.5	0.81	7.58	339.73	0.18	0.20	2.98	0.60	10.66	11.84
5.5	1.10	8.68	486.12	1.43	0.41	1	0.41	11.08	12.25
6.5	1.08	9.75	697.51	1.43	0.85	1	0.85	11.92	13.10
7.5	0.94	10.69	521.34	0.75	0.47	1.74	0.82	12.39	13.92
8.5	1.27	11.96	620.08	1.19	0.67	1	0.67	13.06	14.59
9.5	1.21	13.18	583.16	0.94	0.59	1.63	0.96	13.66	15.55
10.5	1.56	14.74	966.24	1.66	1.62	1	1.62	15.28	17.18
11.5	1.25	15.99	519.25	0.54	0.47	1.94	0.91	15.75	18.09
12.5	1.74	17.73	1093.19	2.11	2.08	1	2.08	17.83	20.17
13.5	1.68	19.41	1337.02	1.22	3.11	1	3.11	20.94	23.28
14.5	0.70	20.10	369.22	0.28	0.24	2.66	0.63	21.18	23.91
15.5	0.49	20.59	200.41	0.54	0.07	2.48	0.17	21.25	24.08
16.5	0.99	21.58	489.13	2.44	0.42	1	0.42	21.66	24.50
17.5	0.69	22.28	384.06	0.79	0.26	1.87	0.48	21.92	24.98
18.5	1.11	23.39	603.15	1.57	0.63	1	0.63	22.55	25.61
19.5	0.59	23.98	224.99	0.37	0.09	2.71	0.24	22.64	25.85
20.5	0.57	24.55	224.08	1.00	0.09	2.15	0.19	22.73	26.04
21.5	1.54	26.09	1033.76	4.61	1.86	1	1.86	24.59	27.90
22.5	0.76	26.84	314.99	0.30	0.17	2.67	0.46	24.76	28.36
23.5	1.23	28.08	586.44	1.86	0.60	1	0.60	25.36	28.96
24.5	1.48	29.56	882.97	1.51	1.36	1	1.36	26.71	30.31
25.5	1.30	30.87	707.93	0.80	0.87	1.57	1.37	27.59	31.69
26.5	1.27	32.13	580.25	0.82	0.59	1.65	0.97	28.17	32.66
27.5	1.60	33.73	976.60	1.68	1.66	1	1.66	29.83	34.32
28.5	1.47	35.20	780.96	0.80	1.06	1.54	1.63	30.89	35.95
29.5	1.11	36.31	404.12	0.52	0.28	2.10	0.60	31.18	36.54
...	...	...	...	...	...	...	...	...	...
119.5	0.21	55.43	443.22	1.22	0.34	1	0.34	47.24	56.03

